Liquidity and Information Aggregation in Large Markets*

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Abstract

We study a financial market with information production and free entry of strategic traders. Equilibrium information aggregation is determined by a sufficient statistic that depends on the information technology only through the marginal cost of information at the prior. In contrast to Grossman and Stiglitz (1980), equilibrium prices may successfully aggregate information, and we provide necessary and sufficient conditions for full revelation. Critical to this efficiency result is a novel two-way positive feedback loop between information and liquidity, which matters despite the market being large and competitive. We explore two normative implications of our model: (i) information efficiency need not harm retail traders nor require excessive information expenditures; and (ii) market designs that segment retail traders are inefficient.

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1 Introduction

How well do competitive financial markets aggregate information? The answer matters for central questions in economics, including whether decentralized markets allocate resources efficiently. Unfortunately, according to the prevalent view, markets cannot efficiently aggregate information: if prices reveal information, investors may prefer to learn from prices rather than acquire information themselves (Grossman and Stiglitz, 1980). At the same time, it remains unclear how severely information aggregation fails, or which primitives determine the extent of the failure. We revisit the question of information aggregation and obtain two novel findings. Our analysis delivers a *possibility* result—the wisdom of the crowd can hold—as well as a *precise characterization* of information aggregation as a function of the information technology.

Our starting point is a strategic trading model à la Kyle (1985, 1989) and Lambert et al. (2018). To these frameworks, we add two key ingredients: free entry, which is a hallmark of competition, and smooth information costs. Free entry implies that the number of traders grows large in equilibrium. Smooth information costs allow traders to be "informationally small." Traders endogenously acquire information, but less so as entry makes markets increasingly competitive. Despite traders being informationally small, both the aggregate amount of information and price informativeness can grow arbitrarily large. We obtain necessary and sufficient conditions for information efficiency in the free-entry equilibrium. In cases where information aggregation is imperfect, we provide a precise characterization of price informativeness, and other market outcomes, entirely as a function of the marginal cost of the first piece of information.

Model and results. Our baseline model features a single risky asset traded by n ex-ante identical and risk-neutral strategic traders, as well as noise traders and an uninformed, competitive, market-maker. Strategic traders have access to an information technology allowing them to learn about the asset value at a cost. Concretely, the information technology is represented by the cost of obtaining a signal with precision z above the prior, given by $c(z)$. In contrast with most of the strategic trading literature, we allow $c(z)$ to be smooth. After acquiring information, strategic traders submit market orders, anticipating their price impact, and competitive market-makers set a break-even price upon observing the order flow. We are interested in the free-entry equilibrium, that is, when the number of traders is determined by a zero ex-ante profits condition.

A fundamental property in our model is that the information technology only matters via $\chi := c'(0)$—the marginal cost of precision at the prior. The reasoning traces

precisely to free entry and smooth information costs. First, given any finite number of strategic traders n, there remain positive profits from collecting information and exploiting it in the market. Free entry allows another strategic trader to enter; smoothness of $c(z)$ allows him to economize on information costs to compete with the existing traders. Indeed, upon facing a new entrant, all traders cut back on information expenditures, due to a more informative market that drags down their trading revenues. This enter-and-compete process continues *ad infinitum*. Therefore, the unique free-entry equilibrium must feature an ever-growing number of strategic traders ($n \to \infty$) who individually collect an ever-shrinking quantity of information ($z_i \to 0$ for each i). In other words, our market consists of a large number of "informationally small" traders. Given informational smallness, $c(z)$ only matters via its local derivative at zero, $\chi := c'(0)$.

Leveraging this property, our main substantive result precisely characterizes the free-entry equilibrium for any χ. We show that price informativeness is an increasing function of the following sufficient statistic:

$$\Gamma := \frac{1}{2}\Big(\frac{\text{fundamental volatility} \times \text{noise trader volatility}}{\chi}\Big)^2.$$

The fact that informativeness is increasing in both fundamental volatility and noise is intuitive: both uncertainties allow strategic traders to "conceal" their information when trading, leading to greater information rents, thereby augmenting incentives for information acquisition. Such a tractable characterization is typically unavailable in small or concentrated markets, where details of the entire information cost function matter.

Surprisingly, and in contrast to Grossman and Stiglitz (1980), prices can be fully-revealing in the free-entry limit. We prove that this occurs *if and only if* $\chi = 0$. When prices are arbitrarily revealing, information-acquisition incentives are virtually absent. Therefore, it is immediate that a *necessary* condition for efficiency is that a small piece of information can be obtained at a vanishingly small marginal cost. The key contribution is to show this is also *sufficient*: as long as $\chi = 0$, then regardless of the other details of the cost function, prices become arbitrarily informative as the market grows. Although each individual trader acquires virtually no information, the aggregate level of information is high. The wisdom of the crowd obtains. At its source, this result depends on a positive two-way feedback between informativeness and liquidity, described next.

The role of liquidity. In our model, traders acknowledge market illiquidity. This is not the source of our results. On the contrary, traders acting strategically works against information efficiency: price impact curbs aggressive trades, which mitigates

the flow of information into the market. If traders (schizophrenically) were to disregard their price impact, they would impound more information into prices. Why, then, are informationally-efficient markets possible in our model? It turns out that markets with perfect information aggregation *are also* markets with perfect liquidity.

To explain this, we isolate a two-way positive feedback loop between price informativeness and market liquidity, displayed in Figure 1. On the one hand, improvements in liquidity always foster price informativeness, for two well-understood reasons: traders are more aggressive with information they already have, and, knowing that information is more profitable, traders acquire even more. On the other hand, in a sufficiently informative market, price informativeness encourages liquidity. The idea is that market-makers, knowing that almost all order flow reflects fundamental value, become confident in setting the price regardless of order flow—which means price impact is low.

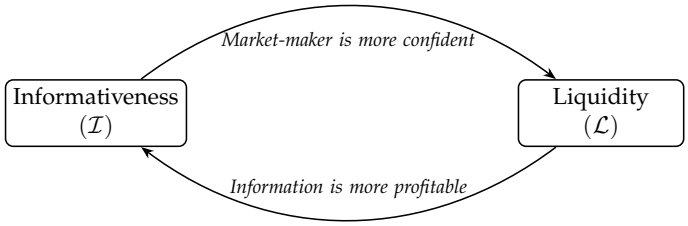


Figure 1: The informativeness–liquidity feedback loop.

This feedback loop is essential to information aggregation. We demonstrate this formally by separating trader-perceived price impact from actual price impact: if the trader-perceived price impact does not vanish, price informativeness can never be perfect. Thus, perfect information aggregation is necessarily accompanied by perfect liquidity.

Conversely, under partial information aggregation, illiquidity issues do not vanish. In such cases where illiquidity persists, all of our comparative statics align with small-market strategic-trading models, even though our market is large and profits small.

These results contrast starkly with Noisy Rational Expectations Equilibrium (NREE) models. As the dominant paradigm, NREE models say that competitive financial markets are informationally-inefficient and perfectly liquid (at least from the perspective of the model's traders, who take prices as given). Our model provides an entirely different view. Competitive markets can never be both inefficient and liquid: they are always *either* informationally-efficient *or* illiquid.

Normative implications. We highlight two implications for competitive financial markets: (1) higher price informativeness need not harm retail traders or aggregate effi-

ciency; (2) market designs that segment retail traders can be inefficient.

First, consider the costs of an informationally-efficient market. To arrive at such a market, two things occur: lots of information is acquired, and lots of strategic trade takes place. Information expenditures are dead-weight losses in our model. At the same time, informed order flow generates transfers from noise traders to sophisticated traders, an important distributional consequence. We show that, in an informationally-efficient market, neither concern is significant. That is, aggregate information expenditures and aggregate trading revenue collapse to zero as the market becomes arbitrarily informative. This result traces back to liquidity: in an information-rich market, market-makers can confidently set prices and absorb order flow, so they induce fierce competition across strategic traders, which competes aggregate trading revenues down to zero.

Second, we document that an increase in noise unambiguously raises both liquidity and informativeness. While this relationship appears in some studies of small or concentrated markets (Admati and Pfleiderer, 1988b; Kyle, 1989; Banerjee and Breon-Drish, 2022), it is novel to large, competitive markets. In fact, in NREE models, an increase in noise leads to *lower* informativeness. The discrepancy can be traced to the improvement in liquidity, which has consequences in our model but not in NREE models.

This result suggests that segmenting retail orders can be inefficient. Currently, a large share of retail orders flow to intermediaries called wholesalers, who may execute those orders outside of large exchanges (Dyhrberg et al., 2025; Ernst et al., 2023). The typical justification is that wholesalers are willing to provide discounts to retail orders, which are largely uninformed. Our model suggests caution: removing retail order flow from the exchange removes noise, lowering incentives for information production and thereby reducing price informativeness. A conventional NREE-style model would conclude the opposite, that segmentation improves price efficiency (an example of such a result, whereby "dark pools" reduce on-exchange noise and thereby improve price discovery, can be found in Zhu, 2014).[1]

Extensions and generality. Our results do not depend on risk-neutrality or the trading protocol. Although our baseline model studies risk-neutral traders who submit market orders (Kyle, 1985), our conclusions continue to hold when traders are risk-averse and submit limit orders (Kyle, 1989). These extensions preserve both the possibility of information aggregation and a precise characterization of price informativeness.

Finally, any "wisdom of the crowd" result requires dispersed information. Our base-

[1]Some theoretical research argues that such order flow segmentation may harm liquidity (Parlour and Rajan, 2003; Parlour and Seppi, 2003), but adverse consequences for informativeness are less explored.

line model follows this tradition by assuming that traders' signals are truth plus independent errors, so the errors wash out in the aggregate. Surprisingly, this assumption is, to some degree, innocuous: we extend the model to allow traders to choose their information sources, thereby allowing correlated signals. Despite the presence of incentives to coordinate signals with other traders, we prove that the unique equilibrium features independent sources—dispersed information is the natural equilibrium outcome.

Related literature. Our paper follows the tradition of modeling strategic financial markets (Kyle, 1985, 1989; Lambert et al., 2018), especially a subset with endogenous information acquisition (Admati and Pfleiderer, 1988a, 1988b; Kyle, 1989; Subrahmanyam, 1991; Froot et al., 1992; Foster and Viswanathan, 1993; Peress, 2010; García and Sangiorgi, 2011; Lee, 2013; Glebkin and Kuong, 2023; Banerjee and Breon-Drish, 2020, 2022; Kacperczyk et al., 2025). This literature is typically applied to small or concentrated markets. Like us, a few exceptions study the competitive limit. Lee and Kyle (2018) study a large-market limit, albeit with exogenous information. They are concerned with what makes markets perfectly competitive, whereas we ask what makes markets efficient. Section 10 of Kyle (1989) allows both free entry and endogenous information, albeit with fixed information costs. In that setup, prices can never be fully-revealing unless the fixed costs vanish. Finally, García and Sangiorgi (2011) introduces a monopolist information seller and finds that risk-sharing demands govern whether equilibrium information is concentrated versus dispersed among many traders. Our paper differs along several dimensions: traders choose their information individually, dispersed informationally-small traders is the generic outcome, and prices can become fully-revealing.

NREE models currently serve as the dominant paradigm for studying price efficiency in competitive markets. By taking free entry seriously and adopting smooth information costs, we arrive at an equilibrium that never looks like an NREE: prices are either highly informationally-efficient, or liquidity issues persist. The NREE literature obtained different results because it considered at most one of our core assumptions at a time. For instance, Grossman and Stiglitz (1980) studies a free-entry equilibrium with fixed information costs, while Verrecchia (1982) adopts smooth information costs but effectively disallows free entry.

Three broad approaches have proved successful in circumventing the Grossman-Stiglitz paradox: adding a source of private value for information (Vives, 2011, 2014); preventing traders from conditioning trade on prices (Milgrom, 1981; Hellwig, 1982; Dubey et al., 1987); or explicitly modeling the price-formation process (Jackson, 1991). Our paper falls closest to the third tradition, despite adopting very different assump-

tions, with a novel emphasis on free entry and information costs.

Finally, we complement the research on information aggregation in strategic environments with common values, such as auctions (Jackson, 2003; Pesendorfer and Swinkels, 2000; Atakan and Ekmekci, 2023) and voting (Martinelli, 2006). Like these papers, we obtain conditions for aggregate levels of acquired information to be high even when individual incentives for learning vanish, but in a canonical financial market.

2 Baseline Model

We study a static strategic trading model with multiple potentially informed traders and costly information acquisition. Information acquisition is decided ex-ante, subject to a cost for signal precision. The baseline model below features risk-neutral traders who submit market orders to a competitive market-maker, following Kyle (1985) and Lambert et al. (2018). Neither risk-neutrality nor the specific trading protocol are fundamental. In Section 4.1, we discuss how our results remain the same if trading occurs through limit orders instead of market orders, and how they extend to the case of risk-aversion.

A single risky asset with random payoff v is traded in a market with n risk-neutral strategic traders. In this market, there is also a representative uninformed market-maker and unmodeled noise traders that submit demand u. We assume that noise trader demand and fundamentals are independently normally distributed with mean normalized to zero: $v \sim \text{Normal}(0, \tau_v^{-1})$, $u \sim \text{Normal}(0, \tau_u^{-1})$, and $u \perp v$.

Before trading, investors have an opportunity to collect information about v in the form of a signal about the security value. Formally, strategic trader i has access to a signal $s_i = v + \varepsilon_i$, where ε_i is the error in the signal, which is normally distributed with variance τ_i^{-1}. At a cost, the investor can choose the variance of the error ε_i. We assume that each trader's error is independent of fundamentals, $\varepsilon_i \perp v$, and that traders' errors are independent of each other, i.e., $\varepsilon_i \perp \varepsilon_j$ for any pair of traders (i, j). In Section 4.2, we revisit the conditional independence of errors and show that it is without loss of generality: by allowing traders to choose their source of information, we show that they would always want to select into distinct signals.

The trading protocol follows Kyle (1985). Each trader submits a market order d_i, and noise trader demand u realizes. The market-maker observes aggregate demand $Y = \sum_i d_i + u$ and sets a break-even price given the information in the order flow:

$$p = \mathbb{E}[v|Y]. \tag{1}$$

If trader i demands d_i units of the asset and market prices are p, her ex-post profit is:

$$d_i\,(v-p)\,.$$

Prior to trading, information is gathered subject to a cost. For convenience, we will let traders choose a normalization of information precision, $z_i \in [0,1]$, which is defined as follows, using the variance of trader i's error, τ_i^{-1}:

$$z_i := \frac{\tau_i}{\tau_i + 2\tau_v}. \tag{2}$$

If $z_i = 0$, then $\tau_i = 0$, and trader i's signal is entirely uninformative. If $z_i = 1$, then $\tau_i = \infty$, and the signal is perfectly informative. Indeed, the precision of an agent's signal is a monotonic function of z_i. We let $c(z_i)$ denote the cost of acquiring normalized precision z_i, where $c : [0,1] \to \mathbb{R}_+$ is a continuously differentiable and strictly convex function satisfying $c(0) = 0$, $c'(0) \geq 0$, and $c'(1-) = \infty$. The latter condition is simply for ease of exposition. As in the entire literature, we assume that information costs are convex in raw precision τ_i, which implies that $c(\cdot)$ is sufficiently convex[2]

Given a collection of precisions, $\mathbf{z} = (z_i)_{i=1}^n$, trader i chooses d_i to maximize expected profits, conditional on her signal realization s_i. Because traders are strategic, they anticipate that their order will change prices, given by a function $\boldsymbol{p}(\cdot, \mathbf{z})$ of the aggregate demand. Moreover, traders take as given the demand functions of other traders, $\boldsymbol{d}_j(\cdot, \mathbf{z})$, which maps trader j's private information to a market order. We let $\tilde{V}_i$ represent i's profits given a collection of precisions conditional on a signal realization. The trader's problem is then:

$$\tilde{V}_i(s_i|\mathbf{z}) := \max_{d_i} \mathbb{E}\left[d_i \left(v - \boldsymbol{p}\Big(\sum_{j\neq i} \boldsymbol{d}_j(s_j, \mathbf{z}) + d_i + u,\ \mathbf{z} \Big) \right) \middle| s_i \right]. \tag{3}$$

Before the trading stage, trader i chooses how much information to collect while taking all other traders' precisions as given. That choice trades off the benefits of information in the subsequent trading equilibrium against the cost of information acquisition:

$$\max_{z\in[0,1]} \mathbb{E}\left[\tilde{V}_i(s_i|z, \mathbf{z}_{-i})\right] - c(z), \tag{4}$$

[2]In particular, let $\tilde{c}(\tau)$ be the convex cost of raw precision. Then, using (2), the cost of normalized precision is $c(z) := \tilde{c}(2\tau_v \frac{z}{1-z})$. Differentiating, $c'(z) = \frac{2\tau_v}{(1-z)^2}\tilde{c}'(\tau)$ and so $c''(z) = (\frac{2\tau_v}{(1-z)^2})^2\tilde{c}''(\tau) + \frac{2}{1-z}c'(z)$. Thus, assuming $\tilde{c}''(\tau) > 0$ is equivalent to assuming $c''(z) > \frac{2}{1-z}c'(z)$.

where $\mathbf{z}_{-i}$ represents the vector of precisions of all other traders. We are now ready to define equilibrium.

Definition 1. *An equilibrium is a collection of precisions,* $\mathbf{z}$*, demand functions* $(\boldsymbol{d}_i)_{i=1}^n$*, with* $\boldsymbol{d}_i : \mathbb{R} \times [0,1]^n \to \mathbb{R}$*, and a price function* $\boldsymbol{p} : \mathbb{R} \to \mathbb{R}$*, such that:*

1. *For any collection of precisions* $\tilde{\mathbf{z}}$*,* $\left\{(\boldsymbol{d}_i(\cdot, \tilde{\mathbf{z}}))_{i=1}^n, \boldsymbol{p}(\cdot, \tilde{\mathbf{z}})\right\}$ *is a linear trading equilibrium. That is:*
   (a) *Traders optimize: for each trader* i*, given prices, other traders' demand functions, and signal realization* s_i*, demand* $\boldsymbol{d}_i(s_i, \tilde{\mathbf{z}})$ *solves (3).*
   (b) *Market-makers break even:* $\boldsymbol{p}(Y, \tilde{\mathbf{z}}) = \mathbb{E}[v|Y]$*,*
   (c) *Demand and prices are linear: there exist constants* $\lambda(\tilde{\mathbf{z}}), \beta_i(\tilde{\mathbf{z}})$*, such that:* $\boldsymbol{p}(Y, \tilde{\mathbf{z}}) = \lambda(\tilde{\mathbf{z}})Y$*, and* $\boldsymbol{d}_i(s_i, \tilde{\mathbf{z}}) = \beta_i(\tilde{\mathbf{z}})s_i$*.*
2. $\mathbf{z}$ *is an information equilibrium. That is,* z_i *solves (4) given* $\mathbf{z}_{-i}$*.*

Regarding the notation above, we emphasize that $\boldsymbol{d}_i$ and $\boldsymbol{p}$ are used to as demand and pricing *functions*, which are defined for an arbitrary vector of precisions $\mathbf{z}$, whereas d_i and p are the equilibrium demand and price (not functions but random variables). While this notation is rather cumbersome in principle, most of the analysis below does not need it. Indeed, we show below that our notation substantially simplifies in terms of a few sufficient summary statistics.

After describing some basic properties of equilibrium for any finite n, the paper then studies the *free-entry* or *competitive* limit of this market. As we will see, this free-entry limit is equivalent to the limit as the number of strategic traders grows large ($n \to \infty$).

2.1 Linear trading equilibrium

For any information choice decision, $\mathbf{z}$, we can compute the equilibrium of the associated trading game. Although the result is a special case of Lambert et al. (2018), it is useful to express the equilibrium using the economic quantities we explore later. To that end, we define two critical statistics. Define *relative price informativeness* as the fraction of the fundamental variance left after observing prices in an equilibrium:[3]

$$I(\mathbf{z}) := 1 - \frac{\text{Var}[v|p, \mathbf{z}]}{\text{Var}[v]}.$$

[3]Many definitions of price informativeness have been used in the literature. We follow the usage in Grossman and Stiglitz (1980).

As $I(\mathbf{z}) \to 1$, prices become perfectly informative, whereas $I(\mathbf{z}) \to 0$ corresponds to zero information revealed by prices. Moreover, we measure *liquidity* as the inverse of the price impact of a marginal noise trade, which is the inverse of Kyle's lambda:

$$L(\mathbf{z}) := \frac{1}{\lambda(\mathbf{z})}.$$

Our first result characterizes the unique linear trading equilibrium through these two variables and trader i's value function.

Lemma 1. *Fix any* $\mathbf{z}$*. There is a unique linear trading equilibrium. In that equilibrium, informativeness, liquidity, and trader* i*'s value are given by*

$$\begin{aligned} I(\mathbf{z}) &= \frac{\mathbf{z}\cdot\mathbf{1}}{1+\mathbf{z}\cdot\mathbf{1}} \\ L(\mathbf{z}) &= \sqrt{\frac{2\tau_v}{\tau_u}}\frac{1+\mathbf{z}\cdot\mathbf{1}}{\sqrt{\mathbf{z}\cdot\mathbf{1}+\mathbf{z}\cdot\mathbf{z}}} \\ V_i(\mathbf{z}) &:= \mathbb{E}[\tilde{V}_i(s_i,\mathbf{z})] = \frac{1}{2\tau_v}(1-I(\mathbf{z}))^2 L(\mathbf{z})\ (z_i+z_i^2) \end{aligned}$$

Define the *aggregate information* in the market as the sum of all normalized precisions, $\mathbf{z}\cdot\mathbf{1}$. Lemma 1 shows that more information in the market mechanically increases price informativeness. Informativeness reaches its maximum of one if and only if aggregate information explodes. Liquidity depends not only on the level, but also on the dispersion of information: when information is more equally shared among traders, the market is more liquid. The relationship between liquidity and the level of aggregate information is more subtle and, in fact, non-monotonic. When there is no information in the market, liquidity is abundant as there is low adverse selection and the market-maker cannot infer anything from the order flow. The price is then fixed at the prior mean, 0, and insensitive to changes in the order flow. On the other extreme, when aggregate information is infinite, liquidity is again plentiful as the price is then pinned down to the fundamental—we discuss this high-information extreme in more detail in Section 3.

Trader i's indirect utility function conveys three economic forces. First, the individual trader benefits from being better informed, as she can use that information to submit informed market orders, which are not perfectly detected by the market-maker due to noise trader demand. Therefore, there is a direct, positive effect of z_i on V_i. Second, a higher informativeness reduces the marginal payoff of information, because equilibrium prices are closer to the real valuation of the asset. Finally, and important for our paper, liquidity also affects the value of information: a deeper market allows traders to leverage

their private knowledge more, as prices do not move as much against their own trades. That means that the value of information is larger in deeper markets.

2.2 Information equilibrium

Next, we describe the information equilibrium, where traders choose $z_i \in [0,1]$ to solve (4). In principle, there could be multiple information equilibria. However, as we shall show, this is not a concern in our model.

First, all equilibria are necessarily symmetric equilibria. The basic intuition is that traders are ex-ante identical, and there is no reason why, with convex information costs, traders should segment into high-informed and low-informed groups.

Lemma 2. *All equilibria are symmetric. That is, equilibrium precisions satisfy* $\mathbf{z} = z^n\mathbf{1}$ *for some scalar* z^n *that depends on the number of traders in the market,* n*.*

Next, focusing without loss on symmetric equilibria, we prove both existence and uniqueness.

Lemma 3. *Assume* $c'(0) < \infty$*. There exists a unique symmetric equilibrium with* $\mathbf{z} = z^n\mathbf{1}$ *and* $z^n \in (0,1)$*.*

The basic intuition for Lemma 3 is that information-acquisition decisions are strategic substitutes. If trader i acquires more precision, it increases aggregate price informativeness, which disincentivizes trader j from acquiring information.

To see this, consider traders' first-order conditions. At a symmetric information equilibrium, we must have

$$\frac{dV_i(\mathbf{z})}{dz_i}\Big|_{\mathbf{z}=z^n\mathbf{1}} \leq c'(z^n), \quad \text{with equality if} \quad z^n > 0. \tag{5}$$

Therefore, the fundamental quantity determining equilibrium is the marginal value of information. Using Lemma 1 and differentiating:

$$\frac{dV_i(z_i,\mathbf{z}_{-i})}{dz_i} = \frac{1}{2\tau_v}(1-I(\mathbf{z}))^2L(\mathbf{z})(1+2z_i) + o_i(\mathbf{z}) \leq \frac{1}{2\tau_v}(1-I(\mathbf{z}))^2L(\mathbf{z})(1+2z_i), \tag{6}$$

where $o_i(\mathbf{z}) := \frac{1}{2\tau_v}z_i(1+z_i)\frac{d}{dz_i}(1-I(\mathbf{z}))^2L(\mathbf{z})$ is a negative function capturing equilibrium effects. The important observation is that, since $(1-I(\mathbf{z}))^2L(\mathbf{z}) \propto \frac{1}{1+\mathbf{z}\cdot\mathbf{1}}\frac{1}{\sqrt{\mathbf{z}\cdot\mathbf{1}+\mathbf{z}\cdot\mathbf{z}}}$, we know that $\frac{dV_i}{dz_i}$ is decreasing in z_j for all $j \neq i$. In other words, information decisions are strategic substitutes.

Finally, for any fixed size-n economy, an information equilibrium always features positive information acquisition, as long as $c'(0) < \infty$. Indeed, notice that liquidity $L(\mathbf{z})$ grows without bound as all agents become uninformed. Meanwhile, the function $o_i(\mathbf{z})$ satisfies $o_i(z\mathbf{1}) \xrightarrow{z\to 0} 0$, so that marginal benefit of information is entirely determined by the explosion in liquidity. In that case, (6) shows that $\lim_{z\to 0} \frac{dV_i(\mathbf{z})}{dz_i}\big|_{\mathbf{z}=z\mathbf{1}} = \infty$, so traders are always willing to pay some cost to acquire a little bit of information. Hence, the first-order condition (5) holds with equality.

2.3 Free entry and large-market equilibrium

The remainder of the paper studies the *free-entry* or *competitive* limit of this market. It turns out that this is equivalent to the limit as the number of strategic traders to grow large ($n \to \infty$). Indeed, traders always make profits net of information costs for every finite n. To see this, first notice from Lemma 1 that $V_i(\mathbf{0}) = 0$ when n is finite. Additionally, recall the properties of information costs—that $c(\cdot)$ is a convex function satisfying $c(0) = 0$. Since $z^n > 0$ in the symmetric equilibrium, as shown above, we thus have that $V_i(z^n\mathbf{1}) - c(z^n) > 0$. This shows that the free-entry limit requires $n \to \infty$. Conversely, we will argue after the next result that trader profits vanish as $n \to \infty$, so that free entry is exhausted in this limit.

A fundamental result is that each trader must become "informationally small" as the market grows. As the proof is simple and enlightening, we include it in the text.

Proposition 1. *Individual information-gathering vanishes asymptotically in the symmetric equilibrium:* $z^n \xrightarrow{n\to\infty} 0$.

Proof. Aggregating traders' ex-ante revenues and using the definition of $Y = \sum_i d_i + u$, we have

$$\sum_{i=1}^n V_i = \sum_{i=1}^n \mathbb{E}[d_i(v - \boldsymbol{p}(Y))] = \mathbb{E}[(Y-u)(v-\boldsymbol{p}(Y))].$$

By the market-maker's break-even pricing condition $\boldsymbol{p}(Y) = \mathbb{E}[v|Y]$, we have the orthogonality result $\mathbb{E}[Y(v - \boldsymbol{p}(Y))] = \mathbb{E}\big[Y\mathbb{E}[v - \mathbb{E}[v|Y] \mid Y]\big] = 0$. Thus,

$$\sum_{i=1}^n V_i = -\mathbb{E}[u(v-\boldsymbol{p}(Y))] \le \sqrt{\mathbb{E}[u^2]}\sqrt{\mathbb{E}\big[(v-\boldsymbol{p}(Y))^2\big]},$$

where the inequality uses Cauchy-Schwarz. Since noise is mean-zero, $\mathbb{E}[u^2] = \tau_u^{-1}$. By the law of iterated expectations and the law of total variance, the latter term satisfies

$$\mathbb{E}\big[(v-\boldsymbol{p}(Y))^2\big] = \mathbb{E}\big[\mathbb{E}\big[(v-\mathbb{E}[v|Y])^2 \mid Y\big]\big] = \mathbb{E}[\mathrm{Var}(v|Y)] \le \tau_v^{-1}.$$

Consequently,

$$\sum_{i=1}^{n} V_i \leq \sqrt{\tau_u^{-1}}\sqrt{\tau_v^{-1}}. \tag{7}$$

Since ex-ante revenues must exceed information costs, $V_i \geq c(z_i)$, and by symmetry, $z_i = z^n$, we obtain

$$c(z^n) \leq \frac{1}{n}\sqrt{\tau_u^{-1}}\sqrt{\tau_v^{-1}}.$$

Since $c(z) \geq c(0) = 0$, we conclude that $z^n \to 0$ as $n \to \infty$. □

Proposition 1 holds because *aggregate* trading revenues are limited by noise volatility and fundamental volatility, as shown in (7). If aggregate trading revenues are bounded, then *per-capita* trading revenues necessarily vanish as the market grows, which implies that individual traders' information expenditures also vanish. With both revenues and costs becoming negligible, profits $V_i(z^n\mathbf{1}) - c(z^n)$ likewise vanish. Consequently, the free-entry limit is, in fact, equivalent to the requirement that $n \to \infty$.

Thus, we have a first insight behind our main results and how they depend on our two novelties: free entry and smooth information costs. Free entry, a hallmark of competition, is equivalent to a large number of traders. With smooth information costs, these large number of traders will necessarily be informationally-small, according to Proposition 1. Ultimately, this suggests that information costs only matter in equilibrium via

$$\chi := c'(0),$$

the marginal cost of information at the prior—i.e. the cost of the first unit of precision.

2.4 Information aggregation

We are interested in price informativeness in the competitive limit, namely as the number of traders, n, grows to infinity.

Definition 2. *Competitive price informativeness is*[4]

$$\mathcal{I} := \lim_{n\to\infty} I(z^n\,\mathbf{1}).$$

We now present a principal result of the paper.

[4]In the proof of Theorem 1, this limit is shown to exist.

Theorem 1. *Competitive informativeness $\mathcal{I} \in [0,1]$ is the unique solution to:*

$$\frac{\mathcal{I}}{(1-\mathcal{I})^3} = \Gamma, \tag{8}$$

$$\textit{where} \quad \Gamma := \frac{1}{2}\frac{1}{\chi^2 \tau_u \tau_v}. \tag{9}$$

Theorem 1 characterizes the asymptotic level of informativeness in the market as a function of three primitives: noise trader uncertainty, τ_u^{-1}, fundamental uncertainty, τ_v^{-1}, and the marginal cost of the first piece of information, χ. Prices naturally are more informative when the marginal cost of information at the prior, χ, is lower. Moreover, prices are more informative either when fundamental uncertainty or noise uncertainty are higher. This characterization is simple, clean, and relatively detail-free with respect to the information cost function, depending only on its derivative at the prior mean, χ. As information costs are difficult to observe, this detail-free characterization of the level of information is advantageous, compared to expressions that rely heavily on the shape of $c(\cdot)$—which arise in the case of NREE models.

The first major implication of Theorem 1 is that price informativeness can take any value, including arbitrarily high values, in stark contrast with NREE models (e.g., Grossman and Stiglitz, 1980). Indeed, as χ varies from 0 to ∞, the sufficient statistic Γ spans the whole real line. Furthermore, arbitrarily high price informativeness can obtain even when it is very expensive to produce a precise signal, so long as the first piece of information is relatively cheap—that is, $\chi = c'(0)$ is all that matters.

Corollary 1. *Prices are asymptotically fully-revealing—i.e. $\mathcal{I} = 1$—if and only if $\chi = 0$.*

Why is $\chi = 0$ necessary and sufficient for price efficiency? First, the necessity of $\chi = 0$ is straightforward: as the level of aggregate information goes to infinity, information swamps the effect of noise and traders cannot hide their informed trades anymore, leading the value of information to decline to zero. If marginal costs of information at the prior were positive, this would eventually lead any particular individual to acquire exactly zero information, which could not lead to fully-revealing prices. On the other hand, sufficiency of $\chi = 0$ is due intuitively to the following: if price informativeness is imperfect, then trading on private information is profitable. With positive profits and zero cost of the first piece of information, individual traders would always want to learn a strictly positive amount. With the number of traders growing to infinity, due to free entry, aggregate information becomes unbounded.

Thus, when the marginal cost of the first piece of information is zero, markets perfectly aggregate dispersed information, despite information being endogenous. This

is in line with recent research on information aggregation in common-value auctions, which also finds that dispersed information can be well-reflected in prices if and only if the marginal cost of information is zero at the prior (Atakan and Ekmekci, 2023).

Remark 1 (Fixed information costs)**.** *For comparison, Appendix D studies our baseline model with fixed information costs. In that case, as is well-understood, an equilibrium with a large quantity of aggregate information can only emerge if the entire cost function vanishes (Kyle, 1989; Vives, 2014). This stands in stark contrast to our result with smooth costs. The intuition for this result is that with fixed costs, the number of informed traders stabilizes as the market becomes large, and so the quantity of aggregate information stabilizes too.*

Remark 2 (NREE models)**.** *For comparison, Appendix E studies a version of an NREE with endogenous information and smooth costs, as in Verrecchia (1982). There, we show that $\mathcal{I} = 1$ generically cannot arise. Instead, informativeness is bounded away from one even if the information cost function satisfies $c'(0) = 0$. A key reason is that traders collect positive amounts of information, $z^* > 0$, which means that the entire cost function $c(z)$ matters—not just the local derivative $c'(0)$. Unless this entire cost function vanishes, informativeness stays bounded.*

3 The Role of Liquidity

So far, we have focused on information aggregation and price informativeness. This section studies liquidity and, more specifically, its role in fostering informative prices. First, we examine liquidity in the benchmark model. Then, we explore two extensions that allow us to isolate the role of liquidity.

3.1 Liquidity in the benchmark model

Our starting point is a strategic model, in which traders internalize their price impact. However, since we ultimately study its large-market limit, a natural conjecture could be that liquidity frictions automatically vanish in the limit. This turns out to be false: illiquidity can persist, and its persistence is indicative of the information environment.

Proposition 2. *Competitive liquidity $\mathcal{L} := \lim_{n\to\infty} L(z^n\,\mathbf{1})$ satisfies*

$$\mathcal{L} = \sqrt{\frac{2\tau_v}{\tau_u}}\frac{1}{\sqrt{\mathcal{I}(1-\mathcal{I})}} = \frac{2\tau_v\chi}{(1-\mathcal{I})^2}$$

An immediate implication of Proposition 2 is that $\mathcal{L} = \infty$ when $\mathcal{I} = 1$. That is, a market with fully-revealing prices is also a market with perfect liquidity. On the other

hand, when prices are only partially-revealing, liquidity is necessarily imperfect. That is, $\mathcal{I} < 1$ corresponds to $\mathcal{L} < \infty$ in Proposition 2.[5] Liquidity frictions can persist even in a large and competitive market.

Combining these results, our competitive equilibrium has *either* (i) a perfectly informative price *or* (ii) an imperfectly liquid market. Accordingly, our equilibrium never coincides with conventional frameworks like NREE which can never have (i) or (ii): an NREE essentially assumes perfect liquidity (no trader ever has price impact) and can never feature fully-informative prices.

Why do perfect liquidity and price efficiency go together? As it turns out, at high levels of price informativeness, there exists a two-way positive feedback loop between $\mathcal{I}$ and $\mathcal{L}$, as depicted in Figure 1. At its core is a strategic complementarity between traders' information-acquisition and market-makers' price-setting. We explain the two directions in turn.

First, consider the impact of informativeness on liquidity. To isolate the impact of a pure change in informativeness, all else equal, consider a shift in the marginal information cost χ. When χ falls, for a fixed level of liquidity, each trader collects more information and trades on it, which increases $\mathcal{I}$. According to Proposition 2, provided $\mathcal{I} > \frac{1}{2}$, the rise in $\mathcal{I}$ then induces an increase in $\mathcal{L}$. The dependence on $\mathcal{I} > \frac{1}{2}$ reflects two opposing forces at play. From the market-maker's perspective, a rise in $\mathcal{I}$ increases adverse selection because he faces more informed trades. But a rise in $\mathcal{I}$ also comes with order flow, so the market-maker sees a better signal about fundamentals. When $\mathcal{I} > \frac{1}{2}$, the second force dominates and $\mathcal{L}$ is rising in $\mathcal{I}$. When informativeness is near-perfect ($\mathcal{I} \to 1$), market-makers are very confident in setting prices based on observed order flow, and so they set prices with minimal price impact, so liquidity explodes ($\mathcal{L} \to \infty$).

Next, consider the impact of liquidity on informativeness. A liquid market allows strategic traders to make profits on their information—this effect is well-understood since Kyle (1985) and Kyle (1989). Therefore, liquid markets encourage more information-acquisition, which ultimately improves price informativeness. To see this incentive for information-acquisition, revisit the marginal value of information in equation (6) but in the large-market limit:

$$\lim_{n\to\infty} \frac{dV_i(z_i, \mathbf{z}_{-i})}{dz_i}\Big|_{\mathbf{z}=z^n\mathbf{1}} = \frac{1}{2\tau_v}(1-\mathcal{I})^2\,\mathcal{L}. \tag{10}$$

As $\mathcal{L}$ increases, all else equal, so does dV_i/dz_i. When liquidity is near-perfect ($\mathcal{L} \to$

[5] This is except for the knife-edge scenario where prices are fully uninformative—$\mathcal{I} = 0$—which occurs when traders acquire no information (e.g., if $\chi = \infty$).

∞), aggregate information-acquisition must become arbitrarily large—informativeness becomes near-perfect ($\mathcal{I} \to 1$). This channel—that better liquidity induces higher price informativeness—was noted by Admati and Pfleiderer (1988b) and is consistent with empirical evidence (Collin-Dufresne and Fos, 2015). The novelty in our paper is that the positive co-movement between liquidity and informativeness continues to survive even in large, competitive markets.[6]

3.2 Fixed liquidity counterfactual

The two-way feedback loop between $\mathcal{I}$ and $\mathcal{L}$ helps explain why it is *possible* to get $\mathcal{I} \to 1$ and $\mathcal{L} \to \infty$. It turns out that perfect liquidity is also *necessary* for price efficiency: without the feedback loop, informationally-efficient markets cannot arise.

To prove this, this section extends the baseline model by decoupling actual and perceived liquidity. This severs the feedback loop in Figure 1. As before, price informativeness affects market-makers' learning and thereby affects *actual* liquidity. Meanwhile, strategic traders' aggression and, hence, ex-ante information-acquisition, is governed by their *perceived* liquidity—changes in actual liquidity do not shift endogenous information and, hence, do not feed back into price informativeness.

More specifically, we assume that traders perceive some exogenous price impact $\tilde{\lambda}_n$, which can depend on the market size n and which differs from the actual price impact $\lambda(\mathbf{z})$. Thus, as the information equilibrium changes, perceptions of market liquidity remain exogenous. Traders then submit orders with their perceived price impact in mind. All other details are identical to the benchmark model. In this extended setting, informationally-efficient markets arise if and only if perceived liquidity explodes.

Proposition 3. *Suppose $\chi < \infty$. Consider any positive sequence of perceived price impacts $(\tilde{\lambda}_n)_{n\geq 1} > 0$. Consider equilibria in which aggregate information explodes, $nz^n \to \infty$. Then, competitive informativeness is at its maximal level and is equal to*

$$\mathcal{I} = \frac{1}{1 + \frac{\tau_v}{\tau_u} \lim_{n\to\infty} \tilde{\lambda}_n^2}.$$

Hence, $\mathcal{I} = 1$ if and only if $\lim_{n\to\infty} \tilde{\lambda}_n = 0$. Furthermore, in this case, competitive liquidity $\mathcal{L}$ exceeds perceived liquidity $\tilde{\mathcal{L}} = \lim_{n\to\infty} \tilde{\lambda}_n^{-1}$ by

$$\mathcal{L} = \frac{1}{\mathcal{I}} \times \tilde{\mathcal{L}}.$$

[6] Another explanation for the positive correlation between adverse selection and liquidity is that informed investors time the market to trade when liquidity is higher (Collin-Dufresne and Fos, 2016).

Proposition 3 demonstrates that illiquidity, in a precise sense, limits information efficiency. Indeed, when the market is perceived to be illiquid, traders do not submit orders as aggressively, which limits how much information can get into prices. Even if traders acquire infinite information in aggregate, price informativeness remains capped.

It is also interesting to study actual liquidity in this world. When traders acquire lots of information, actual liquidity always exceeds perceived liquidity, highlighting that traders are excessively restricting their trades. However, the gap between actual and perceived liquidity is not infinite: capped information efficiency leaves residual uncertainty for the market-maker, who thus remains sensitive to order flow. In this sense, the cap on informativeness puts a cap on equilibrium liquidity, highlighting the fact informativeness continues to positively affect actual liquidity in this extension.

3.3 Growing noise

In the next exercise, we show that the liquidity effect can be surprisingly powerful. We consider an extension in which noise grows with the size of the market, as is commonly studied in the literature.[7] In particular, we let the noise variance in a market with n traders be $\tau_{u,n}^{-1}$ and increasing in n. If this sequence of variances converges to a finite limit as n grows, our previous results apply. We therefore focus on the case in which noise variance diverges and characterize equilibrium for any rate of growth.

With more noise traders, the market becomes deeper for any fixed level of information. Holding information fixed, this liquidity effect is the only effect: examining the trading-equilibrium expressions in Lemma 1, informativeness $I(\mathbf{z})$ is independent of noise τ_u^{-1}, while liquidity $L(\mathbf{z})$ is increasing in noise τ_u^{-1} for any $\mathbf{z}$.[8] Of course, as discussed in previous sections, the improvement in liquidity induces strategic traders to acquire more information, which ultimately raises price informativeness as well. This feedback into price informativeness can be extremely powerful if noise traders continue to enter as markets grow. In fact, when noise volatility grows without bound, information efficiency is not only possible; it is required.

[7] The assumption that noise grows with n is often an important assumption on the way to obtaining a competitive limit (see Hellwig, 1980; Verrecchia, 1982; Kyle, 1989; García and Urošević, 2013; and Kovalenkov and Vives, 2014). For instance, in Hellwig (1980) and Verrecchia (1982), noise variance grows at the rate of n^2, because each trader is assumed to enter with a random supply of shares whose variance rises at a rate of n.

[8] That $I(\mathbf{z})$ is independent of τ_u^{-1} is well-known in the Kyle (1985) framework. With exogenous precisions, informativeness is invariant to noise trading because two effects cancel: noise traders make prices less informative, but strategic traders are then able to trade more aggressively on pre-existing information.

Proposition 4. *Suppose* $\chi < \infty$. *If* $\tau_{u,n}^{-1} \to \infty$, *then* $\mathcal{I} = 1$ *and* $\mathcal{L} = \infty$. *Moreover, a free-entry limit exists if* $\limsup_{n\to\infty} \frac{\tau_{u,n}^{-1}}{n^3} = 0$.

Figure 2 illustrates how informativeness $\mathcal{I}$ varies with large-market noise $\lim_n \tau_{u,n}^{-1}$. When noise converges to a finite limit, the information technology tightly modulates informativeness. But when noise grows without bound, so does informativeness.

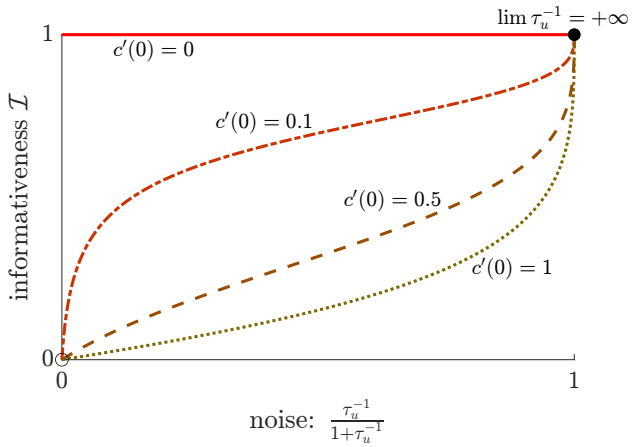


Figure 2: Informativeness against noise for various $\chi = c'(0)$.

4 Key Extensions

4.1 Risk aversion and limit orders

The baseline model builds on Kyle (1985) and assumes traders are *risk-neutral* and submit *market orders*. These assumptions are not critical to our main conclusions. To demonstrate this, we provide a model in which traders are *risk-averse* and submit *limit orders*, à la Kyle (1989). These features also make the extended model more comparable to Noisy Rational Expectations Equilibria (NREE).

In this extended model, traders submit limit orders, in the form of a demand schedule, and markets clear. Given this trading protocol, we eliminate the market-maker from the baseline model. Instead, to keep a throughline from the baseline model to this extended setting, we add a pool of competitive uninformed traders to the model. Specifically, following the approach in Kyle (1989), for any fixed level of informed traders, n, there is free entry of uninformed traders. The free-entry assumption guarantees that there is sufficient uninformed demand to arbitrage away all the public information. In particular, if Y is the aggregate order-flow from informed and noise traders, free entry of

uninformed traders keeps prices unbiased, $p = \mathbb{E}[v|Y]$, thereby serving a role analogous to our baseline model's market-maker.

Next, all traders—informed and uninformed—are assumed to be CARA utility maximizers with risk-aversion γ. That is, the payoff for trader i from demanding d_i at price p is:

$$V_i := \frac{1}{\gamma}\mathbb{E}\left[1 - e^{-\gamma(v-p)d_i}\right].$$

Of course, the informed traders additionally make an ex-ante information choice z_i, subject to cost $c(z_i)$, as in the baseline model.

Appendix B develops this model fully and contains proofs of all its results. Compared to the baseline model, a key difference is that the amount of the information that gets revealed by prices is an equilibrium object. When traders are risk-neutral, exactly 50% of the private information held by traders is leaked through prices. Risk aversion leads to less aggressive trades, and so this fraction is less than 50%. Formally, define:

Definition 3. *Price leakage φ is the fraction of information which gets revealed by prices:*

$$\varphi := \lim_{n\to\infty} \frac{Var^{-1}[v|p] - Var^{-1}[v]}{Var^{-1}[v|p, s_1, ..., s_n] - Var^{-1}[v]}.$$

The numerator in the expression above measures how much more precise the estimate of the fundamental becomes after observing prices. The denominator captures how much more precise that estimate becomes if one observes all the dispersed information in the market. Therefore, the ratio can be understood as the portion of the total precision that is reflected in prices (Lee and Kyle (2018) call φ *relative informational efficiency*).

The following main result extends Theorem 1 and Proposition 2. Note that this generalized model is substantially more complicated than the baseline model, and so we do not provide any formal existence or uniqueness results. Instead, we simply assume a symmetric equilibrium and characterize its free-entry limit.

Theorem 2. *Assume $\gamma\chi < 1$. In the model with risk-averse traders, limit orders, and free entry of uninformed traders, competitive informativeness $\mathcal{I}$ is the unique solution to:*

$$\frac{\mathcal{I}}{(1-\mathcal{I})^3}\frac{1-\varphi}{2\varphi^2} = \Gamma,$$

where $\Gamma := \frac{1}{2}\frac{1}{\chi^2\tau_u\tau_v}$ and where $\varphi = \frac{1}{2} - \frac{1}{2}\frac{\gamma\chi}{1-\mathcal{I}}$. Moreover, liquidity satisfies:

$$\mathcal{L} = \sqrt{\frac{\tau_v}{\tau_u(1-\varphi)}}\frac{1}{\sqrt{\mathcal{I}(1-\mathcal{I})}}.$$

If instead $\gamma\chi \geq 1$, then there is no information acquisition in equilibrium.

Theorem 2 shows that our results obtained in the risk-neutral model are not knife-edge. First, price informativeness depends on the same statistic Γ as before. This statistic is no longer a "sufficient statistic," because we also require the endogenous leakage variable φ. But together, Γ and φ are sufficient to determine $\mathcal{I}$. Since price leakage satisfies $\varphi \leq \frac{1}{2}$ and is decreasing in γ, we have that informativeness $\mathcal{I}$ is decreasing in γ. Markets are less informative with risk-averse traders.

Second, the dependence on information cost χ is identical to the risk-neutral model. When $\chi = 0$, prices become fully-revealing, and liquidity becomes perfect:

Corollary 2. *In the model with risk aversion and limit orders, $\mathcal{I} = 1$ and $\mathcal{L} = \infty$ when $\chi = 0$.*

By contrast, when $\chi > 0$, price informativeness and liquidity are imperfect. Thus, informativeness and liquidity are perfect together and imperfect together. Consequently, all the main implications of the baseline model continue to hold in this extended setting.

In fact, when $\gamma \to 0$, the model becomes identical to the baseline risk-neutral model, discussed in the following remark.

Remark 3 (Limit orders in the risk-neutral model)**.** *If we take $\gamma \to 0$, then we recover the risk-neutral model but with limit orders rather than market orders. This distinction is immaterial: by comparing Theorem 2 with $\gamma \to 0$ against the earlier results (Theorem 1 and Proposition 2), we see that $\mathcal{I}$ and $\mathcal{L}$ are identical across the two settings. Thus, the trading protocol becomes irrelevant in competitive markets.*

4.2 Correlated information choice

Our previous analysis, as well as most of the literature on informed trading, assumes that the signals traders observe are conditionally independent from each other—a necessary condition for our "wisdom-of-the-crowd". A natural question is why errors should be independent to begin with. To address this concern, we extend our baseline model so that traders can choose to learn from the same information sources and, as a consequence, to observe conditionally correlated signals. Intuitively, correlated signals may benefit traders. Indeed, by learning about what others have observed, an investor is able to better predict order flow to bet against aggregate demand. There is a trade-off, however: by partially consuming the same information as others, investors also give up a slice of their local monopoly over information. The main result in this section is that the information monopoly loss prevents the existence of any equilibrium with conditionally correlated signals.

Our model is identical to the baseline, except for the information technology available to traders. In particular, there exists a countable number of information sources, consisting of independent Brownian motions, $(\varepsilon_t^{\xi})_{t\in[0,\infty]}$, indexed by $\xi \in \mathbb{N}$. A strategic trader's information choice then consists of an information source, ξ_i, and of a precision, $\tau_i := 1/T_i$, such that the signal trader i observes is:

$$s_i = v + \varepsilon_{T_i}^{\xi_i}.$$

We define a block $b_n^{\xi} := \{i : \xi_i = \xi\}$ as the set of traders who choose information source ξ. As in our baseline model, information costs are a function of precisions alone, so the choice of information source is free. The timeline of the game is the same as our baseline model except that, at the outset, each trader simultaneously chooses an information source as well as a precision level. The solution concept is identical.

Some comments on the information technology are in order. First, it nests our baseline model. Indeed, if a trader is in a singleton block $|b^{\xi}| = 1$, then his signal is conditionally independent of all the other trader's signals, and his error precision is $1/T_i$. In particular, the equilibrium of our baseline game consists of an outcome in which all traders choose different information sources and, because the equilibrium is symmetric, the same T_i. Second, two traders sharing the same information source have conditionally correlated signals. Finally, the less precise of the players who share an information source observes a signal that is a garbling of the other player's. Their information sets are thus nested. An interpretation of the information technology is that if two traders choose the same dataset and spend the same effort analyzing it, they will receive the same information about the asset value. However, if one spends more effort—measured by precision, $1/T$—than the other, then her information will be a strict refinement of her competitor's.

Proposition 5 (Conditional independence is robust)**.** *In any equilibrium,* $\sup_{\xi\in\mathbb{N}} |b^{\xi}| = 1$.

Proposition 5 shows that in all possible equilibria, each trader picks a different information source: trader's signals are conditionally independent. The proof of this result, while lengthy and therefore contained in Appendix C, relies on a central observation: if multiple traders share the same information source, the one with the lowest precision always has incentives to deviate. As we previously discussed, sharing an information source comes with a trade-off. On one hand, it allows the traders to predict each other's orders, which is helpful when trying to disguise information in the order flow. On the other hand, traders give up monopoly rents on the fraction of their information that is shared. It turns out that, in equilibrium, the benefits from demand forecasting accrue

only to traders who maintain some information monopoly—which never is the case for the one with the lowest precision. In addition to implying robustness of our findings, this result provides a microfoundation to the ubiquitous assumption of conditionally independent signals in information games.

5 Implications

In this section we study two normative implications of our model. First, we show that trader profits are non-monotonic in information technology—specifically, profits can decline as information costs fall and price informativeness rises. Hence, there is not necessarily a trade-off between market efficiency and retail trader surplus. Second, we show that more noise trading increases price informativeness, a result that can hold in small or concentrated markets but is typically flipped in large, competitive markets. Hence, market designs that segment retail and informed traders can be inefficient. For generality, we demonstrate these implications in the extended model with risk-averse traders and limit orders, as studied in Section 4.1.

5.1 Aggregate profits and costs

We examine a novel implication of our model that relates price informativeness to both distributional outcomes and aggregate efficiency. To start, we define two measures in our free-entry limit.

Definition 4. *Aggregate (certainty-equivalent) trading revenues and information costs are*

$$\mathcal{R} := \lim_{n\to\infty} \sum_{i=1}^{n} -\gamma^{-1}\log(1-\gamma V_i)$$
$$\mathcal{C} := \lim_{n\to\infty} \sum_{i=1}^{n} c(z_i).$$

First, consider (certainty-equivalent) aggregate trading revenues $\mathcal{R}$. While trading revenues do serve to provide incentives for information-acquisition, they are ultimately transfers from noise traders. One potential distributional concern is that high price informativeness arises only when sophisticated traders aggressively extract large revenues from less sophisticated retail investors.

Second, consider information costs $\mathcal{C}$. In this model, price informativeness serves no allocative role, and so we do not compute a welfare benefit of market efficiency.

However, we are able to compute the costs associated with price informativeness, namely the sum of information expenditures. A potential efficiency concern is that high price informativeness requires a lot of information expenditure in aggregate.

Proposition 6. *In the free-entry limit of symmetric equilibria, we have*

$$\mathcal{R} = \mathcal{C} = \frac{\chi}{2\varphi}\frac{\mathcal{I}}{1-\mathcal{I}} = \frac{1}{2(1-\varphi)}\frac{\tau_u^{-1}}{\mathcal{L}}.$$

Consequently, as $\chi \to 0$ and markets become informationally-efficient ($\mathcal{I} \to 1$ and $\mathcal{L} \to \infty$), both trading revenues and information costs vanish in aggregate.

Proposition 6 contains two results simultaneously. First, since $\mathcal{R} = \mathcal{C}$, aggregate ex-ante profits are zero in our competitive equilibrium. This is not obvious: free entry only ensures that individual (risk-adjusted) profits are zero. Nevertheless, aggregate ex-ante profits are zero precisely because traders become informationally-small. Consequently, our competitive equilibrium, in some sense, minimizes the losses to noise traders, capping them at the level of information expenditures. The two concerns motivating this analysis—losses to noise traders and dead-weight losses of information-acquisition—collapse into a single statistic.

Second, the expression $\mathcal{R} = \mathcal{C} = \frac{\chi}{2\varphi}\frac{\mathcal{I}}{1-\mathcal{I}}$ captures the two forces in aggregate profits. Each trader pays χ per unit of information, and the total amount of information is $\lim_{n\to\infty} nz^n = \frac{1}{2\varphi}\frac{\mathcal{I}}{1-\mathcal{I}}$. As information costs rise, each trader pays more individually, but traders acquire less in the aggregate. The pattern in aggregate profits depends on the balance of these two forces.

Third, to determine which force dominates, notice that the pattern in profits is governed by liquidity, in that $\mathcal{R} = \mathcal{C} = \frac{\tau_u^{-1}}{2(1-\varphi)}\mathcal{L}^{-1}$. Recall that price impact $\mathcal{L}^{-1}$ displays an inverse U-shaped pattern in χ, with $\mathcal{L}^{-1} = 0$ when either $\chi = 0$ or $\chi \geq \gamma^{-1}$. Price impact is zero at these extremes because the competitive fringe is confident in how they set prices when they know all order flow is informed or uninformed. For instance, when $\chi = 0$, prices are set equal to fundamental value in equilibrium. Aggressive trading, induced by a very liquid market, is key to informational efficiency, but it also implies that traders fiercely compete away each others' profits. Thus, revenues and costs inherit exactly the pattern displayed by price impact. Figure 3 illustrates this inverse-U relation between information expenditures $\mathcal{C}$ and information costs χ.[9]

[9]While $\mathcal{R} = \mathcal{C}$, raw trading revenues are higher than $\mathcal{R}$ because they do not contain a risk-adjustment. However, at the endpoints—when $\chi \to 0$ or $\chi \to \gamma^{-1}$—the risk-adjustment becomes immaterial. Indeed, when $\chi \to 0$, prices become fully-revealing, and so $\text{Var}[(v-p)d] \to 0$. At the other end, when $\chi \to \gamma^{-1}$, no

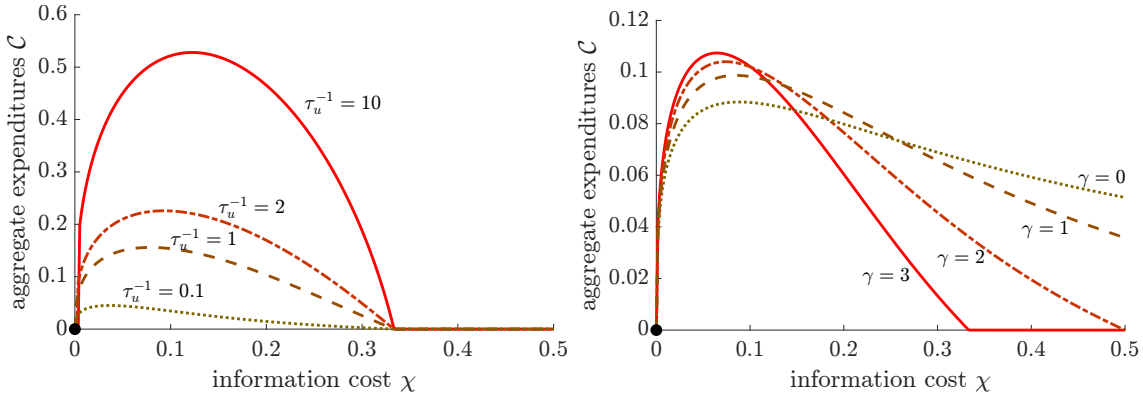


Figure 3: Information expenditures against χ for various τ_u^{-1} and various γ.

A consequence of this connection between liquidity and aggregate profits is that, contrary to common wisdom, there does not need to be a trade-off between informativeness and retail trader's surplus. When the marginal information cost χ is high, a trade-off exists: easier information acquisition raises informativeness but also reduces liquidity. The decline in liquidity raises industry profits at the expense of retail traders. However, when χ is low, even lower χ simultaneously increases informativeness and makes the market more liquid, which leads informed investors to trade more aggressively and compete away their profits, to the benefit of retail traders.

Meanwhile, when χ is very low, aggregate expenses on information vanish despite an exploding quantity of information. Consequently, informationally-efficient markets need not be encumbered by dead-weight losses of information-acquisition either.

5.2 Noise and informativeness

Our second implication concerns the impact of noise. In some existing literature on strategic trading with endogenous information, it has been shown that price informativeness can increase in noise trading (see Admati and Pfleiderer, 1988b; Kyle, 1989; Banerjee and Breon-Drish, 2022). This literature obtains such results with a small number of informed traders, either by assumption or by assuming fixed information costs.[10] By contrast, in NREE models—the common benchmark for large, competitive markets—

information is collected, meaning strategic traders submit no demands, and so $\mathrm{Var}[(v-p)d] \to 0$ again. Therefore, a very similar inverse-U pattern holds for raw trading revenues as well.

[10] One exception is Theorem 10.5 of Kyle (1989), which studies a "monopolistic competition" model with large n. In that framework, it is assumed that as n grows, an informed trader's precision vanishes at rate n^{-1} and, simultaneously, that the fixed cost of becoming informed also vanishes at rate n^{-1}. Judiciously (exogenously) picking these rates permits a large-market equilibrium with dispersed informed traders.

informativeness is always decreasing in noise (Verrecchia, 1982).[11] Our model is built upon strategic-trading foundations but ultimately becomes a large, dispersed market. Naturally, one wonders: how does noise affect the properties of our market?

It turns out that our market, despite being large and competitive, aligns with the strategic trading literature and conflicts with NREE.[12]

Proposition 7. *In the free-entry limit of symmetric equilibria, we have* $\frac{d\mathcal{I}}{d\tau_u^{-1}} > 0$ *and* $\frac{d\mathcal{L}}{d\tau_u^{-1}} > 0$.

Let us unpack the discrepancy between how noise affects our equilibrium versus NREE. An increase in noise has three effects on equilibrium informativeness. The first effect, present in both NREE models and in strategic trading models, is direct: an increase in noise traders makes prices mechanically less informative. Second, this reduction in informativeness increases the marginal value of information by creating profitable opportunities—deviations between the price and the fundamental value—which lead to an increase in information acquisition. This second force, also present in NREE, is never enough to compensate for the mechanical deterioration of informativeness.

The third force—a liquidity effect that we have emphasized repeatedly—reverses that deterioration, leading to higher informativeness. An increase in noise deepens the market, making the trader's *marginal* trade more profitable, as prices move less against her. Because she is able to trade more aggressively and make more profits, the trader is willing to acquire even more information. On aggregate, prices become more informative precisely because the market becomes more liquid.

To see the liquidity effect clearly, reconsider the marginal value of information:

$$\lim_{n\to\infty} \frac{dV_i(z_i, \mathbf{z}_{-i})}{dz_i}\Big|_{\mathbf{z}=z^n\mathbf{1}} = \frac{\varphi}{\tau_v}(1-\mathcal{I})^2\,\mathcal{L}\,. \tag{11}$$

(Notice that equation (11) generalizes (10) from the risk-neutral model. Indeed, as $\gamma \to 0$, information leakage satisfies $\varphi \to \frac{1}{2}$, so (11) and (10) coincide.) A rise in τ_u^{-1} increases $\mathcal{L}$ and thus increases the marginal value of information. This then leads to higher informativeness $\mathcal{I}$ and, though higher $\mathcal{I}$, smaller information leakage $\varphi = \frac{1}{2} - \frac{1}{2}\frac{\gamma\chi}{1-\mathcal{I}}$. These equilibrium effects dampen the incentive to collect information, reducing $\frac{dV_i}{dz_i}$, but as equilibrium effects, they are never enough to offset the direct effect of improved liquidity. Hence, a noise-induced liquidity improvement always raises informativeness.

[11] We also prove this in an NREE counterpart to our model, in Appendix E. Specifically, Proposition E.1, part (i), shows that $\mathcal{I}$ is always decreasing in τ_u^{-1} for any valid information cost function.

[12] Using a broader set of comparative statics in Appendix B.3, we go on to contrast our model with NREE in other ways as well. For example, Appendix B.4 demonstrates that price informativeness is always increasing in price volatility, whereas we prove the relationship is almost-always flipped in NREE.

This distinction between our model and NREE is relevant for several applications. One example is the increasing routing of retail order-flow to off-exchange venues. To the extent that a large part of retail volume is executed off-exchange, the uncertainty of on-exchange noise trading is reduced. NREE suggests that this segmentation brings no trade-offs, because exchange prices should become more informative, and retail traders benefit from better deals with wholesalers. Our model suggests caution: the decrease in noise should deteriorate the information quality on the exchange. This deterioration is difficult to detect simply by comparing on-exchange and off-exchange pricing, not only because off-exchange transactions can be required to occur at "better" prices than on-exchange transactions (e.g., Regulation NMS), but more importantly because the counterfactual without segmentation is unobserved. Our model suggests that an integrated market would have more informative prices both on- and off-exchange.

6 Conclusion

We re-examine information efficiency in a strategic trading model, assuming free entry and smooth information costs. Whereas informationally-efficient markets are commonly thought to be an impossibility, we overturn this conventional wisdom and identify four pillars jointly supporting information efficiency. First, the market must be large and competitive, which therefore requires free entry ($n \to \infty$). Second, the marginal cost of information must be zero at the prior ($\chi \to 0$). Third, market liquidity must be arbitrarily high ($\mathcal{L} \to \infty$). Fourth, information must be dispersed, which requires signal errors to be independent ($\varepsilon_i \perp \varepsilon_j$). The first two pillars are necessary assumptions. The latter two pillars are, in a sense we make precise, natural equilibrium outcomes. For this reason, our theorems state that $\chi = 0$ is necessary and sufficient for perfect efficiency in the free-entry limit.

Going beyond perfect efficiency, our main contribution is an explicit characterization of the competitive market. This allows us to develop two policy implications. First, strategic traders' revenues and information costs both fall as the market approaches efficiency. This result implies that informationally-efficient markets need not hurt retail traders nor require excessive investment in information. Second, price informativeness is increasing in noise trading, which conflicts with the literature on Noisy Rational Expectations Equilibria (NREE). Interpreting noise as retail trade, this result implies that market designs which segment or dampen retail trade can be inefficient.

While our model abstracts from many important and realistic features, our aim is not to explain a host of empirical patterns at this stage. Instead, we revisit the foundations.

Our model should be viewed as a new competitive benchmark that is distinct—both positively and normatively—from the conventional NREE benchmark in which investors take prices as given. To address richer patterns in competitive markets, future research could extend our framework to include such features as heterogeneous information technologies, learning about noise, risk-sharing considerations, and the like. Importantly, our framework, being simple and tractable, can serve as a viable foundation atop which these additional features can be built.

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A Proofs and Derivations

A.1 Linear trading equilibrium (Proof of Lemma 1)

The existence and uniqueness of a linear equilibrium is a direct consequence of Theorem 1 in Lambert et al. (2018).

Because throughout this proof $\mathbf{z}$ is fixed, we call, with a slight abuse of notation $\boldsymbol{p} = \boldsymbol{p}(\cdot, \mathbf{z})$, and, for each trader i, $\boldsymbol{d}_i = \boldsymbol{d}_i(\cdot, \mathbf{z})$—that is, we drop $\mathbf{z}$ from the arguments of these functions.

To characterize the linear equilibrium, notice that, under the equilibrium strategies, the pricing function from the point of view of trader i is:

$$\boldsymbol{p}(Y) = \lambda\, Y = \lambda d_i + \lambda \sum_{j \neq i} \beta_j s_j + \lambda u = \lambda d_i + \lambda \sum_{j \neq i} \beta_j v + \lambda \sum_{j \neq i} \beta_j \varepsilon_j + \lambda u.$$

Taking that into account, the first-order-condition for trader i determines:

$$d_i = \frac{1}{2\lambda} \left(1 - \lambda \sum_{j \neq i} \beta_j\right) \mathbb{E}[v|s_i] = \frac{1}{2\lambda} \left(1 - \lambda \sum_{j \neq i} \beta_j\right) \frac{\tau_i}{\tau_v + \tau_i} s_i,$$

where the second equality follows from Bayesian updating given joint normality of (v, s_i) and $\varepsilon_i \perp v$. Recall that in a linear equilibrium, $d_i = \beta_i s_i$. Define $B = \sum_j \beta_j$, and $\kappa_i = \frac{\tau_i}{\tau_i + \tau_v}$. The equation above is then equivalent to:

$$\beta_i = \frac{1}{2\lambda} \left(1 - \lambda (B - \beta_i)\right) \kappa_i, \tag{A.1}$$

which allows us to solve for β_i as a function of λ and B:

$$\beta_i = \underbrace{\frac{\kappa_i}{2 - \kappa_i}}_{z_i} \frac{1 - \lambda B}{\lambda}.$$

By summing both sides up and solving for B:

$$B = \frac{\mathbf{z} \cdot \mathbf{1}}{1 + \mathbf{z} \cdot \mathbf{1}} \frac{1}{\lambda}, \tag{A.2}$$

and plugging into the expression of β_i:

$$\beta_i = \frac{z_i}{1 + \mathbf{z} \cdot \mathbf{1}} \frac{1}{\lambda}. \tag{A.3}$$

We now obtain an expression for λ by using the market-maker's breakeven condition.

To that goal, define the unbiased signal from aggregate demand:

$$\tilde{Y} = \frac{Y}{B} = v + \underbrace{\frac{\sum_j \beta_j \varepsilon_j + u}{B}}_{\eta},$$

with $\eta \perp v$ normally distributed with precision

$$\tau_\eta := \text{Var}[\eta]^{-1} = \frac{B^2}{\sum_j \beta_j^2 \tau_j^{-1} + \tau_u^{-1}}.$$

Then, by standard Bayesian updating for joint normal variables:

$$\mathbb{E}[v|Y] = \mathbb{E}[v|\tilde{Y}] = \frac{\frac{B^2}{\sum_j \beta_j^2 \tau_j^{-1} + \tau_u^{-1}}}{\frac{B^2}{\sum_j \beta_j^2 \tau_j^{-1} + \tau_u^{-1}} + \tau_v} \tilde{Y} = \frac{B}{B^2 + \tau_v \left(\sum_j \beta_j^2 \tau_j^{-1} + \tau_u^{-1} \right)} Y,$$

where the first equality holds because Y and $\tilde{Y}$ contain the same information.

Because $\mathbb{E}[v|Y] = p = \lambda Y$, we obtain an expression for λ:

$$\lambda = \frac{B}{B^2 + \tau_v \left(\sum_j \beta_j^2 \tau_j^{-1} + \tau_u^{-1} \right)}.$$

Towards a closed-form solution for λ, we will obtain simplifications for parts of the expression above. First, using $\tau_i^{-1} = \frac{1}{2\tau_v} \frac{1-z_i}{z_i}$, and equation (A.3):

$$\sum_j \beta_j^2 \tau_j^{-1} = \frac{1}{2} \frac{\mathbf{z} \cdot \mathbf{1} - \mathbf{z} \cdot \mathbf{z}}{(1 + \mathbf{z} \cdot \mathbf{1})^2 \lambda^2} \tau_v^{-1}.$$

Plugging the equation above and equation (A.2) in the expression for λ, one can solve for λ to obtain:

$$\frac{1}{L(\mathbf{z})} = \lambda = \sqrt{\frac{\tau_u}{2\tau_v}} \frac{\sqrt{\mathbf{z} \cdot \mathbf{1} + \mathbf{z} \cdot \mathbf{z}}}{1 + \mathbf{z} \cdot \mathbf{1}},$$

which established our result on liquidity, and $\beta_i = \sqrt{\frac{2\tau_v}{\tau_u}} \frac{z_i}{\sqrt{\mathbf{z} \cdot \mathbf{1} + \mathbf{z} \cdot \mathbf{z}}}$.

We now characterize price informativeness. Because prices contain the same information as Y, which contains the same information as $\tilde{Y}$, $\text{Var}[v|p] = \text{Var}[v|\tilde{Y}]$. By standard Bayesian updating for jointly normal variables:

$$\text{Var}[v|\tilde{Y}]^{-1} = \tau_v + \tau_\eta = \tau_v (1 + \mathbf{z} \cdot \mathbf{1}).$$

But then:

$$I(\mathbf{z}) = 1 - \frac{\text{Var}[v|\tilde{Y}]}{\text{Var}[v]} = \frac{\mathbf{z}\cdot\mathbf{1}}{1+\mathbf{z}\cdot\mathbf{1}},$$

as we wanted to prove.

All that is left is to calculate the ex-ante utility. For that, note that $\mathbb{E}[p|s_i] = \lambda\mathbb{E}[Y|s_i] = \lambda\sum_{j\neq i}\beta_j\kappa_i s_i + \lambda\beta_i s_i$. Moreover, by equation (A.1), $2\lambda\beta_i = (1-\lambda\sum_{j\neq i}\beta_j)\kappa_i$. Thus:

$$\mathbb{E}[\boldsymbol{d}_i(s_i)(v-\boldsymbol{p}(Y))|s_i] = \beta_i s_i^2\,(2\lambda\beta_i - \lambda\beta_i) = \lambda\beta_i^2 s_i^2.$$

Taking ex-ante utility and recalling that, ex-ante, s_i has mean zero, we have $\mathbb{E}[s_i^2] = \text{Var}[s_i] = \tau_v^{-1} + \tau_i^{-1}$, which delivers, after algebra:

$$V_i(\mathbf{z}) = \lambda\beta_i^2(\tau_v^{-1}+\tau_i^{-1}) = \sqrt{\frac{1}{2\tau_v\tau_u}}\frac{1}{1+\mathbf{z}\cdot\mathbf{1}}\frac{z_i+z_i^2}{\sqrt{\mathbf{z}\cdot\mathbf{1}+\mathbf{z}\cdot\mathbf{z}}}, \tag{A.4}$$

which can be rewritten as in the proposition's statement. This concludes the proof. ■

A.2 Concavity of value function (Statement and Proof of Lemma A.1)

Lemma A.1. *Let $\mathbf{z}\in[0,1]^n$ be arbitrary, with $\mathbf{z}\neq\mathbf{0}$. The function $V_i(\mathbf{z})$ is strictly globally concave in raw precision $\tau_i = 2\tau_v\frac{z_i}{1-z_i}$.*

Proof of Lemma A.1

Recall the expression for V_i in (A.4). Note that for concavity, the constant $\sqrt{2\tau_v\tau_u}$ is irrelevant so we can ignore for the rest of the proof. We now substitute $z_i = \frac{\tau_i}{\tau_i+2\tau_v}$ from (2) and clean up the result to obtain (after much algebra)

$$V_i \propto \frac{\frac{2\tau_i(\tau_i+\tau_v)}{(\tau_i+2\tau_v)^2}}{\frac{D_1(\tau_i)}{\tau_i+2\tau_v}\cdot\sqrt{\frac{D_2(\tau_i)}{(\tau_i+2\tau_v)^2}}} = \frac{2\tau_i(\tau_i+\tau_v)}{D_1(\tau_i)\sqrt{D_2(\tau_i)}} \tag{A.5}$$

where

$$D_1(\tau_i) := (2+b)\tau_i + 2\tau_v(1+b), \qquad D_2(\tau_i) := (2+q)\tau_i^2 + 2\tau_v(1+2q)\tau_i + 4q\tau_v^2,$$

$$b := \sum_{j\neq i} z_j \;\geq 0, \qquad q := \sum_{j\neq i}\left(z_j+z_j^2\right) \;\geq 0,$$

Note that b, q are both nonnegative since each $z_j \in (0,1)$.

The algebra for (A.5) is as follows. To obtain the numerator, note that

$$z_i(1+z_i) = \frac{\tau_i}{\tau_i+2\tau_v} \cdot \frac{2\tau_i+2\tau_v}{\tau_i+2\tau_v} = \frac{2\tau_i(\tau_i+\tau_v)}{(\tau_i+2\tau_v)^2}.$$

To obtain the first term in the denominator, note that

$$1+\mathbf{z}\cdot\mathbf{1} = 1+z_i+b = \frac{2\tau_i+2\tau_v}{\tau_i+2\tau_v}+b = \frac{2\tau_i+2\tau_v+b\,(\tau_i+2\tau_v)}{\tau_i+2\tau_v} = \frac{D_1(\tau_i)}{\tau_i+2\tau_v}$$

To obtain the final term inside the square-root, note that

$$\mathbf{z}\cdot\mathbf{1}+\mathbf{z}\cdot\mathbf{z} = z_i(1+z_i)+q = \frac{2\tau_i(\tau_i+\tau_v)}{(\tau_i+2\tau_v)^2}+q = \frac{2\tau_i(\tau_i+\tau_v)+q(\tau_i+2\tau_v)^2}{(\tau_i+2\tau_v)^2} = \frac{D_2(\tau_i)}{(\tau_i+2\tau_v)^2},$$

Substituting these results, we obtain (A.5).

Starting from (A.5), we differentiate V_i twice with respect to τ_i and combine all terms over a common denominator to obtain

$$V_i''(\tau_i) = \frac{\mathrm{Num}(\tau_i)}{D_1(\tau_i)^3\, D_2(\tau_i)^{5/2}},$$

where $\mathrm{Num}(\tau_i)$ is a degree-5 polynomial in τ_i. Since $\tau_v > 0$ and $b, q \geq 0$, both $D_1(\tau_i) > 0$ and $D_2(\tau_i) > 0$ for every $\tau_i > 0$, so the denominator $D_1(\tau_i)^3 D_2(\tau_i)^{5/2}$ is strictly positive. The sign of $V_i''(\tau_i)$ is therefore entirely determined by the sign of $\mathrm{Num}(\tau_i)$.

Carrying out the differentiation and collecting powers of τ_i,

$$\mathrm{Num}(\tau_i) = \sum_{m=0}^{5} \kappa_m\, \tau_i^m,$$

with

$$\begin{aligned}
\kappa_5 &= -4\tau_v\,(b+2)(q+2)\big(3bq+3b+4q+2\big),\\
\kappa_4 &= -2\tau_v^2\,(b+2)\big(48bq^2+108bq+33b+56q^2+116q+26\big),\\
\kappa_3 &= -2\tau_v^3\,(b+2)\big(144bq^2+240bq+33b+152q^2+244q+30\big),\\
\kappa_2 &= -8\tau_v^4\,(b+1)\big(48bq^2+62bq+4b+100q^2+118q+7\big),\\
\kappa_1 &= -8\tau_v^5\,(b+1)\big(24bq^2+32bq+b+64q^2+52q+1\big),\\
\kappa_0 &= -64\,q\,\tau_v^6\,(b+1)(b+2q+1).
\end{aligned}$$

Since $\tau_v > 0$ and $b, q \geq 0$, we have $\kappa_m < 0$ for all m. For $\tau_i > 0$, every term $\kappa_m \tau_i^m < 0$, and

so $\text{Num}(\tau_i) < 0$. Consequently,

$$V_i''(\tau_i) < 0 \quad \text{for all } \tau_i > 0.$$

The same holds if $\tau_i = 0$ but $\tau_j > 0$ for some j (since then $b, q > 0$), which is ensured by the assumption that $\mathbf{z} \neq \mathbf{0}$. This completes the proof. ■

A.3 No asymmetric equilibria (Proof of Lemma 2)

Writing trader i's problem in terms of raw precision, each trader solves $\max_{\tau_i \geq 0} V_i(\tau_i) - \tilde{c}(\tau_i)$. In these terms, an equilibrium is a profile $\boldsymbol{\tau}^* = (\tau_1^*, \ldots, \tau_n^*)$ such that

$$V_i'(\tau_i^*) - \tilde{c}'(\tau_i^*) = 0 \qquad \text{for all } i = 1, \ldots, n,$$

where $V_i'(\tau_i^*)$ denotes the derivative of V_i with respect to τ_i, holding τ_j^* $(j \neq i)$ fixed. From Lemma 3 below, we know there exists a unique symmetric equilibrium, with $\tau_i^* = \tau_j^*$ for all i, j. Our goal here is to show that this is the only equilibrium—i.e., there are no asymmetric equilibria.

Define the two population totals

$$B := \mathbf{z} \cdot \mathbf{1} = \sum_{i=1}^{n} z_i, \qquad Q := \mathbf{z} \cdot \mathbf{1} + \mathbf{z} \cdot \mathbf{z} = \sum_{i=1}^{n} (z_i + z_i^2),$$

and the aggregates of other players besides i:

$$B_{-i} := B - z_i, \qquad Q_{-i} := Q - z_i - z_i^2.$$

Fixing τ_j $(j \neq i)$, the value function reduces, as shown in equation (A.5), to

$$V_i(\tau_i; B_{-i}, Q_{-i}) \propto \frac{N(\tau_i)}{D_1(\tau_i)\sqrt{D_2(\tau_i)}}, \qquad N(\tau_i) := 2\tau_i(\tau_i + \tau_v),$$

$$D_1(\tau_i) := (2 + B_{-i})\tau_i + 2\tau_v(1 + B_{-i}), \qquad D_2(\tau_i) := (2 + Q_{-i})\tau_i^2 + 2\tau_v(1 + 2Q_{-i})\tau_i + 4Q_{-i}\tau_v^2.$$

Since V_i depends on rivals only through (B_{-i}, Q_{-i}), this function is identical for all i, and so we write it as $V(\tau; b, q)$. Then, agent i's first-order condition reads

$$V'(\tau_i; B_{-i}, Q_{-i}) - \tilde{c}'(\tau_i) = 0.$$

We will use this to summarize equilibrium with a single function. Define

$$\Xi(\tau; B, Q) \;:=\; V'\Big(\tau;\; B - z(\tau),\; Q - \big(z(\tau) + z(\tau)^2\big)\Big) \;-\; \tilde{c}'(\tau),$$

where $z(\tau) := \tau/(\tau + 2\tau_v)$ is the mapping from raw precision to normalized precision. The domain of Ξ is the τ_i such that the rival statistics B_{-i} and Q_{-i} are nonnegative, i.e.,

$$\mathcal{D}(B,Q) := \Big\{\tau > 0 \;:\; B - z(\tau) \geq 0 \;\text{ and }\; Q - z(\tau) - z(\tau)^2 \geq 0\Big\}.$$

In this notation, any equilibrium $\boldsymbol{\tau}^*$ necessarily satisfies

$$\Xi(\tau_i^*; B^*, Q^*) = 0, \qquad \forall i, \tag{A.6}$$

where $B^* = \sum_i z(\tau_i^*)$, $Q^* = \sum_i (z(\tau_i^*) + z(\tau_i^*)^2)$, and $\tau_i^* \in \mathcal{D}(B^*, Q^*)$. We now show that equation (A.6) has at most one root. To do this, it suffices to show that $\Xi(\cdot; B, Q)$ is strictly decreasing on $\mathcal{D}(B,Q)$, which is proved in the next lemma.

Lemma A.2. *For every $B, Q \geq 0$, it holds that $\dfrac{d\Xi}{d\tau}(\tau; B, Q) < 0$ for all $\tau \in \mathcal{D}(B,Q)$.*

Proof. Write $b(\tau) := B - z(\tau)$, $q(\tau) := Q - z(\tau) - z(\tau)^2$ (both ≥ 0 on $\mathcal{D}(B,Q)$ by definition), so that $\Xi(\tau; B, Q) = V'(\tau; b(\tau), q(\tau)) - \tilde{c}'(\tau)$. By the chain rule, using $b'(\tau) = -z'(\tau)$ and $q'(\tau) = -z'(\tau)\big(1 + 2z(\tau)\big)$,

$$\frac{d\Xi}{d\tau} = \frac{\partial}{\partial \tau} V'(\tau; b, q) \;-\; z'(\tau)\Big[\frac{\partial}{\partial b} V'(\tau; b, q) + \big(1 + 2z(\tau)\big)\frac{\partial}{\partial q} V'(\tau; b, q)\Big] \;-\; \tilde{c}''(\tau).$$

The first two terms combine into

$$\frac{\partial}{\partial \tau} V'(\tau; b, q) - z'(\tau)\Big[\frac{\partial}{\partial b} V' + \big(1 + 2z(\tau)\big)\frac{\partial}{\partial q} V'\Big] = \frac{\mathrm{Num}(\tau)}{(\tau + 2\tau_v)^3 D_1(\tau)^3 D_2(\tau)^{5/2}},$$

where $\text{Num}(\tau) = \sum_{k=0}^{8} \gamma_k \tau^k$ with

$$
\begin{aligned}
\gamma_8 &= -4\tau_v(b+2)(q+2)(3bq+3b+4q+2),\\
\gamma_7 &= -2\tau_v^2\big(84b^2q^2+207b^2q+114b^2+266bq^2+620bq+320b+204q^2+444q+216\big),\\
\gamma_6 &= -4\tau_v^3\big(252b^2q^2+501b^2q+225b^2+758bq^2+1450bq+648b+558q^2+1036q+476\big),\\
\gamma_5 &= -8\tau_v^4\big(420b^2q^2+661b^2q+239b^2+1202bq^2+1860bq+688b+852q^2+1319q+502\big),\\
\gamma_4 &= -16\tau_v^5\big(420b^2q^2+515b^2q+148b^2+1150bq^2+1422bq+414b+790q^2+995q+287\big),\\
\gamma_3 &= -32\tau_v^6\big(252b^2q^2+241b^2q+54b^2+670bq^2+660bq+142b+452q^2+452q+91\big),\\
\gamma_2 &= -64\tau_v^7\big(84b^2q^2+67b^2q+11b^2+226bq^2+182bq+26b+154q^2+120q+15\big),\\
\gamma_1 &= -128\tau_v^8\big(12b^2q^2+11b^2q+b^2+38bq^2+28bq+2b+28q^2+17q+1\big),\\
\gamma_0 &= -256\,\tau_v^9\,q(b+1)(b+2q+1).
\end{aligned}
$$

Every γ_k is -1 times a sum of terms with strictly positive coefficients in $b,q \geq 0$, so $\gamma_k < 0$ for all $k > 0$. Thus, $\text{Num}(\tau) < 0$ for all $\tau > 0$. Since $(\tau+2\tau_v)^3 D_1(\tau)^3 D_2(\tau)^{5/2} > 0$ on $\mathcal{D}(B,Q)$ (where $b,q \geq 0$), the combined term is strictly negative. Finally, $\tilde{c}''(\tau) > 0$, so subtracting it only reinforces negativity, and hence

$$\frac{d\Xi}{d\tau} = \frac{\text{Num}(\tau)}{(\tau+2\tau_v)^3 D_1(\tau)^3 D_2(\tau)^{5/2}} - \tilde{c}''(\tau) < 0. \qquad \square$$

By Lemma A.2, equation (A.6) has at most one root in $\mathcal{D}(B^*,Q^*)$. This proves that all equilibria must be symmetric, i.e.,

$$\tau_1^* = \tau_2^* = \cdots = \tau_n^*.$$

Thus, the proof is complete. (Also, this proves that the entire equilibrium is unique, since Lemma 3 showed that there exists a single symmetric equilibrium.) ■

A.4 Existence and uniqueness of symmetric information equilibrium (Proof of Lemma 3)

First, we demonstrate the existence of $z^n \in (0,1)$ such that all traders' first-order conditions hold. Define the function

$$f_n(z) := \frac{dV_i(\mathbf{z})}{dz_i}\Big|_{\mathbf{z}=z\mathbf{1}} - c(z).$$

(Note this function is independent of i because we are evaluating dV_i/dz_i at a symmetric information allocation.) Since we use it repeatedly, we re-write equation (6) fully:

$$\frac{dV_i(\mathbf{z})}{dz_i} = \frac{1}{2\tau_v}(1 - I(\mathbf{z}))^2 L(\mathbf{z})(1 + 2z_i) + o_i(\mathbf{z}), \tag{A.7}$$

$$\text{where} \quad o_i(\mathbf{z}) := \frac{z_i(1+z_i)}{\sqrt{2\tau_u\tau_v}} \frac{d}{dz_i}\Big(\frac{1}{1+\mathbf{z}\cdot\mathbf{1}} \frac{1}{\sqrt{\mathbf{z}\cdot\mathbf{1}+\mathbf{z}\cdot\mathbf{z}}}\Big) \tag{A.8}$$

Note that $o_i(z\mathbf{1}) \to 0$ as $z \to 0$ and $o_i(z\mathbf{1})$ remains finite as $z \to 1$.

Evaluating equation (A.7) at $\mathbf{z} = z\mathbf{1}$ and taking $z \to 0$, we have

$$\lim_{z\to 0} \frac{dV_i(\mathbf{z})}{dz_i}\Big|_{\mathbf{z}=z\mathbf{1}} = \lim_{z\to 0} \frac{1}{2\tau_v}(1 - I(z\mathbf{1}))^2 L(z\mathbf{1}) = +\infty,$$

since $\lim_{z\to 0} I(z\mathbf{1}) = 0$ and $\lim_{z\to 0} L(z\mathbf{1}) = +\infty$ (see Lemma 1). At the same time, recall that $c'(0) < \infty$ by assumption. Consequently, we have $f_n(0+) = +\infty$.

Next, using (A.7), and noting that $L(\mathbf{1}) = (1+n)\sqrt{\tau_v}/\sqrt{n\tau_u}$ is finite when n is finite, we have that $\lim_{z\to 1} \frac{dV_i(\mathbf{z})}{dz_i}\big|_{\mathbf{z}=z\mathbf{1}}$ is finite. In addition, recall the assumption that $c'(1-) = +\infty$. Consequently, we have $f_n(1-) = -\infty$.

Since $f_n(\cdot)$ is continuous, there exists at least one $z^n \in (0,1)$ such that $f_n(z^n) = 0$. Thus, all traders' FOCs (5) hold with equality at such points z^n.

To complete the proof of existence, it remains to check global optimality of $z_i = z^n$. That is, trader i does not wish to deviate to any other level of precision. To do this, it is more convenient to transform the precision-choice problem (4) into a choice of τ_i rather than z_i. In that case, Lemma A.1 above shows that trader i's value function V_i is globally concave in raw precision τ_i. Since the information cost is convex in raw precision, i.e., $\tilde{c}(\tau_i)$ is convex, this implies that (4) is a concave problem, and so $z_i = z^n$ is the unique optimal solution.

Finally, we prove uniqueness. Our goal is to show that $f_n'(z) < 0$ at $z = z^n$ and so there can only be one such z^n. Notice that

$$\frac{d}{dz}\frac{dV_i(\mathbf{z})}{dz_i}\Big|_{\mathbf{z}=z\mathbf{1}} = \Big(\frac{d^2V_i(\mathbf{z})}{dz_i^2} + \sum_{j\neq i}\frac{d^2V_i(\mathbf{z})}{dz_i dz_j}\Big)\Big|_{\mathbf{z}=z\mathbf{1}}$$

Thus, we have that

$$f_n'(z) = \Big[\frac{d}{dz_i}\Big(\frac{dV_i(\mathbf{z})}{dz_i} - c'(z_i)\Big) + \sum_{j\neq i}\frac{d}{dz_j}\Big(\frac{1+2z_i}{2\tau_v}(1-I(\mathbf{z}))^2 L(\mathbf{z}) + o_i(\mathbf{z})\Big)\Big]\Big|_{\mathbf{z}=z\mathbf{1}}.$$

From (A.8), note that

$$o_i(\mathbf{z}) = -\frac{z_i(1+z_i)}{\sqrt{2\tau_u\tau_v}}\frac{1}{1+\mathbf{z}\cdot\mathbf{1}}\frac{1}{\sqrt{\mathbf{z}\cdot\mathbf{1}+\mathbf{z}\cdot\mathbf{z}}}\Big(\frac{1}{1+\mathbf{z}\cdot\mathbf{1}}+\frac{1}{2}\frac{1+2z_i}{\mathbf{z}\cdot\mathbf{1}+\mathbf{z}\cdot\mathbf{z}}\Big),$$

which is clearly decreasing in z_j for each $j \neq i$, i.e., $\frac{d}{dz_j}o_i(\mathbf{z}) < 0$. Similarly, we have that

$$\frac{1+2z_i}{2\tau_v}(1-I(\mathbf{z}))^2 L(\mathbf{z}) = \frac{1+2z_i}{\sqrt{2\tau_u\tau_v}}\Big(\frac{1}{1+\mathbf{z}\cdot\mathbf{1}}\frac{1}{\sqrt{\mathbf{z}\cdot\mathbf{1}+\mathbf{z}\cdot\mathbf{z}}}\Big)$$

is decreasing in z_j, for each $j \neq i$. Finally, because we already established global optimality, the second-order condition holds for trader i, and hence $\frac{d}{dz_i}\big(\frac{dV_i(\mathbf{z})}{dz_i} - c'(z_i)\big) < 0$ at $\mathbf{z} = z^n\mathbf{1}$. Therefore, all the terms are negative when $z = z^n$, and so

$$f_n'(z^n) < 0.$$

This completes the proof. ■

A.5 Characterizing the symmetric equilibrium (Statement and Proof of Lemma A.3)

In this section we characterize the symmetric information equilibrium objects for any finite n, and consider their asymptotic limits.

Lemma A.3. *Let z^n denote a symmetric equilibrium information acquisition in the economy with n strategic traders, in which $p^n = \lambda^n Y^n$ and $d_i^n = \beta^n s_i$ represent prices and (symmetric) demand on the equilibrium path. Then,*

$$\begin{aligned}
\lambda^n &= \sqrt{\frac{\tau_u}{2\tau_v}}\sqrt{\frac{nz^n(1+z^n)}{(1+nz^n)^2}} \\
Y^n &= \sqrt{\frac{2\tau_v}{\tau_u}}\sqrt{\frac{nz^n}{1+z^n}}\Big(\frac{1}{n}\sum_{i=1}^n s_i\Big) + \tau_u^{-1/2}e_u \\
p^n &= \frac{nz^n}{1+nz^n}\Big(\frac{1}{n}\sum_{i=1}^n s_i\Big) + \sqrt{\frac{\tau_v^{-1}}{2}}\frac{\sqrt{nz^n(1+z^n)}}{1+nz^n}e_u,
\end{aligned}$$

where $e_u = u/\tau_u^{-1/2} \sim Normal(0,1)$. Assume $Z^* := \lim_{n\to\infty} nz^n \in [0,\infty]$, the large-$n$ limit of

aggregate information, exists. Then, we have that

$$\frac{1}{n}\sum_{i=1}^{n} s_i \to v + \sqrt{\frac{\tau_v^{-1}}{2Z^*}}e_s,$$

in distribution, where $e_s \sim Normal(0,1)$ *is independent of* v *and* u*. Consequently, the large-*n *limiting equilibrium objects are, in distribution,*

$$\begin{aligned}
\lim_{n\to\infty} \lambda^n &= \sqrt{\frac{\tau_v^{-1}}{2\tau_u^{-1}}}\frac{\sqrt{Z^*}}{1+Z^*} \\
\lim_{n\to\infty} \Upsilon^n &= \sqrt{2\tau_u^{-1}}\left[\sqrt{Z^*}\frac{v}{\tau_v^{-1/2}} + \frac{e_s+e_u}{\sqrt{2}}\right] \\
\lim_{n\to\infty} p^n &= \tau_v^{-1/2}\frac{\sqrt{Z^*}}{1+Z^*}\left[\sqrt{Z^*}\frac{v}{\tau_v^{-1/2}} + \frac{e_s+e_u}{\sqrt{2}}\right].
\end{aligned}$$

Proof of Lemma A.3

In a symmetric equilibrium, $\mathbf{z} = z^n\mathbf{1}$, and so $\mathbf{z}\cdot\mathbf{1} = nz^n$ and $\mathbf{z}\cdot\mathbf{z} = n(z^n)^2$. Substituting these results into the expressions obtained in the proof of Lemma 1 for λ^n and β^n

$$\begin{aligned}
\lambda^n &= \sqrt{\frac{\tau_u}{2\tau_v}}\sqrt{\frac{nz^n(1+z^n)}{(1+nz^n)^2}}, \\
\beta^n &= \frac{1}{n}\sqrt{\frac{2\tau_v}{\tau_u}}\sqrt{\frac{nz^n}{1+z^n}}.
\end{aligned}$$

Then, noting $e_u := u/\tau_u^{-1/2} \sim \text{Normal}(0,1)$, we have

$$\begin{aligned}
\Upsilon^n &= \sqrt{\frac{2\tau_v}{\tau_u}}\sqrt{\frac{nz^n}{1+z^n}}\frac{\sum_i s_i}{n} + \tau_u^{-1/2}e_u, \\
p^n &= \frac{nz^n}{1+nz^n}\frac{\sum_i s_i}{n} + \sqrt{\frac{\tau_v^{-1}}{2}}\frac{\sqrt{nz^n(1+z^n)}}{1+nz^n}e_u.
\end{aligned}$$

We will now take the limit $n\to\infty$. First, note the fact that $s = v + \varepsilon_i$ in a symmetric equilibrium, where

$$\varepsilon \sim \mathcal{N}\left(0, \frac{1}{2}\tau_v^{-1}\frac{1-z^n}{z^n}\right).$$

Therefore,

$$\frac{\sum_i \varepsilon_i}{n} \sim \text{Normal}\left(0, \frac{1}{2}\tau_v^{-1}\frac{1-z^n}{nz^n}\right)$$

and so using the definition $Z^* := \lim_{n\to\infty} nz^n$ and the result that $z^n \to 0$ (Proposition 1), we have by the Central Limit Theorem,

$$\frac{\sum_i s_i}{n} \to v + \sqrt{\frac{\tau_v^{-1}}{2Z^*}}e_s$$

in distribution, where $e_s \sim \text{Normal}(0,1)$. Consequently, regardless of whether Z^* is finite or infinite, we have the following limits in distribution,

$$Y^n \to \sqrt{2\tau_u^{-1}}\left(\sqrt{Z^*}\frac{v}{\tau_v^{-1/2}} + \frac{e_s+e_u}{\sqrt{2}}\right)$$
$$p^n \to \tau_v^{-1/2}\frac{\sqrt{Z^*}}{1+Z^*}\left(\sqrt{Z^*}\frac{v}{\tau_v^{-1/2}} + \frac{e_s+e_u}{\sqrt{2}}\right)$$

This proves Lemma A.3. ■

A.6 Informativeness (Proof of Theorem 1)

The case of $\chi = \infty$ (and therefore $\Gamma = 0$) is immediate: since $\frac{dV_i(z\ \mathbf{1})}{dz_i} < \infty$ for all $z > 0$, the only possible solution to equation (5) is $z^n = 0$ for each n, hence $\mathcal{I} = 0$.

Next, we prove the case $\chi = 0$ (and therefore $\Gamma = \infty$). By Proposition 1, we know that z^n is a positive sequence that converges to zero. Assume a bounded subsequence of nz^n exists. Consider a subsequence such that $n_t z^{n_t} \to Z < \infty$—the existence of which is a consequence of the existence of a bounded subsequence. Then, we have $\lim_{t\to\infty} \frac{dV_i(z^{n_t}\ \mathbf{1})}{dz_i} > 0$. On the other hand, the fact that $z^{n_t} \to 0$ implies by the monotone convergence theorem that $c'(z^{n_t}) \to \chi = 0$. But, along that subsequence, we have that (5) holds with equality, which contradicts the fact that $\frac{dV_i(z^{n_t}\ \mathbf{1})}{dz_i}$ and $c'(z^{n_t})$ have different limit points. Therefore, each subsequence of nz^n must be unbounded, and $nz^n \to \infty$, which proves that $\mathcal{I} = 1$.

Finally, we prove the claim for $\chi \in (0,\infty)$—and so $\Gamma \in (0,\infty)$. By Proposition 1, we have $c'(z^n) \to \chi > 0$, which implies $\frac{dV_i(z^{n_t}\ \mathbf{1})}{dz_i} \to \chi$ for each subsequence $(n_t)_{t\geq 0}$, by equation (5). If $z^{n_t} \to 0$ and $n_t z^{n_t} \to \infty$, $\frac{dV_i(z^{n_t}\ \mathbf{1})}{dz_i} \to 0$, which contradicts the equality, which means that each subsequence is bounded, and therefore the sequence nz^n is itself bounded. Take any converging subsequence $(n_t)_{t\geq 0}$ and define $Z := \lim_{t\to\infty} n_t z^{n_t}$. We

then obtain:

$$\begin{aligned}\chi &= \lim_{t\to\infty} \frac{dV_i(z^{n_t}\,\mathbf{1})}{dz_i} \\ &= \lim_{t\to\infty} \sqrt{\frac{1}{2\tau_v\tau_u}}\frac{1}{1+n_t z^{n_t}}\frac{1}{\sqrt{n_t z^{n_t}+n_t(z^{n_t})^2}} \\ &= \sqrt{\frac{1}{2\tau_v\tau_u}}\frac{1}{1+Z}\frac{1}{\sqrt{Z}},\end{aligned}$$

which can be reorganized to obtain: $Z(1+Z)^2 = \Gamma$. Because there is a unique positive root Z, for this equation, call it Z^*, we conclude that all converging subsequences converge to Z^*. Because nz^n was shown to be bounded, in the first place, then $nz^n \to Z^*$ (i.e., the limit exists and is Z^*). Finally, we know that $I(z^n\mathbf{1}) = \frac{nz^n}{1+nz^n}$, so that $nz^n = \frac{I(z^n\mathbf{1})}{1-I(z^n\mathbf{1})}$. Since $\mathcal{I} = \lim_{n\to\infty} I(z^n\mathbf{1})$, we thus have $Z^* = \frac{\mathcal{I}}{1-\mathcal{I}}$. Combining this with $Z^*(1+Z^*)^2 = \Gamma$, we obtain the equality in the statement of the theorem. ■

A.7 Liquidity (Proof of Proposition 2)

Substitute $z^n\mathbf{1} = \mathbf{z}$ into $L(\mathbf{z})$ from Lemma 1 and then take the limit $n \to \infty$ to get

$$\mathcal{L} = \lim_{n\to\infty} L(z^n\mathbf{1}) = \sqrt{\frac{2\tau_v}{\tau_u}}\lim_{n\to\infty}\frac{1+nz^n}{\sqrt{nz^n}\sqrt{1+z^n}} = \sqrt{\frac{2\tau_v}{\tau_u}}\frac{1+Z^*}{\sqrt{Z^*}},$$

where $Z^* := \lim_{n\to\infty} nz^n$ (this limit is shown to exist in the proof of Theorem 1). Note that the result holds whether or not Z^* is finite, since $z^n \to 0$ by Proposition 1. Combining the expression for $\mathcal{L}$ with $\mathcal{I} = \frac{Z^*}{1+Z^*}$, we obtain the result. ■

A.8 Fixed perceived liquidity (Proof of Proposition 3)

We repeat the same derivations as in the proof of Lemma 1, except $\lambda(\mathbf{z})$ is replaced by $\tilde{\lambda}_n$. First, the expressions for B and β_i in (A.2) and (A.3) are replaced by

$$\begin{aligned}B &= \frac{\mathbf{z}\cdot\mathbf{1}}{1+\mathbf{z}\cdot\mathbf{1}}\frac{1}{\tilde{\lambda}_n} \\ \beta_i &= \frac{z_i}{1+\mathbf{z}\cdot\mathbf{1}}\frac{1}{\tilde{\lambda}_n}\end{aligned}$$

Second, we introduce again the unbiased demand signal $\tilde{Y} = Y/B = v + \eta$, where $\eta := \frac{1}{B}(\sum_j \beta_j \varepsilon_j + u)$ has precision

$$\begin{aligned}\tau_\eta &= \frac{B^2}{\sum_i \beta_i^2 \tau_i^{-1} + \tau_u^{-1}} \\ &= \frac{(\mathbf{z}\cdot\mathbf{1})^2 \tau_v}{(1+\mathbf{z}\cdot\mathbf{1})^2 \tilde{\lambda}_n^2 \frac{\tau_v}{\tau_u} + \frac{1}{2}(\mathbf{z}\cdot\mathbf{1} - \mathbf{z}\cdot\mathbf{z})}\end{aligned}$$

where the second line substitutes β_i, B, and $\tau_i^{-1} = \frac{1}{2\tau_v}\frac{1-z_i}{z_i}$ (recall the latter is the mapping from an individual's scaled precision z_i to her overall precision τ_i). By using Bayesian updating to solve for $\mathbb{E}[v|Y]$, and then setting the result equal to $p = \lambda Y$, we obtain the same expression for *actual* price impact λ, which after substituting β_i, τ_i^{-1}, and B becomes

$$\begin{aligned}\lambda &= \frac{B}{B^2 + \tau_v\left(\sum_i \beta_i^2 \tau_i^{-1} + \tau_u^{-1}\right)} \\ &= \tilde{\lambda}_n \frac{(\mathbf{z}\cdot\mathbf{1})(1+\mathbf{z}\cdot\mathbf{1})}{(\mathbf{z}\cdot\mathbf{1})^2 + \frac{1}{2}(\mathbf{z}\cdot\mathbf{1} - \mathbf{z}\cdot\mathbf{z}) + \frac{\tau_v}{\tau_u}(1+\mathbf{z}\cdot\mathbf{1})^2\tilde{\lambda}_n^2}.\end{aligned} \tag{A.9}$$

Indeed, if we impose rational expectations and set $\tilde{\lambda}_n = \lambda$, then we obtain the same expression from Lemma 1.

Now, we characterize price informativeness. Because prices contain the same information as Y, which contains the same information as $\tilde{Y}$, $\mathrm{Var}[v|p] = \mathrm{Var}[v|\tilde{Y}]$. Thus, we have

$$I(\mathbf{z}) = 1 - \frac{\mathrm{Var}[v|\tilde{Y}]}{\mathrm{Var}[v]} = \frac{\tau_\eta}{\tau_v + \tau_\eta},$$

where the second equality uses $\mathrm{Var}[v|\tilde{Y}]^{-1} = \tau_v + \tau_\eta$, which holds by standard Bayesian updating for jointly normal variables. Substituting τ_η from above into this expression, we obtain

$$\begin{aligned}I(\mathbf{z}) &= \frac{(\mathbf{z}\cdot\mathbf{1})^2}{(\mathbf{z}\cdot\mathbf{1})^2 + \frac{1}{2}(\mathbf{z}\cdot\mathbf{1} - \mathbf{z}\cdot\mathbf{z}) + \frac{\tau_v}{\tau_u}(1+\mathbf{z}\cdot\mathbf{1})^2\tilde{\lambda}_n^2} \\ &= \frac{\lambda}{\tilde{\lambda}_n}\frac{\mathbf{z}\cdot\mathbf{1}}{1+\mathbf{z}\cdot\mathbf{1}}\end{aligned}$$

From the second expression, we see that $I(\mathbf{z})$ coincides with the expression in Lemma 1 when rational expectations ($\lambda = \tilde{\lambda}_n$) is imposed.

Finally, the unique equilibrium is symmetric (by Lemma 2 and Lemma 3), so that

$\mathbf{z} = z^n\mathbf{1}$. Free entry implies that $n \to \infty$ (as discussed in the text) and $z^n \to 0$ (the latter by an identical argument to Proposition 1). We characterize competitive price informativeness $\mathcal{I} = \lim_{n\to\infty} I(z^n\mathbf{1})$. Indeed, let $Z := \lim_{n\to\infty} nz^n$. Then,

$$\mathcal{I} = \frac{Z^2}{Z^2 + \frac{1}{2}Z + \frac{\tau_v}{\tau_u}(1+Z)^2 \lim_{n\to\infty}\tilde{\lambda}_n^2} \leq \frac{1}{1 + \frac{\tau_v}{\tau_u}\lim_{n\to\infty}\tilde{\lambda}_n^2},$$

the second inequality due to $Z \leq \infty$. The last term, which is $\mathcal{I}$ when $Z = \infty$, is strictly less than 1 if and only if $\lim_{n\to\infty}\tilde{\lambda}_n > 0$.

Finally, the comparison between actual competitive liquidity $\mathcal{L} = \lim_{n\to\infty}\frac{1}{\lambda(z^n\mathbf{1})}$ and perceived liquidity $\tilde{\mathcal{L}} = \lim_{n\to\infty}\frac{1}{\tilde{\lambda}_n}$ is obtained by taking $nz^n \to \infty$ in (A.9). ■

A.9 Growing noise (Proof of Proposition 4)

We know the following first-order condition has to hold in the unique information equilibrium (symmetric by Lemma 2 and Lemma 3):

$$\frac{dV_i(z^n\mathbf{1})}{dz_i} = c'(z^n).$$

Starting from the value function

$$V_i(\mathbf{z}) = \sqrt{\frac{\tau_v^{-1}\tau_{u,n}^{-1}}{2}}\frac{1}{1+\mathbf{z}\cdot\mathbf{1}}\frac{z_i + z_i^2}{\sqrt{\mathbf{z}\cdot\mathbf{1} + \mathbf{z}\cdot\mathbf{z}}},$$

calculated in Lemma 1, we obtain its derivative when evaluated at the symmetric z^n:

$$\begin{aligned}\frac{dV_i(z^n\mathbf{1})}{dz_i} &= \frac{V_i(z^n\mathbf{1})}{z^n}\left(\frac{1+2z^n}{1+z^n} - \frac{z^n}{1+nz^n} - \frac{z^n + 2(z^n)^2}{nz^n + n(z^n)^2}\right)\\ &= \frac{V_i(z^n\mathbf{1})}{z^n}\left(\frac{n-1}{n}\frac{1+2z^n}{1+z^n} - \frac{z^n}{1+nz^n}\right).\end{aligned}$$

Notice that all the accumulation points of the sequence in parentheses are in a bounded strictly positive set. Indeed, because

$$0 \leq \frac{z^n}{1+nz^n} \leq \frac{1}{n},$$

we must have:

$$1 \leq \liminf \frac{1+2z^n}{1+z^n} = \liminf \left(\frac{n-1}{n} \frac{1+2z^n}{1+z^n} - \frac{z^n}{1+nz^n} \right)$$
$$\leq \limsup \left(\frac{n-1}{n} \frac{1+2z^n}{1+z^n} - \frac{z^n}{1+nz^n} \right) = \limsup \frac{1+2z^n}{1+z^n} \leq 2.$$

As a consequence, the limiting behavior of $\frac{dV_i(z^n\mathbf{1})}{dz_i}$ follows from the limiting behavior of $\frac{V_i(z^n\mathbf{1})}{z^n}$. Henceforth, we want to study:

$$\frac{V_i(z^n\mathbf{1})}{z^n} = \sqrt{\frac{\tau_v^{-1}\tau_{u,n}^{-1}}{2}} \frac{1}{1+nz^n} \frac{\sqrt{1+z^n}}{\sqrt{nz^n}}.$$

We consider the case of $\chi > 0$ (the case $\chi = 0$ follows easily since aggregate information grows even faster in that case). From the first-order condition and the results above, we then have $\liminf \frac{V_i(z^n\mathbf{1})}{z^n} = k \liminf c'(z^n)$, where $k \geq \frac{1}{2}$. Since $c'(z^n) \geq \chi$ for all n (since c is convex), we therefore have, asymptotically:

$$\liminf \sqrt{\tau_{u,n}^{-1}} = K \liminf((nz^n)^{1/2} + (nz^n)^{3/2})$$

where $K \geq k\chi \frac{\sqrt{2\tau_v}}{\limsup \sqrt{1+z^n}}$ is a positive constant.

If $\tau_{u,n}^{-1}$ grows unbounded, the dominant term in the right-hand side is the one with the higher exponent and so:

$$nz^n \sim \tau_{u,n}^{-1/3}.$$

Because nz^n explodes, it immediately follows that $\mathcal{I} = 1$ and $\mathcal{L} = \infty$.

Regarding the existence of a free-entry limit in this growing-noise case, suppose that $\limsup n^{-3}\tau_{u,n}^{-1} = 0$ as assumed in the proposition. Then, from $nz^n \sim \tau_{u,n}^{-1/3}$, we have that $z^n \to 0$. At the same time, $nz^n \sim \tau_{u,n}^{-1/3}$ also implies $\frac{V_i(z^n\mathbf{1})}{z^n} \sim \bar{v}$ for some positive constant $\bar{v}$. Combining these two results, $V_i(z^n\mathbf{1}) \to 0$. Since we also have $c(z^n) \to c(0) = 0$, this shows that traders make zero profits, i.e., $\lim V_i(z^n\mathbf{1}) - c(z^n) = 0$. Thus, a free-entry equilibrium exists. This concludes our proof.

While not required by the proposition, we also consider the case if $\tau_{u,n}^{-1}$ decreases to zero. In that case, the dominant term in the right-hand-side is the one with the lower exponent and so:

$$nz^n \sim \tau_{u,n}^{-1}.$$

In that case, it immediately follows that $\mathcal{I} = 0$ while $\mathcal{L}$ remains bounded. ■

B A Model of Limit Orders and Risk Aversion

We maintain the assumption that there is one risky asset with payoff $v \sim \mathcal{N}(0, \tau_v^{-1})$, noise trader demand is $u \sim \mathcal{N}(0, \tau_u^{-1})$, and traders observe signal $s_i = v + \varepsilon_i$, with $\varepsilon_i \sim \mathcal{N}(0, \tau_i^{-1})$. Moreover, $u \perp v \perp \varepsilon_i$ for all i, and $\varepsilon_i \perp \varepsilon_j$, for all i, j. Trader i's expected payoff of buying d_i units of the asset at price p is:

$$V_i = \frac{1 - \mathbb{E}\left[e^{-\gamma(v-p)d_i}\right]}{\gamma}.$$

Besides the payoff difference, there are two other changes relative to the baseline model. First, the trader submits a demand schedule, specifying for each possible price p the amount $d_i(p)$ of the asset he demands, taking into account market-clearing. Second, there is a competitive fringe—also understood as free entry—of uninformed traders. This assumption is often used in the literature (Bernhardt and Taub, 2006; Vives, 2014), and it leads to a simplification of the model: market clearing is substituted by the unbiasedness of prices. Formally, given aggregate demand $Y = \sum_i d_i + u$, for all informed traders i:

$$p = \mathbb{E}[v|Y].$$

Finally, when choosing ex-ante scaled precision z_i, a strategic trader maximizes

$$U_i = \frac{1 - \mathbb{E}\left[e^{-\gamma[(v-p)d_i - c(z_i)]}\right]}{\gamma}.$$

We put the cost inside the exponential so that it has the same interpretation—information expenditures—as the baseline risk-neutral model.

Note in advance that the first-order condition for precision says that $U_i'(z_i) = 0$. But since $U_i = \frac{1 - e^{\gamma c(z_i)}[1 - \gamma V_i]}{\gamma}$, we may write the FOC in terms of V_i as

$$e^{\gamma c(z_i)} \left(V_i'(z_i) - c'(z_i)[1 - \gamma V_i(z_i)]\right) = 0.$$

This simplifies when traders are informationally small, as will occur in the large-market limit. Indeed, when $z_i \to 0$, it will turn out that $d_i \to 0$, and so $V_i(0) = 0$. Therefore, the information FOC in the large-market limit can be written simply as

$$V_i'(0) = c'(0),$$

exactly as in the baseline risk-neutral model.

B.1 Proof of Theorem 2

We proceed in four steps. We first characterize the finite-n linear trading equilibrium. We then specialize to a semi-symmetric profile in which trader 1 (without loss of generality) deviates in information precision and compute, exactly, the marginal value of information at $\tau_1 = 0$. Next, we take the competitive limit while holding aggregate precision fixed. Finally, we rewrite the limiting equilibrium conditions in terms of price informativeness and information leakage.

Fix a profile of signal precisions $\boldsymbol{\tau} = (\tau_i)_{i=1}^n$. In a linear equilibrium, trader i submits

$$d_i = B_i s_i + C_i Y, \tag{B.1}$$

and the price satisfies

$$p = \lambda Y. \tag{B.2}$$

In the confines of a linear equilibrium, this permits all limit orders, since Y characterizes p and enters d_i.

Define the aggregate variables:

$$\bar{B} := \sum_j B_j, \qquad \bar{C} := \sum_j C_j, \qquad \bar{D} := \sum_j B_j^2 \tau_j^{-1} + \tau_u^{-1}.$$

Finally, let κ_p be the precision of the fundamental conditional on price alone, let κ_i be the precision of the fundamental after observing prices and s_i, and let τ_{η_i} be the additional precision provided by prices conditional on observing s_i. That is,

$$\kappa_p := \mathrm{Var}(v|p)^{-1}, \tag{B.3}$$

$$\kappa_i := \mathrm{Var}(v \mid p, s_i)^{-1}, \tag{B.4}$$

and

$$\tau_{\eta_i} := \mathrm{Var}(v \mid p, s_i)^{-1} - \mathrm{Var}(v \mid s_i)^{-1} = \kappa_i - \tau_v - \tau_i. \tag{B.5}$$

B.1.1 Linear trading equilibrium

We start by characterizing linear trading equilibria.

Lemma B.1. *Fix a profile of signal precisions $\boldsymbol{\tau}$. Any linear trading equilibrium satisfies:*

$$B_i \left(\lambda \frac{\kappa_i}{1 - \sum_{j \neq i} C_j} + \gamma \right) + \frac{B_i}{\bar{B} - B_i} \tau_{\eta_i} = \tau_i, \tag{B.6}$$

$$C_i\left(\lambda\frac{\kappa_i}{1-\sum_{j\neq i}C_j}+\gamma\right)=\frac{1-\bar{C}}{\bar{B}-B_i}\tau_{\eta_i}-\lambda\kappa_i, \tag{B.7}$$

and

$$\lambda=\frac{\bar{B}(1-\bar{C})}{\tau_v\bar{D}+\bar{B}^2}. \tag{B.8}$$

Proof of Lemma B.1

Fix d_i, the demand of trader i, and assume all other traders submit linear demand, as given by (B.1). Aggregation then leads to

$$Y=\frac{1}{1-\sum_{j\neq i}C_j}\left(\sum_{j\neq i}B_js_j+u+d_i\right).$$

Hence, using the pricing equation (B.2),

$$\frac{\partial p}{\partial d_i}=\lambda\frac{1}{1-\sum_{j\neq i}C_j}.$$

By the usual mean-variance representation of trader i's utility, and taking pricing impacts into account, we can solve for the optimal demand

$$d_i=\frac{\mathbb{E}[v\mid p,s_i]-p}{\gamma\,\mathrm{Var}(v\mid p,s_i)+\lambda\frac{1}{1-\sum_{j\neq i}C_j}}. \tag{B.9}$$

To obtain the Gaussian filtering equations, define

$$W_i:=\frac{(1-\bar{C})Y-B_is_i}{\bar{B}-B_i}=v+\eta_i,$$

where

$$\eta_i:=\frac{\sum_{j\neq i}B_j\varepsilon_j+u}{\bar{B}-B_i}.$$

Then $\eta_i\perp(v,\varepsilon_i)$ and

$$\mathrm{Var}(\eta_i)^{-1}=\frac{(\bar{B}-B_i)^2}{\sum_{j\neq i}B_j^2\tau_j^{-1}+\tau_u^{-1}}. \tag{B.10}$$

Thus,

$$\kappa_i:=\mathrm{Var}(v\mid p,s_i)^{-1}=\tau_v+\tau_i+\tau_{\eta_i},$$

which implies, by our definition (B.5), that $\tau_{\eta_i}=\mathrm{Var}(\eta_i)^{-1}$. By standard Gaussian filter-

ing,

$$\mathbb{E}[v \mid p, s_i] = \frac{\tau_i}{\kappa_i} s_i + \frac{\tau_{\eta_i}}{\kappa_i} W_i.$$

Substituting the posterior mean and variances into (B.9) and matching coefficients gives (B.6) and (B.7).

To close the equilibrium, all that is missing is to characterize λ. For that, we use Gaussian filtering on the order flow. Normalize aggregate order flow as

$$\widetilde{Y} := \frac{1 - \bar{C}}{\bar{B}} Y = v + \eta, \qquad \bar{\eta} := \frac{\sum_j B_j \varepsilon_j + u}{\bar{B}}.$$

Note that

$$\tau_{\bar{\eta}} := \mathrm{Var}(\bar{\eta})^{-1} = \frac{\bar{B}^2}{\bar{D}}. \tag{B.11}$$

Because $p = \mathbb{E}[v \mid Y] = \mathbb{E}[v \mid \widetilde{Y}]$, Gaussian filtering implies

$$p = \frac{\frac{\bar{B}^2}{\bar{D}}}{\tau_v + \frac{\bar{B}^2}{\bar{D}}} \frac{1 - \bar{C}}{\bar{B}} Y.$$

Setting $p = \lambda Y$ by (B.2) and then reorganizing leads to (B.8). ■

Lemma B.2. *Fix a profile of signal precisions $\boldsymbol{\tau}$. Define*

$$R_i := \frac{1}{\gamma} \frac{\lambda}{1 - \sum_{j \neq i} C_j}. \tag{B.12}$$

In a linear trading equilibrium, trader i's ex-ante trading payoff is:

$$V_i = \frac{1}{\gamma} \left[1 - \left(1 + \left(\frac{1}{\kappa_p} - \frac{1}{\kappa_i} \right) \frac{\frac{1}{\kappa_i} + 2R_i}{\left(\frac{1}{\kappa_i} + R_i \right)^2} \right)^{-1/2} \right]. \tag{B.13}$$

Proof of Lemma B.2

Conditional on (p, s_i), $v - p$ is Gaussian with mean $\mathbb{E}[v - p \mid p, s_i]$ and precision κ_i. Hence

$$\mathbb{E}_v \left[e^{-\gamma (v-p) d_i(p)} \mid p, s_i \right] = \exp \left\{ -\gamma \mathbb{E}[v - p \mid p, s_i] d_i(p) + \frac{\gamma^2}{2} \kappa_i^{-1} d_i(p)^2 \right\}.$$

Rewrite (B.9), which is the first order condition for d_i, as

$$\mathbb{E}[v - p \mid p, s_i] = \gamma(\kappa_i^{-1} + R_i)d_i(p).$$

Substituting this equality into the preceding expression gives

$$\mathbb{E}_v\left[e^{-\gamma(v-p)d_i(p)} \mid p, s_i\right] = \exp\left\{-\frac{\gamma^2}{2}(\kappa_i^{-1} + 2R_i)d_i(p)^2\right\}.$$

Consequently, trader i's ex-ante gross utility satisfies

$$\gamma V_i = 1 - \mathbb{E}\left[\exp\left\{-\frac{\gamma^2}{2}(\kappa_i^{-1} + 2R_i)d_i(p)^2\right\}\right].$$

Under a linear equilibrium, $d_i(p)$ is an ex-ante mean-zero Gaussian random variable. Since κ_i^{-1} and R_i are constants, the expectation above is the moment-generating function of a squared Gaussian. Therefore,

$$\gamma V_i = 1 - \left[1 + \gamma^2(\kappa_i^{-1} + 2R_i)\operatorname{Var}(d_i(p))\right]^{-1/2}.$$

It remains to calculate $\operatorname{Var}(d_i(p))$. From the optimal demand in (B.9),

$$\operatorname{Var}(d_i(p)) = \frac{\operatorname{Var}(\mathbb{E}[v - p \mid p, s_i])}{\gamma^2(\kappa_i^{-1} + R_i)^2}.$$

By the law of total variance,

$$\operatorname{Var}(\mathbb{E}[v - p \mid p, s_i]) = \operatorname{Var}(v - p) - \operatorname{Var}(v \mid p, s_i) = \operatorname{Var}(v \mid p) - \operatorname{Var}(v \mid p, s_i) = \kappa_p^{-1} - \kappa_i^{-1},$$

where we used that $\operatorname{Var}(v - p) = \operatorname{Var}(v - p \mid p) = \operatorname{Var}(v \mid p)$, since the conditional variance of joint Gaussians is constant. Thus,

$$\operatorname{Var}(d_i(p)) = \frac{\kappa_p^{-1} - \kappa_i^{-1}}{\gamma^2(\kappa_i^{-1} + R_i)^2}.$$

Substituting into gross utility yields the equation in the statement of the lemma. ■

B.1.2 Free-entry limit of symmetric equilibria

We now let $\boldsymbol{\tau}^n$ represent a profile of identical precisions with n players: that is $\boldsymbol{\tau}^n = \tau^n \mathbf{1}_n$ for some scalar τ^n. To allow player one to deviate from that profile, we let $\widehat{\boldsymbol{\tau}}^n = \tau^n \mathbf{1}_{n-1}$

be the symmetric precisions of the other $n-1$ traders. Therefore, we write $(\tau_i, \widehat{\boldsymbol{\tau}}^n)$ for the profile in which player i has precision τ_i and all other players have precision τ^n. We refer to this type of profile as "semi-symmetric" for short.

Lemma B.3 (Informational smallness)**.** *Let $\boldsymbol{\tau}^n = \tau^n \mathbf{1}_n$ be the precision profile in a symmetric information equilibrium with n traders. Then, as $n \to \infty$,*

$$\tau^n \longrightarrow 0.$$

Proof of Lemma B.3

Fix a symmetric equilibrium with n informed traders. Let B^n and C^n denote the common individual-trader demand coefficients, and define

$$T^n := n\tau^n, \qquad \bar{B}^n := nB^n, \qquad r^n := \frac{n-1}{n}.$$

For notational convenience, also define

$$X^n := \frac{(\bar{B}^n)^2}{T^n}.$$

Plugging this notation into equation (B.11), the precision of the signal about v contained in aggregate order flow (recall this signal was called $\bar{\eta}$) is

$$\tau^n_{\bar{\eta}} = \frac{(\bar{B}^n)^2}{X^n + \tau_u^{-1}}.$$

Similarly, plugging the same notation into equation (B.10), we have that after any trader i removes her own order from aggregate order flow, the precision of the remaining order-flow signal (recall this signal was called η_i) is

$$\tau^n_{\eta_i} = \frac{(r^n \bar{B}^n)^2}{r^n X^n + \tau_u^{-1}}.$$

Hence,

$$\kappa^n_p = \tau_v + \tau^n_{\bar{\eta}}, \qquad \kappa^n_i = \tau_v + \tau^n_{\eta_i} + \tau^n.$$

Notice that $\tau^n_{\eta_i}$ and thus κ^n_i are independent of i, and so going forward we drop the i subscripts from these.

We first compute the difference between the trader's private posterior precision and

the public posterior precision. Since

$$\kappa^n - \kappa_p^n = \tau^n + \tau_\eta^n - \tau_{\bar{\eta}}^n,$$

direct substitution gives

$$\kappa^n - \kappa_p^n = \tau^n + \frac{(r^n \bar{B}^n)^2}{r^n X^n + \tau_u^{-1}} - \frac{(\bar{B}^n)^2}{X^n + \tau_u^{-1}} = \tau^n \frac{\tau_u^{-2}}{(r^n X^n + \tau_u^{-1})(X^n + \tau_u^{-1})}. \tag{B.14}$$

Next, define the price-impact term perceived by an informed trader as

$$H^n := \lambda^n \frac{\kappa^n}{1 - (n-1)C^n}.$$

Multiplying (B.6) by n,

$$T^n = \bar{B}^n \left(H^n + \gamma + \frac{\tau_\eta^n}{r^n \bar{B}^n} \right).$$

Since

$$\frac{\tau_\eta^n}{r^n \bar{B}^n} = \frac{r^n \bar{B}^n}{r^n X^n + \tau_u^{-1}},$$

we obtain

$$H^n + \gamma = \frac{T^n}{\bar{B}^n} - \frac{r^n \bar{B}^n}{r^n X^n + \tau_u^{-1}} = \frac{\tau_u^{-1}}{r^n X^n + \tau_u^{-1}} \sqrt{\frac{T^n}{X^n}}. \tag{B.15}$$

By Lemma B.2, we can write payoffs as

$$V_i^n = \frac{1}{\gamma} \left[1 - (1 + A^n \Delta^n)^{-1/2} \right],$$

where

$$\Delta^n := \frac{1}{\kappa_p^n} - \frac{1}{\kappa^n}.$$

and

$$A^n = \kappa^n \frac{1 + 2H^n/\gamma}{(1 + H^n/\gamma)^2}.$$

For $q \geq 0$,

$$\frac{1 + 2q}{(1+q)^2} \leq \frac{2}{1+q}.$$

Hence, It follows that

$$A^n \Delta^n = \frac{\kappa^n - \kappa_p^n}{\kappa_p^n} \frac{1 + 2H^n/\gamma}{(1 + H^n/\gamma)^2} \leq \frac{\kappa^n - \kappa_p^n}{\kappa_p^n} \frac{2\gamma}{H^n + \gamma}.$$

Using (B.14), (B.15), and $T^n = n\tau^n$:

$$A^n\Delta^n \leq \frac{2\gamma\tau_u^{-1}\sqrt{X^nT^n}}{n(X^n+\tau_u^{-1})\kappa_p^n}.$$

Also,

$$\begin{aligned}(X^n+\tau_u^{-1})\kappa_p^n &= (X^n+\tau_u^{-1})\left(\tau_v + \frac{T^nX^n}{X^n+\tau_u^{-1}}\right)\\ &= \tau_v(X^n+\tau_u^{-1}) + T^nX^n\\ &\geq \tau_v\tau_u^{-1} + T^nX^n = \tau_v\tau_u^{-1} + (\bar{B}^n)^2.\end{aligned}$$

We thus obtain

$$A^n\Delta^n \leq \frac{2\gamma\tau_u^{-1}\bar{B}^n}{n\left(\tau_v\tau_u^{-1} + (\bar{B}^n)^2\right)}.$$

Using

$$\sup_{x\geq 0}\frac{x}{a+x^2} = \frac{1}{2\sqrt{a}},$$

with $a = \tau_v\tau_u^{-1}$, we conclude that

$$A^n\Delta^n \leq \frac{\gamma}{n\sqrt{\tau_u\tau_v}}.$$

Finally, because

$$1-(1+y)^{-1/2} \leq \frac{y}{2} \qquad \text{for } y \geq 0,$$

the trader's gross value satisfies

$$V_i^n \leq \frac{1}{2n\sqrt{\tau_u\tau_v}}.$$

A trader can always choose zero information and obtain zero net payoff. Therefore equilibrium optimality implies

$$0 \leq c\left(\frac{\tau^n}{\tau^n+2\tau_v}\right) \leq V_i^n \leq \frac{1}{2n\sqrt{\tau_u\tau_v}},$$

Since $c(0) = 0$ and $c(z) > 0$ for every $z > 0$, it follows that $\tau^n \longrightarrow 0$. ■

Lemma B.4. *Consider a sequence of semi-symmetric profiles $\boldsymbol{\tau}^n$ such that $\lim_{n\to\infty}(n-1)\tau^n = T \in (0,\infty)$. Then:*

$$\lim_{n\to\infty} \frac{\partial}{\partial \tau_i} V_i(0, \widehat{\boldsymbol{\tau}}^n) = \frac{\bar{B}}{2T(\tau_v + \tau_{\bar{\eta}})}, \tag{B.16}$$

where $\bar{B}$ and $\tau_{\bar{\eta}}$ are the unique positive solutions to:

$$\tau_{\bar{\eta}} = \frac{\bar{B}^2}{\frac{\bar{B}^2}{T} + \tau_u^{-1}}, \qquad \textit{and} \qquad T = \gamma \bar{B} + 2\tau_{\bar{\eta}}.$$

Proof of Lemma B.4

Without loss, let suppose trader 1 is the deviating player. We start by establishing the derivative of trader 1's B_1 coefficient when $\tau_1 = 0$ while the other traders are using the symmetric precision $\tau_i = \tau^n$ for $i = 2, \ldots, n$. Let B^n, C^n denote the common trading coefficients of traders $2, \ldots, n$, and define the finite-n aggregates

$$\bar{B}^n := (n-1)B^n, \qquad \bar{C}^n := (n-1)C^n, \qquad T^n := (n-1)\tau^n.$$

Note these definitions differ from the previous lemma, because they aggregate only traders $2, \ldots, n$ and exclude trader 1. Both $\bar{B}^n$ and $\bar{C}^n$ may depend on τ_1, as we will see.

From (B.6), the equation for coefficient B_1 is

$$B_1 \left(\lambda \frac{\kappa_1}{1 - \bar{C}^n} + \gamma \right) + \frac{B_1}{\bar{B}^n} \tau_{\eta_1} = \tau_1. \tag{B.17}$$

For any other trader $i \neq 1$, by symmetry of precisions and (B.6):

$$B^n \left(\lambda \frac{\kappa^n}{1 - \frac{n-2}{n-1}\bar{C}^n - C_1} + \gamma \right) + \frac{B^n}{\frac{n-2}{n-1}\bar{B}^n + B_1} \tau_\eta^n = \tau^n,$$

where, from equations (B.5) and (B.10) specialized to this semi-symmetric situation, we have

$$\kappa_1 = \tau_v + \tau_1 + \tau_{\eta_1}, \qquad \kappa^n = \tau_v + \tau^n + \tau_\eta^n,$$

with

$$\tau_{\eta_1} = \frac{(\bar{B}^n)^2}{(\bar{B}^n)^2/T^n + \tau_u^{-1}}, \qquad \tau_\eta^n = \frac{\left(\frac{n-2}{n-1}\bar{B}^n + B_1\right)^2}{B_1^2/\tau_1 + \frac{n-2}{n-1}(\bar{B}^n)^2/T^n + \tau_u^{-1}}.$$

Moreover,

$$\bar{D} = \frac{B_1^2}{\tau_1} + \frac{(\bar{B}^n)^2}{T^n} + \frac{1}{\tau_u}, \qquad \bar{B} = B_1 + \bar{B}^n.$$

We now evaluate the semi-symmetric equilibrium at $\tau_1 = 0$. We will write various

objects as functions of τ_1 and evaluate them at $\tau_1 = 0$. Equation (B.17) implies

$$B_1(0) = 0.$$

Thus, $\bar{B}(0) = \bar{B}^n(0)$. At $\tau_1 = 0$, trader 1 has no private information, so

$$\kappa_1(0) = \kappa_p(0) = \tau_v + \tau^n_{\bar{\eta}}(0) = \tau_v + \frac{\bar{B}^n(0)^2}{\bar{D}(0)} = \frac{\tau_v \bar{D}(0) + \bar{B}^n(0)^2}{\bar{D}(0)}, \tag{B.18}$$

where $\tau^n_{\bar{\eta}}$ is the precision of the error in the price—defined by equation (B.11) and here also superscripted by n to distinguish it from its limiting value—which we use to obtain the third equality.

Equation (B.7) for trader 1 also gives $C_1(0) = 0$. Indeed, using (B.8) and (B.18), its right-hand side is exactly zero, while the coefficient multiplying C_1 is strictly positive. Thus, at $\tau_1 = 0$, $\bar{C}(0) = \bar{C}^n(0)$.

Differentiate (B.17) with respect to τ_1 at zero. Since $B_1(0) = 0$, all derivatives of the terms multiplying B_1 disappear, giving

$$B_1'(0)\left(\lambda \frac{\kappa_1(0)}{1 - \bar{C}^n(0)} + \gamma + \frac{\tau_{\eta_1}(0)}{\bar{B}^n(0)}\right) = 1.$$

Using the price-impact equation (B.8) and equation (B.18), followed by (B.11), we obtain

$$\lambda \frac{\kappa_1(0)}{1 - \bar{C}^n(0)} = \frac{\bar{B}^n(0)(1 - \bar{C}^n(0))}{\tau_v \bar{D}(0) + \bar{B}^n(0)^2} \frac{1}{1 - \bar{C}^n(0)} \frac{\tau_v \bar{D}(0) + \bar{B}^n(0)^2}{\bar{D}(0)} = \frac{\bar{B}^n(0)}{\bar{D}(0)} = \frac{\tau^n_{\bar{\eta}}(0)}{\bar{B}^n(0)}. \tag{B.19}$$

Finally, notice by comparing (B.11) for $\tau^n_{\bar{\eta}}$ against the equation for τ_{η_1} that we have

$$\tau^n_{\bar{\eta}}(0) = \tau_{\eta_1}(0). \tag{B.20}$$

We conclude that

$$B_1'(0) = \frac{1}{\gamma + 2\tau^n_{\bar{\eta}}(0)/\bar{B}^n(0)}. \tag{B.21}$$

We now proceed to prove the main result. For trader 1,

$$\kappa_1 - \kappa_p = \tau_1 + \tau_{\eta_1} - \tau^n_{\bar{\eta}},$$

where, again,

$$\tau_{\eta_1} = \frac{(\bar{B}^n)^2}{(\bar{B}^n)^2/T^n + \tau_u^{-1}}, \qquad \tau_{\bar{\eta}}^n = \frac{(\bar{B}^n + B_1)^2}{B_1^2/\tau_1 + (\bar{B}^n)^2/T^n + \tau_u^{-1}}.$$

Because $B_1(0) = 0$ and B_1 is differentiable at $\tau_1 = 0$,

$$B_1(\tau_1) = B_1'(0)\tau_1 + o(\tau_1),$$

so

$$\frac{B_1(\tau_1)^2}{\tau_1} = B_1'(0)^2\tau_1 + o(\tau_1).$$

Differentiating $\kappa_1 - \kappa_p$ at zero, the terms involving $\frac{\partial}{\partial \tau_1}\bar{B}^n(0)$ cancel, yielding

$$\left.\frac{\partial(\kappa_1 - \kappa_p)}{\partial \tau_1}\right|_{\tau_1=0} = \left[1 - \frac{\bar{B}^n(0)}{\bar{B}^n(0)^2/T^n + \tau_u^{-1}} B_1'(0)\right]^2. \tag{B.22}$$

Using (B.21),

$$\left.\frac{\partial(\kappa_1 - \kappa_p)}{\partial \tau_1}\right|_{\tau_1=0} = \left(\frac{\gamma + \tau_{\bar{\eta}}^n(0)/\bar{B}^n(0)}{\gamma + 2\tau_{\bar{\eta}}^n(0)/\bar{B}^n(0)}\right)^2.$$

At $\tau_1 = 0$, recall from (B.18) that $\kappa_1 = \kappa_p$. Hence the term $\kappa_p^{-1} - \kappa_1^{-1}$ in the ex-ante payoff equation (B.13) is zero, so derivatives of the remaining multiplicative factor drop out when differentiating the value function. Moreover,

$$\left.\frac{\partial}{\partial \tau_1}\left(\frac{1}{\kappa_p} - \frac{1}{\kappa_1}\right)\right|_{\tau_1=0} = \frac{1}{\kappa_p(0)^2}\left.\frac{\partial(\kappa_1 - \kappa_p)}{\partial \tau_1}\right|_{\tau_1=0}.$$

Finally, (B.12), (B.19), and (B.20) deliver

$$\kappa_p(0)R_1(0) = \frac{1}{\gamma}\frac{\tau_{\bar{\eta}}^n(0)}{\bar{B}^n(0)}.$$

Substitution into (B.13) simplifies to

$$\frac{\partial}{\partial \tau_i} V_i(\tau_i, \widehat{\boldsymbol{\tau}}^n) = \frac{1}{2(\tau_v + \tau_{\bar{\eta}}^n(0))\left(\gamma + 2\tau_{\bar{\eta}}^n(0)/\bar{B}^n(0)\right)}. \tag{B.23}$$

We now take $n \to \infty$ while maintaining $T^n := (n-1)\tau^n \to T$. Consider any subse-

quence along which

$$\bar{B}^n(0) \longrightarrow \bar{B} > 0.$$

Then, by continuity,

$$\tau^n_{\bar{\eta}}(0) \longrightarrow \tau_{\bar{\eta}} := \frac{\bar{B}^2}{\bar{B}^2/T + \tau_u^{-1}}. \tag{B.24}$$

It remains to obtain the limiting aggregate trading equation. At $\tau_1 = 0$, recall that $B_1 = C_1 = 0$. For all other traders,

$$\tau^n_\eta(0) = \frac{\left(\frac{n-2}{n-1}\bar{B}^n\right)^2}{\frac{n-2}{n-1}(\bar{B}^n)^2/T^n + \tau_u^{-1}} \longrightarrow \tau_{\bar{\eta}},$$

and, because $\tau^n = T^n/(n-1) \to 0$,

$$\kappa^n = \tau_v + \tau^n + \tau^n_\eta \longrightarrow \tau_v + \tau_{\bar{\eta}}.$$

We want to take the limit of the price-impact term for traders $2, \ldots, n$, which is the term

$$H^n := \lambda \frac{\kappa^n}{1 - \frac{n-2}{n-1}\bar{C}^n}.$$

Combining (B.7) with the price-impact equation (B.8), using the facts that when $\tau_1 = 0$ we have $B_1 = C_1 = 0$, $\bar{D} = \frac{(\bar{B}^n)^2}{T^n} + \tau_u^{-1}$, and $\tau^n_{\bar{\eta}} = \frac{(\bar{B}^n)^2}{T^n}$, and doing algebra, we obtain the following scalar equation for H^n:

$$H^n\left(H^n + \gamma + \frac{\tau^n_\eta}{\frac{n-2}{n-1}\bar{B}^n}\right) = \frac{\bar{B}^n}{(\bar{B}^n)^2/T^n + \tau_u^{-1}} \frac{\kappa^n}{\kappa_p}(2H^n + \gamma).$$

The coefficients in this equation are bounded along the convergent subsequence, so H^n is bounded. Moreover,

$$\frac{\tau^n_\eta}{\frac{n-2}{n-1}\bar{B}^n} \longrightarrow \frac{\tau_{\bar{\eta}}}{\bar{B}}, \qquad \frac{\bar{B}^n}{(\bar{B}^n)^2/T^n + \tau_u^{-1}} \frac{\kappa^n}{\kappa_p} \longrightarrow \frac{\tau_{\bar{\eta}}}{\bar{B}}.$$

Thus, every limit point H of H^n satisfies

$$H\left(H + \gamma + \frac{\tau_{\bar{\eta}}}{\bar{B}}\right) = \frac{\tau_{\bar{\eta}}}{\bar{B}}(2H + \gamma),$$

or equivalently

$$\left(H - \frac{\tau_{\bar{\eta}}}{\bar{B}}\right)(H + \gamma) = 0.$$

Since $H^n \geq 0$, it follows that

$$H^n \longrightarrow \frac{\tau_{\bar{\eta}}}{\bar{B}}.$$

Multiplying the equations in (B.6) for traders $2,\ldots,n$ by $n-1$ and taking limits therefore gives

$$T = \gamma \bar{B} + 2\tau_{\bar{\eta}}. \tag{B.25}$$

Combining (B.23), (B.24), and (B.25) delivers the result. ■

Lemma B.5 (Local value of information)**.** *Consider a sequence of semi-symmetric profiles $\boldsymbol{\tau}^n$ such that $(n-1)\tau^n \longrightarrow T \in (0,\infty)$. Then, for every fixed $a > 0$,*

$$\frac{V_1(a\tau^n, \widehat{\boldsymbol{\tau}}^n)}{a\tau^n} - \frac{\partial}{\partial \tau_1} V_1(0, \widehat{\boldsymbol{\tau}}^n) \longrightarrow 0.$$

Proof of Lemma B.5

Fix $a > 0$ and consider the semi-symmetric profile $(a\tau^n, \widehat{\boldsymbol{\tau}}^n)$. Let $\bar{B}^n = (n-1)B^n$ denote the aggregate signal coefficient of traders $2,\ldots,n$ in the associated trading equilibrium.

We first establish that the aggregate trading equilibrium converges to the same limit as when trader 1 has zero precision. From the equations for the loadings on signals, (B.6),

$$0 \leq B_1 \leq \frac{a\tau^n}{\gamma}, \qquad 0 \leq \bar{B}^n \leq \frac{(n-1)\tau^n}{\gamma}.$$

Hence $B_1 \to 0$ and $(\bar{B}^n)$ is bounded.

Consider any convergent subsequence,

$$\bar{B}^n \longrightarrow B^\circ.$$

Since $B_1 \to 0$, $\tau^n \to 0$ by Lemma B.3, and $(n-1)\tau^n \to T$, the same limiting argument used in the proof of Lemma B.4 applies to the symmetric traders. In particular,

$$\tau^n_{\eta_i} \longrightarrow \frac{(B^\circ)^2}{(B^\circ)^2/T + \tau_u^{-1}},$$

and the price-impact term in their B-equation converges to

$$\frac{B^\circ}{(B^\circ)^2/T + \tau_u^{-1}}.$$

Taking limits in their aggregate B-equation therefore gives

$$T = \gamma B^\circ + 2\frac{(B^\circ)^2}{(B^\circ)^2/T + \tau_u^{-1}}.$$

As shown in the proof of Lemma B.4, this equation has a unique positive solution. Hence $B^\circ = \bar{B}$, where $\bar{B}$ is the coefficient appearing in the statement of that lemma. Since every convergent subsequence has the same limit,

$$\bar{B}^n \longrightarrow \bar{B}.$$

Consequently,

$$\tau_{\eta_1} \longrightarrow \tau_{\bar{\eta}} := \frac{\bar{B}^2}{\bar{B}^2/T + \tau_u^{-1}}.$$

Next, write

$$B_1 = b^n a \tau^n.$$

Dividing trader 1's equation (B.17) by $a\tau^n$ gives

$$b^n \left(\lambda \frac{\kappa_1}{1 - \bar{C}^n} + \gamma + \frac{\tau_{\eta_1}}{\bar{B}^n} \right) = 1.$$

Applying the same price-impact limit established above,

$$\lambda \frac{\kappa_1}{1 - \bar{C}^n} \longrightarrow \frac{\tau_{\bar{\eta}}}{\bar{B}},$$

so

$$b^n \longrightarrow \frac{1}{\gamma + 2\tau_{\bar{\eta}}/\bar{B}} = \frac{\bar{B}}{T}, \tag{B.26}$$

where the last equality follows from (B.25).

We can now calculate the incremental precision generated by trader 1's private signal. Using the definitions of τ_{η_1} and τ_η already given in the proof of Lemma B.4, with $B_1 = a\tau^n b^n$, we obtain

$$\begin{aligned}
\kappa_1 - \kappa_p &= a\tau^n + \tau_{\eta_1} - \tau_\eta \\
&= a\tau^n \frac{\left(\frac{(\bar{B}^n)^2}{(n-1)\tau^n} + \tau_u^{-1} - \bar{B}^n b^n \right)^2}{\left(\frac{(\bar{B}^n)^2}{(n-1)\tau^n} + \tau_u^{-1} \right) \left(\frac{(\bar{B}^n)^2}{(n-1)\tau^n} + \tau_u^{-1} + a\tau^n (b^n)^2 \right)}.
\end{aligned}$$

Using (B.26) and $\bar{B}^n \to \bar{B}$,

$$\frac{\kappa_1 - \kappa_p}{a\tau^n} \longrightarrow \left(1 - \frac{\tau_{\bar{\eta}}}{T}\right)^2.$$

Moreover,

$$\kappa_1, \kappa_p \longrightarrow \tau_v + \tau_{\bar{\eta}}.$$

Hence

$$\frac{1}{a\tau^n}\left(\frac{1}{\kappa_p} - \frac{1}{\kappa_1}\right) \longrightarrow \frac{\left(1 - \tau_{\bar{\eta}}/T\right)^2}{(\tau_v + \tau_{\bar{\eta}})^2}. \tag{B.27}$$

We now apply the payoff formula in Lemma B.2. Since

$$\kappa_1 R_1 = \frac{1}{\gamma}\lambda \frac{\kappa_1}{1 - \bar{C}^n},$$

we have

$$\kappa_1 R_1 \longrightarrow \frac{1}{\gamma}\frac{\tau_{\bar{\eta}}}{\bar{B}}.$$

Therefore the multiplicative term in (B.13) converges to

$$(\tau_v + \tau_{\bar{\eta}}) \frac{1 + \frac{2}{\gamma}\frac{\tau_{\bar{\eta}}}{\bar{B}}}{\left(1 + \frac{1}{\gamma}\frac{\tau_{\bar{\eta}}}{\bar{B}}\right)^2}.$$

By (B.27), the term inside the outer parentheses in (B.13) is of order τ^n. Thus, using

$$1 - (1 + x)^{-1/2} = \frac{x}{2} + o(x)$$

as $x \to 0$,

$$\frac{V_1(a\tau^n, \widehat{\boldsymbol{\tau}}^n)}{a\tau^n} \longrightarrow \frac{1}{2\gamma(\tau_v + \tau_{\bar{\eta}})} \frac{1 + \frac{2}{\gamma}\frac{\tau_{\bar{\eta}}}{\bar{B}}}{\left(1 + \frac{1}{\gamma}\frac{\tau_{\bar{\eta}}}{\bar{B}}\right)^2} \left(1 - \frac{\tau_{\bar{\eta}}}{T}\right)^2.$$

Using

$$T = \gamma \bar{B} + 2\tau_{\bar{\eta}},$$

the right-hand side simplifies to

$$\frac{\bar{B}}{2T(\tau_v + \tau_{\bar{\eta}})}.$$

But by Lemma B.4,

$$\frac{\partial}{\partial \tau_1} V_1(0, \widehat{\boldsymbol{\tau}}^n) \longrightarrow \frac{\bar{B}}{2T(\tau_v + \tau_{\bar{\eta}})}.$$

Therefore

$$\frac{V_1(a\tau^n, \widehat{\boldsymbol{\tau}}^n)}{a\tau^n} - \frac{\partial}{\partial \tau_1} V_1(0, \widehat{\boldsymbol{\tau}}^n) \longrightarrow 0.$$

■

B.1.3 Characterizing informativeness and liquidity

Finishing the proof of Theorem 2

We start by proving that the equilibrium must satisfy the first-order condition

$$\lim_{n\to\infty} \frac{\partial}{\partial \tau_i} V_i(0, \widehat{\boldsymbol{\tau}}^n) = \frac{\chi}{2\tau_v} \tag{B.28}$$

Toward that goal, let

$$M^n := \frac{\partial}{\partial \tau_1} V_1(0, \widehat{\boldsymbol{\tau}}^n), \qquad \tilde{c}(\tau) := c\left(\frac{\tau}{\tau + 2\tau_v}\right),$$

where the change of variables on c is to make χ compatible with the benchmark model, in which we wrote costs in terms of z.

By Lemma B.5,

$$V_1(a\tau^n, \widehat{\boldsymbol{\tau}}^n) = a\tau^n M^n + o(\tau^n), \qquad a = 1, 2.$$

Since $\tau^n \to 0$ and $c(0) = \tilde{c}(0) = 0$,

$$\tilde{c}(a\tau^n) = a\tau^n \tilde{c}'(0) + o(\tau^n), \qquad \tilde{c}'(0) = \frac{\chi}{2\tau_v}.$$

Optimality of τ^n against the deviation to zero then implies

$$\tau^n \left(M^n - \tilde{c}'(0)\right) + o(\tau^n) \geq 0,$$

and therefore

$$\liminf_{n\to\infty} M^n \geq \tilde{c}'(0).$$

Optimality against the deviation to $2\tau^n$ similarly implies

$$\tau^n \left(M^n - \tilde{c}'(0)\right) + o(\tau^n) \geq 2\tau^n \left(M^n - \tilde{c}'(0)\right) + o(\tau^n),$$

so

$$\limsup_{n\to\infty} M^n \leq \tilde{c}'(0).$$

Hence

$$\lim_{n\to\infty}\frac{\partial}{\partial\tau_1}V_1(0,\widehat{\boldsymbol{\tau}}^n)=\frac{\chi}{2\tau_v}.$$

We now use this first-order condition to calculate the aggregate informativeness in equilibrium. Define

$$Z:=\frac{\tau_{\bar{\eta}}}{\tau_v},\qquad \varphi:=\lim_{n\to\infty}\frac{\mathrm{Var}(v\mid p)^{-1}-\mathrm{Var}(v)^{-1}}{\mathrm{Var}(v\mid s_1,\ldots,s_n)^{-1}-\mathrm{Var}(v)^{-1}}=\frac{\tau_{\bar{\eta}}}{T}.$$

Price informativeness is

$$\mathcal{I}=1-\frac{\mathrm{Var}(v\mid p)}{\mathrm{Var}(v)}=\frac{\tau_{\bar{\eta}}}{\tau_v+\tau_{\bar{\eta}}}=\frac{Z}{1+Z}.$$

Moreover, $T=\tau_v\frac{Z}{\varphi}$. From the definition of $\tau_{\bar{\eta}}$,

$$\bar{B}^2=\frac{Z}{1-\varphi}\frac{\tau_v}{\tau_u}. \tag{B.29}$$

Dividing (B.25) by T gives

$$1-2\varphi=\gamma\frac{\bar{B}}{T}. \tag{B.30}$$

Using (B.29) and $T=\tau_v Z/\varphi$, this is equivalently

$$1-2\varphi=\frac{\gamma\varphi}{\sqrt{1-\varphi}}\frac{1}{\sqrt{Z\tau_v\tau_u}}. \tag{B.31}$$

Next, (B.16) and (B.30) imply

$$\lim_{n\to\infty}\frac{\partial}{\partial\tau_i}V_i(0,\widehat{\boldsymbol{\tau}}^n)=\frac{1}{2\gamma\tau_v}\frac{1-2\varphi}{1+Z}.$$

Hence, by (B.28)

$$\frac{1}{2\gamma\tau_v}\frac{1-2\varphi}{1+Z}=\frac{\chi}{2\tau_v}. \tag{B.32}$$

Equivalently,

$$1-2\varphi=\gamma\chi(1+Z). \tag{B.33}$$

Since $1+Z=(1-\mathcal{I})^{-1}$,

$$\varphi=\frac{1}{2}\left(1-\frac{\gamma\chi}{1-\mathcal{I}}\right). \tag{B.34}$$

Combining (B.31) with (B.32) gives

$$Z(1+Z)^2 = \Gamma \frac{2\varphi^2}{1-\varphi}, \qquad \Gamma := \frac{1}{2\chi^2 \tau_u \tau_v}. \tag{B.35}$$

Since $Z = \mathcal{I}/(1-\mathcal{I})$, this is

$$\frac{\mathcal{I}}{(1-\mathcal{I})^3} \frac{1-\varphi}{2\varphi^2} = \Gamma, \tag{B.36}$$

which is the desired characterization.

For uniqueness, substitute (B.33) into (B.35):

$$Z(1+Z)^2 = \frac{\Gamma}{2} \frac{[1-\gamma\chi(1+Z)]^2}{1+\gamma\chi(1+Z)}. \tag{B.37}$$

On

$$0 < Z < \frac{1}{\gamma\chi} - 1,$$

the left-hand side is strictly increasing in Z, while the right-hand side is strictly decreasing. At $Z = 0$ the left-hand side is zero and the right-hand side is positive when $\gamma\chi < 1$. At $Z = (\gamma\chi)^{-1} - 1$, the right-hand side is zero and the left-hand side is positive. Hence there is a unique positive solution whenever $\gamma\chi < 1$.

Equation (B.33) also shows that a positive-information solution cannot exist when $\gamma\chi \geq 1$: since $\varphi \geq 0$ and $Z > 0$,

$$1 - 2\varphi \leq 1 \quad \text{whereas} \quad \gamma\chi(1+Z) > 1.$$

Thus there is no positive-information competitive equilibrium in that case.

Finally, liquidity is the inverse price response to noise demand:

$$\mathcal{L} = \left(\frac{\partial p}{\partial u} \right)^{-1}.$$

From (B.8),

$$\frac{\partial p}{\partial u} = \frac{\lambda}{1-C} = \frac{\bar{B}}{\tau_v D + \bar{B}^2}$$

in the competitive limit. Therefore

$$\mathcal{L} = \frac{\tau_v D + \bar{B}^2}{\bar{B}} = \frac{D(\tau_v + \tau_{\bar{\eta}})}{\bar{B}}.$$

Using $\tau_{\bar{\eta}} = \bar{B}^2/D$ and (B.29),

$$\mathcal{L} = \sqrt{\frac{\tau_v}{\tau_u(1-\varphi)}}\frac{1+Z}{\sqrt{Z}} = \sqrt{\frac{\tau_v}{\tau_u(1-\varphi)}}\frac{1}{\sqrt{\mathcal{I}(1-\mathcal{I})}}.$$

This establishes the stated liquidity formula. ■

B.2 Aggregate profits and costs (Proof of Proposition 6)

In the free-entry limit of symmetric equilibria, we know that traders become informationally-small ($z^n \to 0$). Thus, using a Taylor approximation at $z^n = 0$, and noting that $V_i(0) = 0$, we have

$$\mathcal{R} = \lim_{n\to\infty} nV_i'(0)z^n + no(z^n), \tag{B.38}$$

where the notation $o(z)$ means that $\lim_{z\to 0}\frac{o(z)}{z} = 0$. Doing the same for aggregate costs, we have

$$\mathcal{C} = \lim_{n\to\infty} nc'(0)z^n + no(z^n), \tag{B.39}$$

Since the terms $no(z^n)$ vanish as $n \to \infty$, we may use the precision FOC $V_i'(0) = c'(0)$ to obtain that $\mathcal{R} = \mathcal{C}$. Thus, aggregate ex-ante profits are zero in the free-entry limit.

Next, we use $\chi = c'(0)$ and also use (2) to convert from scaled precision z^n to raw precision $\tau^{*,n}$:

$$\mathcal{R} = \mathcal{C} = \chi \lim_{n\to\infty} \frac{n\tau^{*,n}}{\tau^{*,n} + 2\tau_v} = \frac{\chi}{2\tau_v}\lim_{n\to\infty} n\tau^{*,n},$$

where we used that $\tau^{*,n} \to 0$. In the proof of Theorem 2, we showed that the aggregate precision $n\tau^{*,n}$ satisfies $\lim_{n\to\infty} n\tau^{*,n} = \lim_{n\to\infty} T^n = \frac{\tau_v}{\varphi}\frac{\mathcal{I}}{1-\mathcal{I}}$. Thus,

$$\mathcal{R} = \mathcal{C} = \frac{\chi}{2\varphi}\frac{\mathcal{I}}{1-\mathcal{I}}$$

which is the first result of the proposition.

Finally, we use the results of Theorem 2 to rewrite this expression. We replace $\chi^2 = \frac{1}{2}\Gamma^{-1}\tau_u^{-1}\tau_v^{-1}$, and then use the expressions for $\mathcal{I}$ and $\mathcal{L}$ to obtain

$$\mathcal{R} = \mathcal{C} = \frac{1}{2(1-\varphi)}\frac{\tau_u^{-1}}{\mathcal{L}}$$

as desired. ■

B.3 Comparative statics (including Proof of Proposition 7)

In addition to informativeness and liquidity, we also introduce price volatility, which is commonly studied in the literature because it is easier to observe in the data.

Definition 5. Price volatility $\mathcal{V}$ *is the standard deviation of the asymptotic price, i.e.,*

$$\mathcal{V} := Std\big[\lim_{n\to\infty} p^n\big]$$

The next proposition collects all the comparative statics for the generalized model with risk-averse traders and limit orders. The proof follows. Proposition 7 in the text is a direct corollary of this result.

Proposition B.1. *In the model with risk-averse traders, limit orders, and free entry of uninformed traders, the following comparative statics hold:*

- *Price informativeness increases in fundamental volatility, increases in noise volatility, and decreases in information costs—i.e.,* $\frac{d\mathcal{I}}{d\tau_v^{-1}}, \frac{d\mathcal{I}}{d\tau_u^{-1}} > 0$ *and* $\frac{d\mathcal{I}}{d\chi} < 0$*;*
- *Liquidity decreases in fundamental volatility, increases in noise volatility, and is non-monotonic in information costs—i.e.,* $\frac{d\mathcal{L}}{d\tau_v^{-1}} < 0 < \frac{d\mathcal{L}}{d\tau_u^{-1}}$ *and* $\frac{d\mathcal{L}}{d\chi} \gtrless 0$*;*
- *Price volatility increases in fundamental volatility, increases in noise volatility, and decreases in information costs—i.e.,* $\frac{d\mathcal{V}}{d\tau_v^{-1}}, \frac{d\mathcal{V}}{d\tau_u^{-1}} > 0$ *and* $\frac{d\mathcal{V}}{d\chi} < 0$*.*

Informativeness. Rewrite (**??**) as:

$$Z(1+Z)^2 \frac{1+\gamma\chi(1+Z)}{(1-\gamma\chi(1+Z))^2} = \frac{1}{2}\Gamma.$$

The right-hand-side is decreasing in τ_u, and the left-hand-side is increasing in Z. Thus, Z decreases in τ_u, hence so does $\mathcal{I} = \frac{Z}{1+Z}$. The same holds for τ_v, since it enters Γ symmetrically to τ_u. Finally, since the right-hand-side is decreasing in χ for fixed Z, we also have that Z is decreasing in χ.

Volatility. Because of uninformed trader free-entry, prices satisfy $p = \mathbb{E}[v|Y]$, and we have the following identity:

$$\begin{aligned}\text{Var}[p] = \text{Var}[\mathbb{E}[v|Y]] = \text{Var}[v] - \mathbb{E}[\text{Var}[v|Y]] = \text{Var}[v] - \mathbb{E}[\text{Var}[v|p]] \\ = \text{Var}[v] - \text{Var}[v|p] = \tau_v^{-1}\mathcal{I},\end{aligned}$$

where the second equality holds by the Law of total variance, the third equality follows from $p = \lambda Y$, the fourth equality from $\text{Var}[v|p]$ being independent of the realization of p by joint normality, and the last equality from the definition of $\mathcal{I}$. Therefore, $\mathcal{V} = \sqrt{\tau_v^{-1}\mathcal{I}}$.

Now, we have already shown informativeness is increasing with τ_u^{-1} and τ_v^{-1} and χ^{-1}. Given these comparative statics for $\mathcal{I}$, the comparative statics of $\mathcal{V}$ follow directly.

Liquidity. Rewriting equation (**??**) as a function of $\mathcal{I} = \frac{Z}{1+Z}$, we obtain the following:

$$\frac{\mathcal{I}}{(1-\mathcal{I})^3} = \frac{1}{2}\frac{1}{\chi^2\tau_u\tau_v}\frac{(1-\frac{\gamma\chi}{1-\mathcal{I}})^2}{1+\frac{\gamma\chi}{1-\mathcal{I}}}. \tag{B.40}$$

Note that the final ratio can be written:

$$\frac{(1-\frac{\gamma\chi}{1-\mathcal{I}})^2}{1+\frac{\gamma\chi}{1-\mathcal{I}}} = \frac{(1-\gamma\chi-\mathcal{I})^2}{(1-\mathcal{I})(1+\gamma\chi-\mathcal{I})}.$$

Plugging the above in equation (B.40) and solving for τ_v:

$$\tau_v = \frac{1}{2\chi^2\tau_u}\frac{(1-\mathcal{I})^2(1-\gamma\chi-\mathcal{I})^2}{\mathcal{I}(1+\gamma\chi-\mathcal{I})}. \tag{B.41}$$

Therefore, using the expressions for liquidity and φ in Theorem 2:

$$\mathcal{L} = \sqrt{\frac{2\tau_u^{-1}}{\tau_v^{-1}}}\frac{1}{\sqrt{\mathcal{I}(1-\mathcal{I})(1+\frac{\gamma\chi}{1-\mathcal{I}})}} = \frac{\tau_u^{-1}}{\chi}\frac{(1-\mathcal{I})(1-\gamma\chi-\mathcal{I})}{\mathcal{I}(1+\gamma\chi-\mathcal{I})}. \tag{B.42}$$

Finally, notice that $\mathcal{L}$ is decreasing as a function of $\mathcal{I}$, all else equal. Indeed, as $\mathcal{I}$ increases, $\frac{1-\mathcal{I}}{\mathcal{I}}$ decreases. Moreover, we see that

$$\frac{1-\gamma\chi-\mathcal{I}}{1+\gamma\chi-\mathcal{I}} = 1 - 2\gamma\chi\frac{1}{1+\gamma\chi-\mathcal{I}}$$

is also decreasing in $\mathcal{I}$.

We are now ready to prove the result for τ_v. First, notice that, in (B.42), $\mathcal{L}$ does

not depend on τ_v directly. If τ_v^{-1} increases, we have proved $\mathcal{I}$ increases, and thus $\mathcal{L}$ decreases, since we proved $\mathcal{L}$ is decreasing in $\mathcal{I}$ for fixed τ_u^{-1}.

For the τ_u^{-1} result, we start again from (B.40). By the expression for $\mathcal{L}$ above, equation (B.40) can be rewritten as, after taking square root:

$$2\chi\tau_v = \mathcal{L}(1-\mathcal{I})(1-\gamma\chi-\mathcal{I}).$$

Notice that the equality above does not depend on τ_u^{-1} directly. If τ_u^{-1} grows, $\mathcal{I}$ increases, so the terms depending on $\mathcal{I}$ decrease. So to maintain equality, $\mathcal{L}$ has to increase.

Finally, $\mathcal{L}$ is non-monotonic in χ, since $\mathcal{L} \to \infty$ as $\chi \to 0$ and as $\chi \to \infty$. ■

B.4 Volatility-informativeness relation

Using Proposition B.1, we now contrast the *volatility-informativeness relation* in our model versus Noisy Rational Expectations Equilibria (NREE). NREE serve as the dominant paradigm for competitive financial markets with endogenous information. The NREE framework we use is detailed in Appendix E—for suitable comparison to our own model, this NREE features endogenous information and smooth information costs, à la Verrecchia (1982).

As Proposition B.1 shows, volatility and informativeness always move in the same direction, no matter which parameter is moving. The cleanliness of this result stems from the expression for volatility $\mathcal{V} = \sqrt{\tau_v^{-1}\mathcal{I}}$.[13] By contrast, in an NREE model, volatility is "almost-always" decreasing in informativeness. Proposition E.1 provides a precise proof of this result and the sense in which the negative relationship holds "almost always." Figure B.1 illustrates an example.

In the main text, following Proposition 7, we have already discussed why τ_u^{-1} impacts $\mathcal{I}$ oppositely in our model versus the NREE. Because τ_u^{-1} naturally increases $\mathcal{V}$ in both models, this explains why $\mathcal{I}$ and $\mathcal{V}$ co-move differently across the two models when τ_u^{-1} is the parameter moving.

Next, we explain the impact of fundamental uncertainty τ_v^{-1}. As fundamental uncertainty rises, $\mathcal{V}$ always rises in both models. At the same time, higher uncertainty induces two opposing forces pertinent to information-acquisition. First, with more risk in the economy, risk-averse traders become more hesitant to trade and collect information. Second, with more uncertainty, there is more potential information to be collected. In NREE models, the first effect always dominates the second, and informativeness is

[13] This simple expression arises partly because prices are unbiased, $p = \mathbb{E}[v|Y]$, due to free entry of uninformed traders.

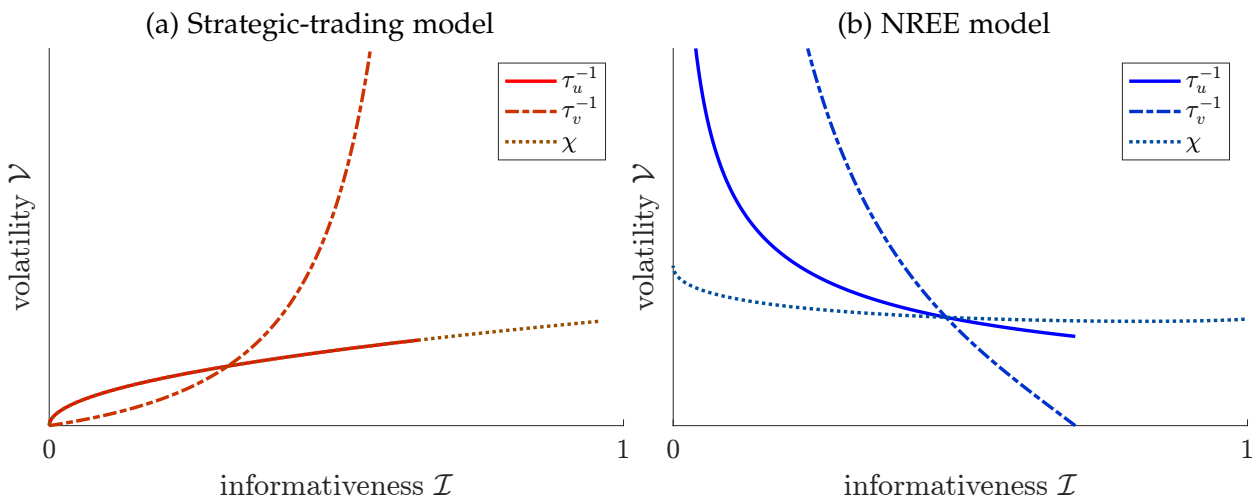


Figure B.1: Informativeness against Volatility, in our strategic-trading model and the NREE. Baseline parameters: $\tau_u^{-1} = 0.5$, $\tau_v^{-1} = 0.5$, and $\gamma = 3$. For the information cost functions, $\chi = 0.1$ in both models. We vary the parameters τ_u^{-1}, τ_v^{-1}, and χ one at a time. In the NREE, we use the cost function $c(z) = \chi(\frac{z}{1-z})$ which satisfies $c'(0) = \chi$ (note, however, that in this example χ also controls the entire level of the cost function in the NREE but only stands for the marginal cost at zero in our strategic-trading model). See Appendix E for more details on the NREE solution.

always falling in fundamental uncertainty (this is proved in Proposition E.1). In our strategic-trading model, the risk effect is small, tracing to a deep feature of our competitive equilibrium—traders are informationally-small and therefore trade small amounts, implying that risk effects are always second-order. Thus, the second effect—that more uncertainty implies more potential information—dominates, and informativeness is always rising in fundamental uncertainty.

C Correlated Information Choice

This entire section is devoted to proving Proposition 5. In this proof we will show that for any number of traders, n, and any block b_n^ξ with $b_n^\xi > 1$, the member of the block choosing the lowest precision wants to deviate to an independent source. Throughout, we fix a number of strategic traders n.

The structure of the proof goes as follows. First, Lemma C.1 characterizes, for an arbitrary assignment of traders to information sources, the unique linear trading equilibrium. We then proceed with three technical results. Lemma C.2 shows that all traders' demands load positively on their own signals, on top of establishing that an equilibrium matrix has non-negative entries. Lemma C.3 solves trader's own-signal loadings in closed form under the two different configurations—deviating versus not deviating. Finally, Lemma C.4 provides bounds for the ratio of loadings before and after the deviation. Using these bounds, the theorem shows that this deviation increases the gross payoff of the deviating trader. Because the deviation maintains the same precision (and, thus, information cost) for all traders, it is a profitable deviation in the information-acquisition game.

To characterize the equilibrium, a number of definitions will be needed. Let Ω be the covariance matrix of the signal errors:

$$\Omega_{ij} := \begin{cases} \min\{T_i, T_j\}, & \xi_i = \xi_j, \\ 0, & \xi_i \neq \xi_j. \end{cases}$$

The signal covariance matrix is

$$\Sigma := \tau_v^{-1}\mathbf{1}\mathbf{1}' + \Omega.$$

Let

$$D := \operatorname{diag}(\tau_v^{-1} + T_1, \ldots, \tau_v^{-1} + T_n), \qquad B := \Omega + D.$$

We will look for a linear equilibrium in which β_i are trading coefficients—β in vector form—and the real number λ is the equilibrium price impact. We define the vector of scaled trading coefficients

$$a := \lambda\beta.$$

Three variables will play a fundamental role going forward:

$$y := B^{-1}\mathbf{1}, \qquad G := \mathbf{1}'y, \qquad \text{and} \qquad H := y'Dy.$$

As we will see, y is the direction of the vector of scaled trading coefficients (representing trader's loadings on their signals), G is a constant that scales such coefficients, and H scales liquidity.

Lemma C.1. *Fix any information configuration $\mathcal{P}$ with associated Ω, and precision choices $\{1/T_i\}_{i=1,\dots,n}$. The unique linear equilibrium has scaled trading coefficients given by*

$$a = \frac{\tau_v^{-1}}{1+\tau_v^{-1}G}\, y, \tag{i}$$

liquidity

$$\lambda = \frac{\tau_v^{-1}}{1+\tau_v^{-1}G}\sqrt{\frac{H}{\tau_u^{-1}}} \tag{ii}$$

and trader i's gross payoff

$$V_i = \frac{\sqrt{\tau_u^{-1}}\,\tau_v^{-1}\,(\tau_v^{-1}+T_i)y_i^2}{(1+\tau_v^{-1}G)\sqrt{H}}. \tag{iii}$$

Proof of Lemma C.1

Fix an information configuration and suppose that the other traders use linear strategies

$$x_j = \beta_j s_j, \qquad j \neq i,$$

while the market-maker uses the linear pricing rule

$$p = \lambda Y, \qquad Y = x_i + \sum_{j\neq i}\beta_j s_j + u, \qquad \lambda > 0.$$

We first derive the traders' first-order conditions and then impose the market-maker's break-even condition.

Step 1: Traders' first-order conditions.

Conditional on observing s_i, trader i chooses an order x_i to maximize

$$\mathbb{E}\left[\left(v - \lambda\left(x_i + \sum_{j\neq i}\beta_j s_j + u\right)\right)x_i \,\middle|\, s_i\right].$$

Since u is independent of s_i and has mean zero, this objective can be written as

$$x_i\left(\mathbb{E}[v \mid s_i] - \lambda\sum_{j\neq i}\beta_j \mathbb{E}[s_j \mid s_i]\right) - \lambda x_i^2.$$

Because $\lambda > 0$, the objective is strictly concave in x_i. Its first-order condition is therefore necessary and sufficient:

$$2\lambda x_i = \mathbb{E}[v \mid s_i] - \lambda \sum_{j \neq i} \beta_j \mathbb{E}[s_j \mid s_i]. \tag{C.1}$$

The vector of signals is jointly Gaussian and centered. Consequently,

$$\mathbb{E}[v \mid s_i] = \frac{\mathrm{Cov}(v, s_i)}{\mathrm{Var}(s_i)} s_i = \frac{\tau_v^{-1}}{\tau_v^{-1} + T_i} s_i.$$

Similarly, for every $j \neq i$,

$$\mathbb{E}[s_j \mid s_i] = \frac{\mathrm{Cov}(s_j, s_i)}{\tau_v^{-1} + T_i} s_i.$$

Substituting these expressions and the conjectured equilibrium order $x_i = \beta_i s_i$ into (C.1), multiplying by $\tau_v^{-1} + T_i$, and recalling the definition of the scaled trading coefficients $a_i := \lambda \beta_i$ gives

$$2(\tau_v^{-1} + T_i) a_i + \sum_{j \neq i} \mathrm{Cov}(s_i, s_j) a_j = \tau_v^{-1}. \tag{C.2}$$

Recalling the definition of Σ and D, the set of first-order-conditions can be written compactly as:

$$(\Sigma + D) a = \tau_v^{-1} \mathbf{1}. \tag{C.3}$$

The matrix $\Sigma + D$ is positive definite. Indeed, Σ is a covariance matrix and is therefore positive semidefinite, while D is positive definite because $\tau_v^{-1} + T_i > 0$ for every i. It follows that $\Sigma + D$ is invertible, so equation (C.3) has a unique solution for a.

Step 2: Solving for the scaled trading coefficients.

Define, as in the statement of the result, $B := \Omega + D$. We may therefore write

$$\Sigma + D = B + \tau_v^{-1} \mathbf{1}\mathbf{1}'.$$

The matrix B is positive definite because Ω is positive semidefinite and D is positive definite. In particular, B is invertible.

Let

$$y := B^{-1}\mathbf{1}, \qquad G := \mathbf{1}'y = \mathbf{1}'B^{-1}\mathbf{1}.$$

Since B is positive definite, $G = \mathbf{1}'B^{-1}\mathbf{1} > 0$. Applying the Sherman–Morrison formula

gives

$$(\Sigma + D)^{-1} = (B + \tau_v^{-1}\mathbf{1}\mathbf{1}')^{-1} = B^{-1} - \frac{\tau_v^{-1}B^{-1}\mathbf{1}\mathbf{1}'B^{-1}}{1 + \tau_v^{-1}\mathbf{1}'B^{-1}\mathbf{1}}.$$

Multiplying this expression by $\mathbf{1}$ yields

$$\begin{aligned}(\Sigma + D)^{-1}\mathbf{1} = (B + \tau_v^{-1}\mathbf{1}\mathbf{1}')^{-1}\mathbf{1} &= B^{-1}\mathbf{1} - \frac{\tau_v^{-1}B^{-1}\mathbf{1}\mathbf{1}'B^{-1}\mathbf{1}}{1 + \tau_v^{-1}\mathbf{1}'B^{-1}\mathbf{1}} \\ &= y - \frac{\tau_v^{-1}G}{1 + \tau_v^{-1}G}y \\ &= \frac{1}{1 + \tau_v^{-1}G}y.\end{aligned}$$

Using equation (C.3), we conclude that

$$a = \tau_v^{-1}(B + \tau_v^{-1}\mathbf{1}\mathbf{1}')^{-1}\mathbf{1} = \frac{\tau_v^{-1}}{1 + \tau_v^{-1}G}y. \tag{i}$$

Step 3: Determining the price-impact coefficient.

The competitive market-maker sets

$$p = \mathbb{E}[v \mid Y].$$

Because (v, Y) is jointly Gaussian and centered, this conditional expectation is linear, with coefficient

$$\lambda = \frac{\mathrm{Cov}(v, Y)}{\mathrm{Var}(Y)} = \frac{\tau_v^{-1}\mathbf{1}'\beta}{\beta'\Sigma\beta + \tau_u^{-1}}.$$

Using $a = \lambda\beta$, the equation above can be rewritten as

$$a'\Sigma a + \tau_u^{-1}\lambda^2 = \tau_v^{-1}\mathbf{1}'a. \tag{C.4}$$

On the other hand, premultiplying the traders' equilibrium condition, (C.3), by a' gives

$$a'\Sigma a + a'Da = \tau_v^{-1}\mathbf{1}'a. \tag{C.5}$$

Comparing equations (C.4) and (C.5), we obtain

$$\tau_u^{-1}\lambda^2 = a'Da.$$

Substituting equation (i) into $a'Da$ gives

$$a'Da = \left(\frac{\tau_v^{-1}}{1+\tau_v^{-1}G}\right)^2 y'Dy = \left(\frac{\tau_v^{-1}}{1+\tau_v^{-1}G}\right)^2 H.$$

Taking the positive square root—the relevant root for the market maker's projection coefficient—yields

$$\lambda = \frac{\tau_v^{-1}}{1+\tau_v^{-1}G}\sqrt{\frac{H}{\tau_u^{-1}}}. \tag{ii}$$

In particular, $\lambda > 0$, which also verifies the strict concavity used in deriving the traders' first-order conditions.

Step 4: Traders' equilibrium payoffs.

Trader i's ex-ante gross payoff is

$$\begin{aligned} V_i &= \mathbb{E}[(v-p)x_i] \\ &= \beta_i \mathbb{E}[vs_i] - \lambda\beta_i \mathbb{E}\left[s_i\left(\sum_{j=1}^n \beta_j s_j + u\right)\right]. \end{aligned}$$

Since u is independent of s_i,

$$\mathbb{E}[s_i u] = 0,$$

and hence

$$V_i = \beta_i\tau_v^{-1} - \lambda\beta_i\left[\beta_i(\tau_v^{-1}+T_i) + \sum_{j\neq i}\beta_j \mathrm{Cov}(s_i,s_j)\right].$$

The individual first-order condition (C.2) can be rewritten to obtain:

$$\tau_v^{-1} - \lambda\left[\beta_i(\tau_v^{-1}+T_i) + \sum_{j\neq i}\beta_j \mathrm{Cov}(s_i,s_j)\right] = \lambda\beta_i(\tau_v^{-1}+T_i).$$

It follows that

$$V_i = \lambda\beta_i^2(\tau_v^{-1}+T_i) = \frac{(\tau_v^{-1}+T_i)a_i^2}{\lambda}.$$

Using equations (ii) and (iii),

$$V_i = (\tau_v^{-1} + T_i)\left(\frac{\tau_v^{-1}}{1+\tau_v^{-1}G}\right)^2 y_i^2 \frac{1+\tau_v^{-1}G}{\tau_v^{-1}}\sqrt{\frac{\tau_u^{-1}}{H}}$$
$$= \frac{\sqrt{\tau_u^{-1}}\,\tau_v^{-1}(\tau_v^{-1}+T_i)y_i^2}{(1+\tau_v^{-1}G)\sqrt{H}}, \tag{iii}$$

concluding the proof. ■

We now obtain a useful property of the loadings y for any block.

Lemma C.2. *Consider a block b with $|b| = m$, and re-enumerate its members such that:*

$$0 < T_1 \le T_2 \le \cdots \le T_m.$$

Let

$$\Omega_b = \left[\min\{T_i, T_j\}\right]_{i,j=1}^m, \qquad D_b = diag(\tau_v^{-1}+T_1, \ldots, \tau_v^{-1}+T_m), \qquad B_b := \Omega_b + D_b.$$

Then,

$$y_b := B_b^{-1}\mathbf{1} \ge 0$$

componentwise. Moreover,

$$B_b^{-1}\Omega_b \ge 0 \tag{C.6}$$

entrywise.

Proof of Lemma C.2

We treat separately the cases in which the precision choices are strictly ordered and in which some precisions coincide.

Case 1: strictly ordered precisions. Suppose first that

$$0 < T_1 < T_2 < \cdots < T_m.$$

Let

$$\Delta_1 := T_1, \qquad \Delta_i := T_i - T_{i-1} > 0, \quad i = 2, \ldots, m.$$

Then, Ω_b admits the factorization

$$\Omega_b = L\Delta L',$$

where

$$L_{ij} = \mathbf{1}\{j \le i\}, \qquad \Delta := \operatorname{diag}(\Delta_1, \ldots, \Delta_m).$$

Consequently,

$$\Omega_b^{-1} = L'^{-1}\Delta^{-1}L^{-1}.$$

The inverse of L is the lower-bidiagonal matrix

$$L^{-1} = \begin{pmatrix} 1 & 0 & \cdots & 0 \\ -1 & 1 & \ddots & \vdots \\ 0 & -1 & \ddots & 0 \\ \vdots & \ddots & \ddots & 1 \end{pmatrix}.$$

Indeed, L^{-1} is the first-difference operator. Direct multiplication gives

$$\Omega_b^{-1} = \begin{pmatrix} \frac{1}{\Delta_1} + \frac{1}{\Delta_2} & -\frac{1}{\Delta_2} & 0 & \cdots & 0 \\ -\frac{1}{\Delta_2} & \frac{1}{\Delta_2} + \frac{1}{\Delta_3} & -\frac{1}{\Delta_3} & \ddots & \vdots \\ 0 & -\frac{1}{\Delta_3} & \ddots & \ddots & 0 \\ \vdots & \ddots & \ddots & \frac{1}{\Delta_{m-1}} + \frac{1}{\Delta_m} & -\frac{1}{\Delta_m} \\ 0 & \cdots & 0 & -\frac{1}{\Delta_m} & \frac{1}{\Delta_m} \end{pmatrix}.$$

Thus Ω_b^{-1} is tridiagonal, with positive diagonal entries and nonpositive off-diagonal entries (it is, then, a Z-matrix).

By direct multiplication, we have

$$\Omega_b^{-1}\mathbf{1}_b = \frac{1}{T_1}e_1 \ge 0, \tag{C.7}$$

where $e_1 = (1, 0, \ldots, 0)'$ is the first elementary vector.

We may write

$$B_b = \Omega_b + D_b = \Omega_b\left(I + \Omega_b^{-1}D_b\right),$$

and thus

$$B_b^{-1}\mathbf{1}_b = \left(I + \Omega_b^{-1}D_b\right)^{-1}\Omega_b^{-1}\mathbf{1}_b. \tag{C.8}$$

Set

$$M := I + \Omega_b^{-1}D_b.$$

Because D_b is a positive diagonal and the off-diagonal entries of Ω_b^{-1} are nonpositive, M is a Z-matrix. By the semi-positivity characterization of nonsingular M-matrices (Berman and Plemmons, 1994, Chapter 6, Theorem 2.3, condition (I27)), a Z-matrix M is a nonsingular M-matrix if there exists $x > 0$ such that $Mx > 0$. We now show that M is an M-matrix by considering

$$x := D_b^{-1}\mathbf{1}_b > 0.$$

We have

$$Mx = D_b^{-1}\mathbf{1}_b + \Omega_b^{-1}\mathbf{1}_b.$$

By (C.7),

$$Mx = D_b^{-1}\mathbf{1}_b + \frac{1}{T_1}e_1 > 0.$$

Therefore, M is an M-matrix, implying

$$M^{-1} = B_b^{-1}\Omega_b \geq 0$$

componentwise, proving (C.6) . Combining this fact with (C.7) and (C.8) gives

$$y_b = B_b^{-1}\mathbf{1}_b = M^{-1}\frac{1}{T_1}e_1 \geq 0.$$

Case 2: weakly ordered precisions. Now suppose only that

$$0 < T_1 \leq T_2 \leq \cdots \leq T_m,$$

so that some precisions may coincide. In this case, Ω_b need not be invertible, and the preceding factorization through Ω_b^{-1} cannot be used directly.

For every $\varepsilon > 0$, define perturbed horizons

$$T_i^\varepsilon := T_i + i\varepsilon, \qquad i = 1, \ldots, m.$$

These horizons are strictly ordered:

$$0 < T_1^\varepsilon < T_2^\varepsilon < \cdots < T_m^\varepsilon.$$

Let

$$\Omega_{b,ij}^\varepsilon := \min\{T_i^\varepsilon, T_j^\varepsilon\}, \qquad D_b^\varepsilon := \operatorname{diag}(V + T_1^\varepsilon, \ldots, V + T_m^\varepsilon),$$

and

$$B_b^\varepsilon := \Omega_b^\varepsilon + D_b^\varepsilon.$$

By the result for strictly ordered horizons,

$$y_b^\varepsilon := (B_b^\varepsilon)^{-1}\mathbf{1}_b \geq 0, \qquad \text{and} \qquad (B_b^\varepsilon)^{-1}\Omega_b^\varepsilon \geq 0$$

entrywise for every $\varepsilon > 0$.

As $\varepsilon \downarrow 0$,

$$\Omega_b^\varepsilon \longrightarrow \Omega_b, \qquad D_b^\varepsilon \longrightarrow D_b, \qquad B_b^\varepsilon \longrightarrow B_b.$$

The matrix B_b is positive definite because Ω_b is positive semidefinite and D_b is positive definite. Hence B_b is invertible.

Matrix inversion is continuous on the set of invertible matrices. Therefore,

$$(B_b^\varepsilon)^{-1} \longrightarrow B_b^{-1},$$

and consequently

$$y_b^\varepsilon = (B_b^\varepsilon)^{-1}\mathbf{1}_b \longrightarrow B_b^{-1}\mathbf{1}_b = y_b.$$

Thus,

$$y_b = B_b^{-1}\mathbf{1}_b \geq 0.$$

Moreover,

$$0 \leq (B_b^\varepsilon)^{-1}\Omega_b^\varepsilon \to B_b^{-1}\Omega_b$$

This proves the result for weakly ordered precisions. ■

Our goal now is to compare two possible configurations. Fix an arbitrary information-source partition $\mathcal{P} = S$, and focus on a particular block b of size m. We will re-enumerate the block such that:

$$T_2 \leq ... \leq T_m \leq T_1.$$

Let $\tilde{b} = b \setminus \{1\}$. We then consider the deviation from S to the alternative partition $I = \{S \setminus b\} \cup \tilde{b} \cup \{1\}$. In words, partition I removes trader 1 from block b and assigns him to an independent, empty information source. We will assume that traders' precisions are kept constant across the configurations I and S. We use the partition name as a superscript to represent variables under that specific information configuration.

Lemma C.3. *Define $r := (T_2, ..., T_m)$. Then, trader 1's loadings are*

$$y_1^S = \frac{1 - \mathbf{1}' B_{\tilde{b}}^{-1} r}{\tau_v^{-1} + 2T_1 - r' B_{\tilde{b}}^{-1} r} \quad \textit{and} \quad y_1^I = \frac{1}{\tau_v^{-1} + 2T_1},$$

whereas other trader's loadings are:

$$y_{-1}^S = B_{\tilde{b}}^{-1} \left(\mathbf{1} - r y_1^S \right) \quad \textit{and} \quad y_{-1}^I = B_{\tilde{b}}^{-1} \mathbf{1}.$$

Proof of Lemma C.3

Under the shared-source configuration, the block loading vector solves

$$\begin{pmatrix} \tau_v^{-1} + 2T_1 & r' \\ r & B_{\tilde{b}} \end{pmatrix} \begin{pmatrix} y_1^S \\ y_{-1}^S \end{pmatrix} = \begin{pmatrix} 1 \\ \mathbf{1} \end{pmatrix}.$$

Thus,

$$(\tau_v^{-1} + 2T_1) y_1^S + r' y_{-1}^S = 1, \tag{C.9}$$

$$r y_1^S + B_{\tilde{b}} y_{-1}^S = \mathbf{1}. \tag{C.10}$$

Since $B_{\tilde{b}}$ is positive definite, it is invertible. Solving (C.10) for y_{-1}^S gives

$$y_{-1}^S = B_{\tilde{b}}^{-1} \left(\mathbf{1} - r y_1^S \right)$$

Substituting this expression into (C.9),

$$(\tau_v^{-1} + 2T_1) y_1^S + r' B_{\tilde{b}}^{-1} \left(\mathbf{1} - r y_1^S \right) = 1.$$

By solving this equation for y_1^S we obtain the expression in the statement.

Now suppose trader 1 moves to an independent source. His covariance with each of the other traders becomes zero, while the submatrix governing those traders is unchanged. Hence the corresponding block matrix is

$$B_b^I = \begin{pmatrix} \tau_v^{-1} + 2T_1 & 0 \\ 0 & B_{\tilde{b}} \end{pmatrix}.$$

The loading vector therefore solves

$$\begin{pmatrix} \tau_v^{-1} + 2T_1 & 0 \\ 0 & B_{\tilde{b}} \end{pmatrix} \begin{pmatrix} y_1^I \\ y_{-1}^I \end{pmatrix} = \begin{pmatrix} 1 \\ \mathbf{1} \end{pmatrix}.$$

The two blocks decouple, so

$$(\tau_v^{-1} + 2T_1) y_1^I = 1, \qquad B_{\tilde{b}} y_{-1}^I = \mathbf{1},$$

concluding the proof. ■

Lemma C.4. *Fix a number of traders n. Consider partitions I and S as previously described. Then, the following bounds hold*

$$1 < \frac{y_1^I}{y_1^S} = \max_{i \in \{1,\dots,n\}} \frac{y_i^I}{y_i^S}.$$

Proof of Lemma C.4

For the proof, define

$$r := (T_2, \dots, T_m)', \qquad w := y_{-1}^I = B_{\tilde{b}}^{-1}\mathbf{1}, \qquad x := B_{\tilde{b}}^{-1} r.$$

We establish the result in four steps. First, we prove the auxiliary result that $w > 0$ componentwise, and $x \geq 0$. Second, we show that $y_1^I / y_1^S > 1$. In the third step, we prove that the equality in the statement holds when the max is taken over $i \in b$—that is, $i = 1, ..., m$. Finally, the fourth step concludes that the equality continues holding once we expand the constraint set to $i \notin b$.

Step 1. $w > 0$ and $x \geq 0$. We start proving $w > 0$ componentwise. To see this, the adjacent-row equations for $B_{\tilde{b}} w = \mathbf{1}$ imply, for $i = 2, \dots, m-1$,

$$(\tau_v^{-1} + T_i) w_i = (\tau_v^{-1} + 2T_{i+1} - T_i) w_{i+1} + (T_{i+1} - T_i) \sum_{j=i+2}^{m} w_j. \tag{C.11}$$

If $w_m = 0$, equation (C.11) applied successively at $i = m-1, m-2, \dots, 2$ implies

$$w_2 = \cdots = w_m = 0,$$

contradicting $B_{\tilde{b}} w = \mathbf{1}$. Hence $w_m > 0$. Since (C.11) has a nonnegative right-hand side,

this then implies $w_i > 0$ for every i.

Next, because T_m is the largest horizon among traders $2,\dots,m$, we have that r is the last column of the Brownian covariance matrix $\Omega_{\tilde{b}}$: i.e., $r = \Omega_{\tilde{b}} e_{m-1}$, where e_{m-1} is the coordinate vector with a one in position $m-1$. By equation (C.6) in Lemma C.2, we conclude that

$$x = B_{\tilde{b}}^{-1} \Omega_{\tilde{b}} e_{m-1} \geq 0.$$

Step 2. $y_1^I / y_1^S > 1$. From Lemma C.3,

$$y_1^I > y_1^S \iff (\tau_v^{-1} + 2T_1)\, r'w > r'x.$$

By symmetry of $B_{\tilde{b}}^{-1}$ (as the inverse of a symmetric matrix),

$$r'w = r' B_{\tilde{b}}^{-1} \mathbf{1} = \mathbf{1}' B_{\tilde{b}}^{-1} r = \mathbf{1}' x.$$

Because $x \geq 0$ and $r_j = T_j \leq T_1$,

$$r'x \leq T_1 \mathbf{1}' x = T_1 r'w.$$

Finally, because $\tau_v^{-1} + 2T_1 > T_1$, and $r'w > 0$, we conclude $y_1^I > y_1^S$.

Step 3. $\max_{i \in b} \frac{y_i^I}{y_i^S} = \frac{y_1^I}{y_1^S}$. We first establish that

$$\frac{x_2}{w_2} \leq \frac{x_3}{w_3} \leq \cdots \leq \frac{x_m}{w_m}. \tag{C.12}$$

For that, note that the last row of $B_{\tilde{b}} w = \mathbf{1}$ gives

$$1 - r'w = (\tau_v^{-1} + T_m) w_m > 0. \tag{C.13}$$

Recalling $\mathbf{1}'x = r'w$, we obtain that $\mathbf{1}'x < 1$. Fix $i \in \{2,\dots,m-1\}$ and let

$$\Delta_i := T_{i+1} - T_i \geq 0.$$

Subtracting adjacent rows of $B_{\tilde{b}} w = \mathbf{1}$ gives

$$(\tau_v^{-1} + T_i) w_i = (\tau_v^{-1} + 2T_{i+1} - T_i) w_{i+1} + \Delta_i \sum_{j=i+2}^{m} w_j.$$

Similarly, subtracting adjacent rows of $B_{\tilde{b}}x = r$ gives

$$(\tau_v^{-1} + T_i)x_i = (\tau_v^{-1} + 2T_{i+1} - T_i)x_{i+1} + \Delta_i \sum_{j=i+2}^{m} x_j - \Delta_i.$$

Multiply the second equation by w_{i+1}, multiply the first by x_{i+1}, and subtract. The common term cancels, yielding

$$(\tau_v^{-1} + T_i)(x_i w_{i+1} - w_i x_{i+1}) = \Delta_i \left[w_{i+1} \sum_{j=i+2}^{m} x_j - w_{i+1} - x_{i+1} \sum_{j=i+2}^{m} w_j \right].$$

Since $x_{i+1} \geq 0$, $w_j > 0$, and

$$\sum_{j=i+2}^{m} x_j < \mathbf{1}'x < 1,$$

the expression in square brackets is strictly negative whenever $\Delta_i > 0$, and is zero when $\Delta_i = 0$. Hence

$$x_i w_{i+1} \leq w_i x_{i+1}.$$

Because $w_i, w_{i+1} > 0$,

$$\frac{x_i}{w_i} \leq \frac{x_{i+1}}{w_{i+1}},$$

which proves (C.12).

We now compare x_m/w_m to trader 1's proportional loading gain, y_1^I/y_1^S. The last row of $B_{\tilde{b}}x = r$ gives

$$T_m - r'x = (\tau_v^{-1} + T_m)x_m. \tag{C.14}$$

Combining (C.14) with (C.13),

$$\frac{x_m}{w_m} = \frac{T_m - r'x}{1 - r'w}. \tag{C.15}$$

On the other hand, by Lemma C.3

$$\frac{1}{y_1^S} - \frac{1}{y_1^I} = \frac{(\tau_v^{-1} + 2T_1)\, r'w - r'x}{1 - r'w}.$$

Therefore

$$\frac{x_m}{w_m} \leq \frac{1}{y_1^S} - \frac{1}{y_1^I} \iff T_m \leq (\tau_v^{-1} + 2T_1)\, r'w. \tag{C.16}$$

To verify the second inequality in (C.16), (C.13) implies $(\tau_v^{-1} + 2T_m)w_m \leq 1$, and hence,

using (C.13) again,

$$\begin{aligned} r'w &= 1 - (\tau_v^{-1} + T_m)w_m \\ &\geq 1 - \frac{\tau_v^{-1} + T_m}{\tau_v^{-1} + 2T_m} \\ &= \frac{T_m}{\tau_v^{-1} + 2T_m}. \end{aligned}$$

It follows that

$$(\tau_v^{-1} + 2T_1)\, r'w \geq (\tau_v^{-1} + 2T_1)\, \frac{T_m}{\tau_v^{-1} + 2T_m} \geq T_m,$$

since $T_1 \geq T_m$, which proves the required right-inequality (C.16).

Combining the left-inequality in (C.16) with (C.12), we obtain, for $j = 2, ..., m$

$$y_j^S = w_j \leq \frac{y_1^I}{y_1^S}(w_j - x_j y_1^S) = \frac{y_1^I}{y_1^S} y_j^S,$$

where the first and last equalities follows from Lemma C.3. We have then established $\max_{i \in b} \frac{y_i^I}{y_i^S} = \frac{y_1^I}{y_1^S}$.

Step 4: extend to $i \notin b$. For every trader j outside the affected block, the corresponding matrix block of B is unchanged, and therefore

$$\frac{y_j^I}{y_j^S} = 1 < \frac{y_1^I}{y_1^S},$$

which concludes the proof. ■

Proof of Proposition 5

Suppose, toward a contradiction, that an equilibrium has source partition $\mathcal{P} := S$, containing a block $b \in S$ with $|b| = m \geq 2$. Re-enumerate traders in b such that $b = \{1, 2, ..., m\}$, with $T_2 \leq ... \leq T_m \leq T_1$. Consider a deviation in which trader 1 keeps the same precision $1/T_1$ but moves to an independent information source, creating a singleton block. Let the resulting partition be I—this is the definition of I and S we have provided previously.

We start by comparing the equilibrium variable G under the different configurations:

$$G^I = \sum_j y_j^I \leq \frac{y_1^I}{y_1^S} \sum_j y_j^S = \frac{y_1^I}{y_1^S} G^S, \tag{C.17}$$

where the inequality follows from Lemma C.4, and the non-negativity of y, which is a consequence of Lemma C.2.

We now do the same for equilibrium variable H. Because precisions themselves are unchanged by the deviation, the diagonal matrix D is the same in the two configurations. Therefore,

$$\begin{aligned} H^I = y^{I'} D y^I &= \sum_j (\tau_v^{-1} + T_j)(y_j^I)^2 \\ &\leq \left(\frac{y_1^I}{y_1^S}\right)^2 \sum_j (\tau_v^{-1} + T_j)(y_j^S)^2 = \left(\frac{y_1^I}{y_1^S}\right)^2 H^S, \end{aligned} \tag{C.18}$$

where the inequality follows, once more, from Lemma C.4.

Using the expression for profits in Lemma C.1:

$$\frac{V_1^I}{V_1^S} = \left(\frac{y_1^I}{y_1^S}\right)^2 \frac{1 + \tau_v^{-1} G^S}{1 + \tau_v^{-1} G^I} \sqrt{\frac{H^S}{H^I}}.$$

From (C.17), and the fact that $y_1^I / y_1^S > 1$, which follows from Lemma C.4, $1 + \tau_v^{-1} G^I < \frac{y_1^I}{y_1^S}\left(1 + \tau_v^{-1} G^S\right)$. Consequently,

$$\frac{1 + \tau_v^{-1} G^S}{1 + \tau_v^{-1} G^I} > \frac{y_1^S}{y_1^I}. \tag{C.19}$$

Combining (C.18) and (C.19) into the ratio of profits:

$$\frac{V_1^I}{V_1^S} > \left(\frac{y_1^I}{y_1^S}\right)^2 \frac{y_1^S}{y_1^I} \frac{y_1^S}{y_1^I}. = 1.$$

Because the cost function depends only on precisions, we have established $V_1^I - c(1/T_1) > V_1^S - c(1/T_1)$, and therefore the partition S cannot be an equilibrium configuration. Because S was arbitrary except for $|b| > 1$, it follows that there is no equilibrium with non-singleton (non-empty) blocks. ■

D A Model with Fixed Costs of Information

Now, let's assume the information cost satisfies $\bar{\chi} := c(z) > 0$ for all z. There is a fixed cost of acquiring any information. The fact that there is no additional variable cost simplifies the analysis so that either $z_i = 0$ or $z_i = 1$ for each trader i (none or perfect information).

One concern with fixed costs, suggested by the analysis of Grossman and Stiglitz (1980), is that an equilibrium may fail to exist. We will show that this is not the case here. In fact, there is a well-defined equilibrium for any $\bar{\chi}$. First, we will consider symmetric equilibria, followed by asymmetric equilibria. The key statistic, which is exactly analogous to (9), is

$$\Gamma := \frac{1}{2\bar{\chi}^2\tau_u\tau_v}. \tag{D.1}$$

Whether Γ is greater or less than 2 determines the type of equilibrium. And the level of Γ determines the level of aggregate information in a large market.

In a symmetric equilibrium, the pre-information-cost utility value of each trader, assuming all other traders collect information z^n, is the same as before:

$$V_n(z;z^n) = \sqrt{\frac{\tau_v^{-1}\tau_u^{-1}}{2}}\frac{z(1+z)}{(1+(n-1)z^n+z)\sqrt{(n-1)z^n(1+z^n)+z(1+z)}}.$$

Notice that $V_n(0;0) = V_n(0;1) = 0$. The key criterion for equilibrium is optimality: in an equilibrium with information collection, trader utility must satisfy

$$V_n(1;1) - V_n(0;1) \geq \bar{\chi}, \quad \text{if} \quad z^n = 1. \tag{D.2}$$

On the other hand, for an equilibrium with ignorance, trader utility must satisfy

$$V_n(1;0) - V_n(0;0) \leq \bar{\chi}, \quad \text{if} \quad z^n = 0. \tag{D.3}$$

An asymmetric equilibrium is more complicated. To set it up, let π_n^* denote the equilibrium probability a trader acquires any information. Because each trader only acquires either $z_i = 0$ or $z_i = 1$, aggregate information in the market will be equal to the total number of informed traders, which is the binomial random variable $\text{Binom}(n, \pi_n^*)$. The pre-cost value of a single trader, assuming the fraction of the other $n-1$ traders

acquiring information is π_n, is (abusing notation, we also refer to this function as V_n)

$$V_n(z;\pi_n) = \sqrt{\frac{\tau_v^{-1}\tau_u^{-1}}{2}}\mathbb{E}\Big[\frac{z(1+z)}{(1+(n-1)\pi_n+z)\sqrt{2(n-1)\pi_n+z(1+z)}}\Big],$$

where the expectation is over the possible realizations of π_n. By rational expectations, this random variable has probability distribution $(n-1)\pi_n \sim \text{Binom}(n-1,\pi_n^*)$. In equilibria of the large-n economy, the amount of randomness in π_n vanishes, i.e., $\pi_n - \pi_n^* \to 0$ almost-surely, by the law of large numbers. In a mixed-strategy equilibrium with $\pi_n^* > 0$ being the information probability, it must be the case that

$$V_n(1;\pi_n) - \bar{\chi} = V_n(0;\pi_n) = 0, \quad \text{if} \quad \pi_n \in (0,1),$$

i.e., the trader is indifferent between acquiring information or not.

D.1 Equilibrium information

Based on this characterization, we can prove the following theorem, which is roughly speaking the fixed cost version of Theorem 1.

Theorem D.1. *An equilibrium with fixed information costs always exists in the large-n limit and satisfies the following:*

1. *If $\Gamma \leq 2$, then the equilibrium is symmetric and features $nz^n = 0$ for all n large enough.*
2. *If $\Gamma > 2$, then the equilibrium is asymmetric and features $n\pi_n^* \to \Pi^*$, where Π^* is the unique positive solution to $(\Pi+2)^2(\Pi+1) = 2\Gamma$.*

Proof. 1. Suppose $\Gamma \leq 2$. First, we prove that, assuming a symmetric equilibrium exists, $nz^n = 0$ for all n large enough. This is a direct consequence of $\lim_{n\to\infty} V_n(1;1) = 0$, implying $V_n(1;1) < \bar{\chi}$ for all n large enough, so (D.2) cannot hold. Second, we prove existence of a symmetric equilibrium with $z^n = 0$ for large n. The utility from deviating from $z^n = 0$ is $V_n(1;0) = \frac{1}{2}\sqrt{\tau_v^{-1}\tau_u^{-1}}$. Thus, $V_n(1;0) - V_n(0;0) - \bar{\chi} = \frac{1}{2}\sqrt{\tau_v^{-1}\tau_u^{-1}} - \bar{\chi} \leq 0$, so (D.3) holds, and the equilibrium is confirmed.

2. Suppose $\Gamma > 2$. As a consequence of the argument for claim 1, a symmetric equilibrium cannot occur if $\Gamma > 2$, because in such case neither (D.2) nor (D.3) can hold. Moving to asymmetric equilibria, it is easy to see that $n\pi_n^* \not\to \infty$ as $n \to \infty$. Indeed, if $n\pi_n^* \to \infty$ occurred, then $V_n(1;\pi_n^*) \to 0$, and so $V_n(1;\pi_n) \to 0$ a.s. (recall $\pi_n - \pi_n^* \to 0$

by the law of large numbers). In fact, writing $\Pi^* := \lim_{n\to\infty} n\pi_n^*$ and then evaluating $V_n(1;\pi_n) = \bar{\chi}$ in the large-n limit, we find that Π^* must solve the cubic equation

$$(\Pi+2)^2(\Pi+1) = \frac{\tau_v^{-1}\tau_u^{-1}}{\bar{\chi}^2}.$$

This has a unique positive solution $\Pi^* > 0$ if and only if $\Gamma > 2$. Thus, an asymmetric equilibrium exists if $\Gamma > 2$, but not if $\Gamma \leq 2$, and the aggregate information in this equilibrium is Π^* as $n \to \infty$. □

Remark D.1. *Theorem D.1 proves that equilibrium exists generically. In competitive models like Grossman and Stiglitz (1980), equilibrium non-existence is a problem: neither a fully-informed nor a fully-uninformed equilibrium can exist. But here, we can obtain either of these extreme cases depending on the size of the statistic* Γ*. If* $\Gamma \leq 2$*, we have a fully-uninformed equilibrium. But if* $\Gamma \to \infty$*, we have a fully-informed equilibrium, in the sense that aggregate information becomes maximal. The characterization of the result in terms of* $\Pi^* = \lim_{n\to\infty} n\pi_n^*$ *is intuitive:* Π^* *represents the* number of informed traders *in the large-n limit, which necessarily remains finite for any fixed cost* $\bar{\chi}$*.*

D.2 Asymmetric equilibrium characterization

In this section, we characterize equilibrium for fixed information costs, with $\Gamma > 2$ in the notation of Theorem D.1. In such case, an "asymmetric equilibrium" arises where each trader chooses to be fully-informed with probability π_n^*.

Lemma D.1. *Let* π_n^* *denote the probability of information acquisition in an asymmetric equilibrium with n strategic traders, and let* $\Pi_n := Binom(n, \pi_n^*)$ *be the aggregate number of informed traders. Then,*

$$Y^n = \sqrt{\frac{\tau_u^{-1}}{\tau_v^{-1}}\Pi_n} v + \tau_u^{-1/2} e_u$$
$$p^n = \tau_v^{-1/2}\left(\frac{\Pi_n}{1+\Pi_n}\frac{v}{\tau_v^{-1/2}} + \frac{\sqrt{\Pi_n}}{1+\Pi_n} e_u\right)$$

where $e_u = u/\tau_u^{-1/2} \sim Normal(0,1)$*. Let* $\Pi^* := \lim_{n\to\infty} n\pi_n^* \in [0,\infty]$ *be the large-n limit of*

aggregate information. Then, we have the following limiting equilibrium objects, almost-surely,

$$\lim_{n\to\infty} Y^n = \tau_u^{-1/2}\left[\sqrt{\Pi^*}\frac{v}{\tau_v^{-1/2}} + e_u\right]$$
$$\lim_{n\to\infty} p^n = \tau_v^{-1/2}\frac{\sqrt{\Pi^*}}{1+\Pi^*}\left[\sqrt{\Pi^*}\frac{v}{\tau_v^{-1/2}} + e_u\right]$$

Proposition D.1. *In the large-n limit of asymmetric equilibria, the measures of informativeness and liquidity are given by*

$$\textit{(liquidity)} \quad \mathcal{L} = \sqrt{\frac{\tau_v}{\tau_u}}\frac{1+\Pi^*}{\sqrt{\Pi^*}}$$
$$\textit{(informativeness)} \quad \mathcal{I} = \frac{\Pi^*}{1+\Pi^*}$$

Proof. Suppose in the size-n economy, each trader obtains a perfect signal with probability π_n^*, and otherwise remains uninformed. In this case, $\mathbf{z}$ is a vector of zeros and ones, with the probability of each non-zero entry being an independent Bernoulli draw. Then, $\mathbf{z}\cdot\mathbf{1} = \|\mathbf{z}\|^2 := \Pi_n \sim \text{Binom}(n, \pi_n^*)$. Substituting this into formulas in the proof of Lemma 1, we have

$$\lambda^n = \sqrt{\frac{\tau_v^{-1}}{\tau_u^{-1}}\frac{\Pi_n}{(1+\Pi_n)^2}} \quad \text{and} \quad \sum_{i=1}^n \beta_i s_i = \sqrt{\frac{\tau_u^{-1}}{\tau_v^{-1}}\Pi_n}v,$$

and so

$$Y^n = \sqrt{\frac{\tau_u^{-1}}{\tau_v^{-1}}\Pi_n}v + \tau_u^{-1/2}e_u$$
$$p^n = \tau_v^{-1/2}\left(\frac{\Pi_n}{1+\Pi_n}\frac{v}{\tau_v^{-1/2}} + \frac{\sqrt{\Pi_n}}{1+\Pi_n}e_u\right)$$

To take the limit $n\to\infty$, note that $\Pi_n/(n\pi_n^*)$ converges to 1 almost-surely, by the Strong Law of Large Numbers. Hence, $\Pi_n \to \Pi^*$ almost-surely. Using this fact, we have

$$Y^n \to \tau_u^{-1/2}\left(\sqrt{\Pi^*}\frac{v}{\tau_v^{-1/2}} + e_u\right)$$
$$p^n \to \tau_v^{-1/2}\frac{\sqrt{\Pi^*}}{1+\Pi^*}\left(\sqrt{\Pi^*}\frac{v}{\tau_v^{-1/2}} + e_u\right)$$

This proves Lemma D.1.

Proposition D.1 is proved by combining these results with the definitions of $\mathcal{I}, \mathcal{L}$. □

Remark D.2. *Comparing the result above with Lemma A.3, we see that the limiting objects are very similar across the symmetric and asymmetric equilibria (which recall arise with variable and fixed costs, respectively). In particular, modulo some $\sqrt{2}$ constants, the formulas are identical in distribution. For instance, compare the limiting price in the two equilibrium types:*

$$\text{(symmetric equilibrium)} \quad \lim_{n\to\infty} p^n = \tau_v^{-1/2} \frac{\sqrt{Z^*}}{1+Z^*} \left[\sqrt{Z^*} \frac{v}{\tau_v^{-1/2}} + \frac{e_s + e_u}{\sqrt{2}} \right]$$

$$\text{(asymmetric equilibrium)} \quad \lim_{n\to\infty} p^n = \tau_v^{-1/2} \frac{\sqrt{\Pi^*}}{1+\Pi^*} \left[\sqrt{\Pi^*} \frac{v}{\tau_v^{-1/2}} + e_u \right]$$

Since Z^* and Π^* have similar interpretations, as aggregate information (in units of precision), the formulas are almost identical. The only difference is that the noise $\frac{e_s+e_u}{\sqrt{2}}$ in the symmetric equilibrium price is replaced by e_u in the asymmetric equilibrium price. But since both are standard normal random variables, the prices are identical in distribution.

E Comparison to Competitive NREE

We compare our results to a competitive Noisy Rational Expectations Equilibrium (NREE) setting along the lines of Grossman (1976), Grossman and Stiglitz (1980), Hellwig (1980), and Verrecchia (1982). Because we will ignore strategic considerations, we now need traders to be risk averse. We assume trader i's utility function is of the CARA type:

$$\tilde{V}_i := -\exp\big(-\gamma[d_i(v-p) - c(z_i)]\big),$$

where z_i is the signal precision acquired, and $c(\cdot)$ its cost, analogous to the baseline model. In the trading stage, traders receive their signal $s_i = v + \varepsilon_i$ and solve

$$\max_{d_i} \mathbb{E}\big[\tilde{V}_i \mid s_i, p\big]$$

In the information stage, traders choose their precision z_i by solving

$$\max_{z_i} \mathbb{E}\Big[\max_{d_i} \mathbb{E}\big[\tilde{V}_i \mid s_i, p\big]\Big]$$

As in the baseline model, we assume signal errors ε_i are independent across traders and independent of both the fundamental v and the noise u. Furthermore, recall that the precision z_i is defined to be related to the overall signal variance $\sigma_i := \mathrm{Var}[s_i]$ in that

$$\sigma_i = \sigma(z_i) := \frac{1}{2}\tau_v^{-1}(z_i^{-1} + 1)$$

To keep the proper comparison to the baseline model, traders pay a cost to increase z_i, but the results would be very similar if the cost was defined over the precision of the signal error $(\sigma_i - \tau_v^{-1})^{-1} = \frac{2}{\tau_v^{-1}}\frac{z_i}{1-z_i}$.[14] The equilibrium asset price is determined via the market clearing condition

$$\lim_{n\to\infty} \frac{1}{n}\sum_{i=1}^{n} d_i + u = 0$$

[14] Note that Verrecchia (1982) defines a cost function $\tilde{c}(s)$ over the error precision $s(z) := \frac{2}{\tau_v^{-1}}\frac{z}{1-z}$. We assume that this cost of precision is convex, i.e., $\tilde{c}''(s) \geq 0$. Given our representation of the cost function as $c(z)$, this is implemented simply by putting $\tilde{c}(s) := c(z^{-1}(s)) = c(\frac{\tau_v^{-1}s}{2+\tau_v^{-1}s})$. Note that for $\tilde{c}''(s)$ to be convex, we require that $(1-z)^2c'(z)$ be increasing in z, or equivalently that $(1-z)c''(z)/c'(z) \geq 2$. Finally, notice that $\tilde{c}'(0) = \frac{1}{2}\tau_v^{-1}c'(0)$, so the critical object—the marginal cost at zero precision—is invariant to this transformation.

It is standard in the NREE literature to clear markets based on the *average* demand, by analogy to the continuum limit, rather than the total demand; equivalently, some versions explicitly model the noise u grow with the number of traders (i.e., u is the noise per capita).[15] Finally, conjecture (and later verify) that the equilibrium pricing function takes the form

$$p = \beta_v v + \beta_u u,$$

for some β_v and β_u.

In this setup, joint normality of (v, u, s_i) implies the distribution of (s_i, p) is

$$(s_i, p) \sim \text{Normal}\left(0, \tilde{\Sigma}_i\right),$$
$$\text{where} \quad \tilde{\Sigma}_i := \begin{pmatrix} \sigma_i & \beta_v \tau_v^{-1} \\ \beta_v \tau_v^{-1} & \beta_v^2 \tau_v^{-1} + \beta_u^2 \tau_u^{-1} \end{pmatrix}$$

Furthermore, the conditional distribution of v given (s_i, p) is

$$(v \mid s_i, p) \sim \text{Normal}(\tilde{\mu}_{vi}, \tilde{\sigma}_{vi}),$$

$$\begin{aligned} \text{where} \quad \tilde{\mu}_{vi} &:= \frac{\tau_v^{-1}}{(\sigma_i - \tau_v^{-1})\beta_v^2 \tau_v^{-1} + \sigma_i \beta_u^2 \tau_u^{-1}} \left(\beta_u^2 \tau_u^{-1} s_i + (\sigma_i - \tau_v^{-1}) \beta_v p \right) \\ \tilde{\sigma}_{vi} &:= \tau_v^{-1} - \frac{\tau_v^{-2}}{(\sigma_i - \tau_v^{-1})\beta_v^2 \tau_v^{-1} + \sigma_i \beta_u^2 \tau_u^{-1}} \begin{pmatrix} 1 \\ \beta_v \end{pmatrix}^\top \begin{pmatrix} \beta_v^2 \tau_v^{-1} + \beta_u^2 \tau_u^{-1} & -\beta_v \tau_v^{-1} \\ -\beta_v \tau_v^{-1} & \sigma_i \end{pmatrix} \begin{pmatrix} 1 \\ \beta_v \end{pmatrix} \\ &= \tau_v^{-1} \frac{(\sigma_i - \tau_v^{-1}) \beta_u^2 \tau_u^{-1}}{(\sigma_i - \tau_v^{-1})\beta_v^2 \tau_v^{-1} + \sigma_i \beta_u^2 \tau_u^{-1}} \end{aligned}$$

[15] One could consider an alternative economy in which the noise is given by $u_n \sim \text{Normal}(0, \tau_u^{-1}/n^2)$. In this specification, analogously to our baseline model, write the market clearing condition in levels:

$$\sum_{i=1}^{n} d_i + n u_n = 0$$

The aggregate noise nu_n has constant size τ_u^{-1} by construction. Dividing the market clearing condition by n and taking the limit, we have

$$\lim_{n \to \infty} \left(\frac{1}{n} \sum_{i=1}^{n} d_i + u_n \right) = 0$$

By the law of large numbers, the result is $\lim_{n\to\infty} \frac{1}{n} \sum_{i=1}^{n} d_i = 0$ almost-surely, i.e., the asymptotic economy is noise-less. Therefore, a way to analyze a competitive model analogous to our baseline model is to study the small-noise limit *after* having derived the equilibrium for each τ_u^{-1}.

Then, the standard CARA solution for asset demand is

$$d_i = \frac{\tilde{\mu}_{vi} - p}{\gamma \tilde{\sigma}_{vi}}$$

Before continuing, note that if agents collect a symmetric amount of information, $z_i = z^*$ for all i so that $\sigma_i = \sigma^*$, then market clearing implies by the law of large numbers

$$p = \frac{\tau_v^{-1}}{(\sigma^* - \tau_v^{-1})\beta_v^2\tau_v^{-1} + \sigma^*\beta_u^2\tau_u^{-1}}\left(\beta_u^2\tau_u^{-1}v + (\sigma^* - \tau_v^{-1})\beta_v p\right) + \gamma\tau_v^{-1}\frac{(\sigma^* - \tau_v^{-1})\beta_u^2\tau_u^{-1}}{(\sigma^* - \tau_v^{-1})\beta_v^2\tau_v^{-1} + \sigma^*\beta_u^2\tau_u^{-1}}u$$

Rearranging this expression, we have

$$\begin{aligned} p &= \frac{1}{1 - \frac{\tau_v^{-1}(\sigma^* - \tau_v^{-1})\beta_v}{(\sigma^* - \tau_v^{-1})\beta_v^2\tau_v^{-1} + \sigma^*\beta_u^2\tau_u^{-1}}}\left[\frac{\tau_v^{-1}\beta_u^2\tau_u^{-1}}{(\sigma^* - \tau_v^{-1})\beta_v^2\tau_v^{-1} + \sigma^*\beta_u^2\tau_u^{-1}}v + \gamma\tau_v^{-1}\frac{(\sigma^* - \tau_v^{-1})\beta_u^2\tau_u^{-1}}{(\sigma^* - \tau_v^{-1})\beta_v^2\tau_v^{-1} + \sigma^*\beta_u^2\tau_u^{-1}}u\right] \\ &= \frac{1}{(\sigma^* - \tau_v^{-1})\beta_v^2\tau_v^{-1} + \sigma^*\beta_u^2\tau_u^{-1} - \tau_v^{-1}(\sigma^* - \tau_v^{-1})\beta_v}\left[\tau_v^{-1}\beta_u^2\tau_u^{-1}v + \gamma\tau_v^{-1}(\sigma^* - \tau_v^{-1})\beta_u^2\tau_u^{-1}u\right] \end{aligned}$$

Matching the loadings on v and u with β_v and β_u, respectively, we obtain the system of two equations in the two unknowns (β_v, β_u):

$$\begin{aligned} \beta_v &= \frac{\tau_v^{-1}\beta_u^2\tau_u^{-1}}{(\sigma^* - \tau_v^{-1})\beta_v^2\tau_v^{-1} + \sigma^*\beta_u^2\tau_u^{-1} - \tau_v^{-1}(\sigma^* - \tau_v^{-1})\beta_v} \\ \beta_u &= \frac{\gamma\tau_v^{-1}(\sigma^* - \tau_v^{-1})\beta_u^2\tau_u^{-1}}{(\sigma^* - \tau_v^{-1})\beta_v^2\tau_v^{-1} + \sigma^*\beta_u^2\tau_u^{-1} - \tau_v^{-1}(\sigma^* - \tau_v^{-1})\beta_v} \end{aligned}$$

The solution is

$$\begin{aligned} \beta_v &= \frac{\tau_v^{-1}[1 + \gamma^2\tau_u^{-1}(\sigma^* - \tau_v^{-1})]}{\tau_v^{-1} + \gamma^2\sigma^*\tau_u^{-1}(\sigma^* - \tau_v^{-1})} \\ \beta_u &= \gamma(\sigma^* - \tau_v^{-1})\beta_v \end{aligned}$$

Thus, we have solved (β_v, β_u) in terms of the symmetric equilibrium information acquisition z^*, or equivalently σ^*. These results so far also match the limiting economy of Hellwig (1980) for the case of identical risk aversions.

E.1 Equilibrium information acquisition

Let us now drop i subscripts everywhere, and use the notation $\sigma(z) := \frac{1}{2}\tau_v^{-1}(z^{-1} + 1)$ for the individual-specific signal variance. To solve the information choice, a la Verrecchia

(1982), first compute the ex-ante utility

$$\begin{aligned} V(z) &:= \mathbb{E}\Big[\max_d \mathbb{E}\big[\tilde{V} \mid s,p\big]\Big] \\ &= -\mathbb{E}\Big[\exp\Big(-\frac{1}{2}\frac{(\tilde{\mu}_v - p)^2}{\tilde{\sigma}_v} + \gamma c(z)\Big)\Big] \end{aligned}$$

To compute this unconditional expectation, note that $\tilde{\mu}_v$ is linear in (s,p), which has a normal distribution. The joint density is of (s,p) is

$$\begin{aligned} \varphi(s,p) &= \frac{1}{2\pi} K_0^{1/2} \exp\Big[-\frac{1}{2}\big(k_s s^2 + 2k_{sp} sp + k_p p^2\big)\Big] \\ K_0 &:= \frac{1}{\det(\tilde{\Sigma})} = \frac{1}{(\sigma(z) - \tau_v^{-1})\beta_v^2 \tau_v^{-1} + \sigma(z)\beta_u^2 \tau_u^{-1}} \\ k_s &= \tilde{\Sigma}_{11}^{-1} = K_0(\beta_v^2 \tau_v^{-1} + \beta_u^2 \tau_u^{-1}) \\ k_{sp} &= \tilde{\Sigma}_{12}^{-1} = \tilde{\Sigma}_{21}^{-1} = -K_0 \beta_v \tau_v^{-1} \\ k_p &= \tilde{\Sigma}_{22}^{-1} = K_0 \sigma(z) \end{aligned}$$

Furthermore, write $-\frac{1}{2}\frac{(\tilde{\mu}_v - p)^2}{\tilde{\tau}_v^{-1}} = a(b_s s + b_p p)^2$ where we define

$$\begin{aligned} a &:= -\frac{1}{2\tilde{\sigma}_v} = -\frac{1}{2} K_0^{-1} \frac{1}{(\sigma(z) - \tau_v^{-1})\beta_u^2 \tau_v^{-1} \tau_u^{-1}} \\ b_s &:= K_0 \beta_u^2 \tau_u^{-1} \tau_v^{-1} \\ b_p &:= K_0[(\sigma(z) - \tau_v^{-1})\beta_v \tau_v^{-1} - K_0^{-1}] \end{aligned}$$

Using this expression and the normal density, compute

$$\begin{aligned} \mathbb{E}\Big[\exp\Big(-\frac{1}{2}\frac{(\tilde{\mu}_v - p)^2}{\tilde{\sigma}_v}\Big)\Big] &= \mathbb{E}[e^{a(b_s s + b_p p)^2}] \\ &= \frac{1}{2\pi} K_0^{1/2} \iint e^{a(b_s s + b_p p)^2} e^{-\frac{1}{2}(k_s s^2 + 2k_{sp} sp + k_p p^2)} \mathrm{d}s\mathrm{d}p \\ &= \frac{1}{2\pi} K_0^{1/2} \iint \exp\Big[-\frac{1}{2}\binom{s}{p}^\top \Omega^{-1} \binom{s}{p}\Big] \mathrm{d}s\mathrm{d}p, \end{aligned}$$

where

$$\Omega^{-1} := \begin{pmatrix} k_s - 2ab_s^2 & k_{sp} - 2ab_s b_p \\ k_{sp} - 2ab_s b_p & k_p - 2ab_p^2 \end{pmatrix}$$

Since the integral above involves a normal kernel, we obtain

$$V = -\exp[\gamma c(z)]\frac{K_0^{1/2}}{\det(\Omega^{-1})^{1/2}}$$

After a lengthy amount of algebra, we obtain

$$\Omega^{-1} = \begin{pmatrix} (\sigma(z)-\tau_v^{-1})^{-1} & -(\sigma(z)-\tau_v^{-1})^{-1} \\ -(\sigma(z)-\tau_v^{-1})^{-1} & \tau_v + (\sigma(z)-\tau_v^{-1})^{-1} + \frac{(1-\beta_v)^2}{\beta_u^2\tau_u^{-1}} \end{pmatrix}$$

Therefore,[16]

$$\frac{K_0}{\det(\Omega^{-1})} = \left(\tau_v + (\sigma(z)-\tau_v^{-1})^{-1} + \frac{\beta_v^2}{\beta_u^2\tau_u^{-1}}\right)^{-1}\left(\beta_u^2\tau_u^{-1} + (1-\beta_v)^2\tau_v^{-1}\right)^{-1}.$$

It is equivalent to maximize $-\log(-V)$, so we solve

$$\max_z -\gamma c(z) + \frac{1}{2}\log\left(\tau_v + (\sigma(z)-\tau_v^{-1})^{-1} + \frac{\beta_v^2}{\beta_u^2\tau_u^{-1}}\right) + \frac{1}{2}\log\left(\beta_u^2\tau_u^{-1} + (1-\beta_v)^2\tau_v^{-1}\right)$$

Notice that the final term is irrelevant, given z is absent. Also, at this point let us make the replacement $(\sigma(z)-\tau_v^{-1})^{-1} = 2\tau_v\frac{z}{1-z}$. The FOC for z is

$$\gamma(1-z)^2c'(z)\left[1+2\frac{z}{1-z}+\frac{\beta_v^2\tau_v^{-1}}{\beta_u^2\tau_u^{-1}}\right] \quad \begin{cases} \geq 1, & \text{if } z=0; \\ = 1, & \text{if } z\in(0,1); \\ \leq 1, & \text{if } z=1. \end{cases} \tag{E.1}$$

[16] Note that Verrecchia (1982) makes an algebra mistake in deriving his equation (7). Indeed, in deriving his ex-ante value function, he obtains (after translating into our notation)

$$\frac{K_0}{\det(\Omega^{-1})} = \left(\tau_v + (\sigma(z)-\tau_v^{-1})^{-1} + \frac{\beta_v^2}{\beta_u^2\tau_u^{-1}}\right)^{-1}\frac{1}{\beta_u^2\tau_u^{-1}}.$$

One can check that the mistake originates in his appendix, where he uses (his notation) $a_3 + a_6 = h_0 + s$ rather than the correct expression $a_3 + a_6 = h_0 + s + \frac{(1-\beta)^2}{\gamma^2 V}$. That said, this mistake is inconsequential, because it is multiplicatively separable from the information choice, as can be seen in the subsequent derivations.

In a symmetric equilibrium where agents choose z^*, we may use the expressions for $\beta_v/\beta_u = 2\gamma^{-1}\tau_v z^*/(1-z^*)$ to rewrite (E.1) as the equilibrium condition

$$\gamma(1-z^*)^2 c'(z^*)\Big[1+2\big(\frac{z^*}{1-z^*}\big)+\frac{4}{\gamma^2\tau_u^{-1}\tau_v^{-1}}\big(\frac{z^*}{1-z^*}\big)^2\Big] \quad \begin{cases} \geq 1, & \text{if } z^*=0; \\ =1, & \text{if } z^*\in(0,1); \\ \leq 1, & \text{if } z^*=1. \end{cases} \tag{E.2}$$

again with equality if $z^*>0$. Under some conditions on the cost function, Verrecchia (1982) proves that a solution exists to this condition, hence an NREE exists with endogenous information acquisition. Notice that $\gamma c'(0)<1$ suffices to ensure that the solution necessarily satisfies $z^*>0$.

Let us, for reference, also write the pricing function in terms of z^* rather than the variance σ^*. We have

$$\begin{aligned} p &= \beta_v v + \beta_u u \\ \text{where}\quad \beta_v &= \frac{1+\frac{\gamma^2}{2}\tau_v^{-1}\tau_u^{-1}\frac{1-z^*}{z^*}}{1+\frac{\gamma^2}{4}\tau_v^{-1}\tau_u^{-1}\frac{1-z^*}{z^*}\frac{1+z^*}{z^*}} \\ \beta_u &= \frac{\gamma\tau_v^{-1}}{2}\frac{1-z^*}{z^*}\beta_v \end{aligned} \tag{E.3}$$

In general, this economy cannot feature fully-revealing prices regardless of the marginal cost at zero, $c'(0)$. To see this, imagine holding $\tau_u\in(0,\infty)$, $\tau_v\in(0,\infty)$, and $\gamma>0$ fixed and vary the information cost function $c(\cdot;\chi)$ with some perturbation parameter χ, which then moves $z^*(\chi)$ in equilibrium. Notice from expression (E.3) that fully-revealing prices obtain if and only if $z^*(\chi)\to 1$ as we move χ in a manner that reduces the information cost. But from the (E.2), we can easily prove that $z^*(\chi)\to 1$ requires $c'(1;\chi)\leq\frac{\gamma}{4}\tau_u^{-1}\tau_v^{-1}$ for all χ corresponding to $z^*(\chi)$ close enough to 1. In other words, what matters in an NREE for fully-revealing prices is that each agent obtains a perfectly-informative private signal. This places a strong requirement on the cost function at $z=1$, with the shape of the cost function at $z=0$ being irrelevant. In summary, holding (τ_u,τ_v,γ) fixed, informativeness is bounded above in NREE, even if $c'(0)=0$.

E.2 Two examples: zero and positive marginal cost at zero

We consider two examples: one with $c'(0)=0$ and the second with $c'(0)>0$.

Example 1 (quadratic cost). Consider the cost function

$$c(z) = \frac{\kappa}{2}\left(\frac{z}{1-z}\right)^2. \tag{E.4}$$

This function satisfies $c'(0) = 0$ and therefore is capable of generating infinite price informativeness in our baseline Kyle model in the text. To provide an interpretation, recall that the chosen precision over the signal error ε_i satisfies $s_i = \frac{2}{\tau_v^{-1}}\frac{z_i}{1-z_i}$. Hence, the function (E.4) represents quadratic costs over this signal error precision.

Evaluating the equilibrium condition (E.2) for this case, we have

$$\gamma\kappa\left[\frac{z^*}{1-z^*} + 2\left(\frac{z^*}{1-z^*}\right)^2 + \frac{4}{\gamma^2\tau_u^{-1}\tau_v^{-1}}\left(\frac{z^*}{1-z^*}\right)^3\right] = 1 \tag{E.5}$$

Thus, $z^*/(1-z^*)$ solves a cubic equation. There is exactly one root satisfying $z^* \in (0,1)$.

Example 2 (linear cost). Now, consider the cost function[17]

$$c(z) = \chi\left(\frac{z}{1-z}\right), \quad \text{where} \quad \gamma\chi < 1. \tag{E.6}$$

This function satisfies $c'(0) = \chi > 0$. (One must take caution interpreting comparative statics with respect to χ, because here χ modulates the *entire cost function* in addition to the marginal cost $c'(0)$.) Also note that information costs are "larger" here than for the quadratic case above. Indeed, since the chosen precision over the signal error ε_i satisfies $s_i = \frac{2}{\tau_v^{-1}}\frac{z_i}{1-z_i}$, the function (E.6) represents linear costs over signal error precision.

Evaluating the equilibrium condition (E.2) for this case, we have

$$\gamma\chi\left[1 + 2\left(\frac{z^*}{1-z^*}\right) + \frac{4}{\gamma^2\tau_u^{-1}\tau_v^{-1}}\left(\frac{z^*}{1-z^*}\right)^2\right] = 1 \tag{E.7}$$

Thus, $z^*/(1-z^*)$ solves a quadratic equation. Assuming $\gamma\chi < 1$, there exists a unique positive solution z^*. (If $\gamma\chi > 1$, then there is no positive solution, so $z^* = 0$ is the equilibrium.) This unique solution is

$$\frac{z^*}{1-z^*} = \frac{\gamma\tau_u^{-1}\tau_v^{-1}}{4\chi}\left[-\gamma\chi + \sqrt{(\gamma\chi)^2 + (1-\gamma\chi)\frac{4\chi}{\gamma\tau_u^{-1}\tau_v^{-1}}}\right] \tag{E.8}$$

[17] This cost function is the one which is minimally-convex in signal precisions—see footnote 14.

E.3 Price informativeness, liquidity, and volatility

Analogous to the baseline model, define the following measures of (relative) price informativeness, volatility, and liquidity:

$$\begin{aligned}\mathcal{I} &:= 1 - \frac{\text{Var}[v \mid p]}{\text{Var}[p]} \\ \mathcal{L} &:= [\partial p / \partial u]^{-1} \\ \mathcal{V} &:= \text{Std}[p]\end{aligned}$$

The informativeness and volatility measures are exactly analogous to our baseline model. The liquidity measure deserves discussion. Note that $\mathcal{L}^{-1}$ is the price response to an exogenous increase in demand, whereas in our baseline model it was the loading of price on aggregate demand. The definition is written this way for the NREE because "aggregate demand" is not well-defined here—it is zero by market clearing. But a noise shock represents an exogenous increase in asset demand that must be absorbed by the informed traders, and hence generates a price impact. As an alternative way to justify our definition of $\mathcal{L}^{-1}$, recall that price impact in the baseline model comes out exactly equal to the price loading on the noise shock u. That is, the result for price impact in our baseline model coincides with $\partial p / \partial u$. By contrast, volume is not well-defined in the competitive economy, so we do not examine it.

Using expression (E.3) and properties of the joint normal distribution, compute

$$\mathcal{I} = \frac{\beta_v^2 \tau_v^{-1}}{\beta_u^2 \tau_u^{-1} + \beta_v^2 \tau_v^{-1}} = \frac{1}{1 + \frac{\gamma^2}{4}\left(\frac{1-z^*}{z^*}\right)^2 \tau_u^{-1} \tau_v^{-1}} \tag{E.9}$$

$$\mathcal{L} = \beta_u^{-1} \tag{E.10}$$

$$\mathcal{V} = \sqrt{\beta_u^2 \tau_u^{-1} + \beta_v^2 \tau_v^{-1}} \tag{E.11}$$

E.4 Comparative statics

We have the following proposition characterizing how informativeness, liquidity, and volatility co-move with τ_u^{-1} and τ_v^{-1}. There is no obvious way to obtain comparative statics with information costs (unlike our large-market limit), because the details of the entire information cost function matter for equilibrium.

Proposition E.1. *In the NREE model above, the following hold:*

(i) Informativeness $\mathcal{I}$ is always decreasing with fundamental uncertainty τ_v^{-1} and noise τ_u^{-1}.

(ii) Liquidity $\mathcal{L}$ is always decreasing with fundamental uncertainty τ_v^{-1} and increasing with noise τ_u^{-1}.

(iii) Volatility $\mathcal{V}$ is always increasing with fundamental uncertainty τ_v^{-1}. Provided either equilibrium information-collection z^* is not too close to 1 or the curvature of the information cost function is not too high, $\mathcal{V}$ is also increasing with noise τ_u^{-1}.

Preliminaries. As a preliminary, we use the equilibrium condition (E.2) to examine how information-collection z^* varies with noise and uncertainty. Define the object

$$\nu(z) := (1-z)\frac{c''(z)}{c'(z)}. \tag{E.12}$$

For the information cost to be convex in precision, we require $\nu(z) \geq 2$ for all z (see footnote 14). Using the implicit function theorem and differentiating (E.2), assuming $z^* \in (0,1)$, we have

$$\begin{aligned}\frac{d\log z^*}{d\log \tau_u^{-1}} &= \frac{d\log z^*}{d\log \tau_v^{-1}} \\ &= \frac{4z^*\frac{\tau_u\tau_v}{\gamma^2}(1-z^*)}{4z^*\frac{\tau_u\tau_v}{\gamma^2}\big(2(1-z^*)+z^*\nu(z^*)\big)+(1+z^*)(1-z^*)\nu(z^*)-2z^*(1-z^*)}\end{aligned} \tag{E.13}$$

Since $\nu(z) \geq 2$, we have that

$$0 < \frac{d\log z^*}{d\log \tau_u^{-1}} = \frac{d\log z^*}{d\log \tau_v^{-1}} \leq \frac{1-z^*}{2}\frac{4z^*\frac{\tau_u\tau_v}{\gamma^2}}{4z^*\frac{\tau_u\tau_v}{\gamma^2}+1-z^*} < \frac{1-z^*}{2}. \tag{E.14}$$

We make repeated use of (E.14) below.

For the analysis of liquidity and volatility below, let us also note the following useful relationship. By examining the expressions (E.9), (E.10), and (E.11), we see that

$$1-\mathcal{I} = \frac{\tau_u^{-1}}{(\mathcal{L}\mathcal{V})^2}. \tag{E.15}$$

Consequently, it holds that

$$\begin{aligned}\frac{d\log(1-\mathcal{I})}{d\log \tau_u^{-1}} &= 1-2\Big(\frac{d\log\mathcal{L}}{d\log\tau_u^{-1}}+\frac{d\log\mathcal{V}}{d\log\tau_u^{-1}}\Big)\\ \frac{d\log(1-\mathcal{I})}{d\log \tau_v^{-1}} &= -2\Big(\frac{d\log\mathcal{L}}{d\log\tau_v^{-1}}+\frac{d\log\mathcal{V}}{d\log\tau_v^{-1}}\Big)\end{aligned}$$

Informativeness in NREE. We start by identifying how informativeness varies with τ_u and τ_v in NREE. Using the expression for $\mathcal{I}$ in (E.9), we have that

$$\frac{d\log\mathcal{I}}{d\log\tau_u^{-1}} = \frac{\gamma^2}{2}\frac{1-z^*}{(z^*)^2}\mathcal{I}\tau_v^{-1}\Big[\frac{d\log z^*}{d\log\tau_u^{-1}} - \frac{1-z^*}{2}\Big]$$
$$\frac{d\log\mathcal{I}}{d\log\tau_v^{-1}} = \frac{\gamma^2}{2}\frac{1-z^*}{(z^*)^2}\mathcal{I}\tau_u^{-1}\Big[\frac{d\log z^*}{d\log\tau_v^{-1}} - \frac{1-z^*}{2}\Big]$$

Using the upper bound in (E.14), we have that whenever $z^* \in (0,1)$ it holds that

$$\frac{d\log\mathcal{I}}{d\log\tau_u^{-1}} < 0 \quad \text{and} \quad \frac{d\log\mathcal{I}}{d\log\tau_v^{-1}} < 0$$

as desired. ■

Liquidity in NREE. We now analyze how liquidity $\mathcal{L}$ varies with noise τ_u^{-1} and fundamental uncertainty τ_v^{-1} in NREE.

First, substitute the expression for β_u from (E.3) into the expression for liquidity in (E.10) and simplify terms to write

$$\mathcal{L} = \underbrace{\Big(\frac{D_2}{D_1}\Big)}_{\beta_v^{-1}}\Big(\frac{\gamma}{2}\frac{1-z^*}{z^*}\tau_v^{-1}\Big)^{-1}$$
$$\text{where} \quad D_1 := 1 + \frac{\gamma^2}{2}\tau_u^{-1}\tau_v^{-1}\frac{1-z^*}{z^*}, \quad D_2 := 1 + \frac{\gamma^2}{2}\tau_u^{-1}\tau_v^{-1}\frac{1-z^*}{z^*}\frac{1+z^*}{2z^*}.$$

We first differentiate the β_v term. We have

$$\frac{d\log\beta_v}{d\log\tau_u^{-1}} = \frac{\frac{\gamma^2}{2}\tau_u^{-1}\tau_v^{-1}\frac{1-z^*}{z^*}\big[1 + \frac{d\log(\frac{1-z^*}{z^*})}{d\log\tau_u^{-1}}\big]}{D_1} - \frac{\frac{\gamma^2}{2}\tau_u^{-1}\tau_v^{-1}\frac{1-z^*}{z^*}\frac{1+z^*}{2z^*}\big[1 + \frac{d\log(\frac{1-z^*}{z^*}\frac{1+z^*}{2z^*})}{d\log\tau_u^{-1}}\big]}{D_2}$$

We compute

$$\frac{d\log(\frac{1-z^*}{z^*})}{d\log\tau_u^{-1}} = -\frac{1}{1-z^*}\frac{d\log z^*}{d\log\tau_u^{-1}}$$
$$\frac{d\log(\frac{1+z^*}{2z^*})}{d\log\tau_u^{-1}} = -\frac{1}{1+z^*}\frac{d\log z^*}{d\log\tau_u^{-1}}$$

Substituting and doing some algebra, we have

$$\frac{d\log\beta_v}{d\log\tau_u^{-1}} = \frac{\frac{\gamma^2}{2}\tau_u^{-1}\tau_v^{-1}\frac{1-z^*}{z^*}}{D_1D_2}\left[-\frac{1-z^*}{2z^*} + \frac{1}{z^*}\left(1+\frac{\gamma^2}{4}\tau_u^{-1}\tau_v^{-1}\frac{1-z^*}{z^*}\right)\frac{d\log z^*}{d\log\tau_u^{-1}}\right] \tag{E.16}$$

Therefore,

$$\frac{d\log\mathcal{L}}{d\log\tau_u^{-1}} = \frac{\frac{\gamma^2}{2}\tau_u^{-1}\tau_v^{-1}\frac{1-z^*}{z^*}}{D_1D_2}\left[\frac{1-z^*}{2z^*} - \frac{1}{z^*}\left(1+\frac{\gamma^2}{4}\tau_u^{-1}\tau_v^{-1}\frac{1-z^*}{z^*}\right)\frac{d\log z^*}{d\log\tau_u^{-1}}\right] + \frac{1}{1-z^*}\frac{d\log z^*}{d\log\tau_u^{-1}}$$

Combining terms, and simplifying the expression, we have

$$\frac{d\log\mathcal{L}}{d\log\tau_u^{-1}} = \frac{1}{D_1D_2(z^*)^2}\left[\frac{\gamma^2}{4}\tau_u^{-1}\tau_v^{-1}(1-z^*)^2 + \frac{F(z^*)}{1-z^*}\frac{d\log z^*}{d\log\tau_u^{-1}}\right],$$

where

$$F(z) := z^2 - \frac{\gamma^2}{4}\tau_u^{-1}\tau_v^{-1}(1-z)(1-5z) + \left(\frac{\gamma^2}{2}\tau_u^{-1}\tau_v^{-1}\right)^2(1-z)^2$$

Using the fact that $\frac{d\log z^*}{d\log\tau_u^{-1}} > 0$, we have

$$\frac{d\log\mathcal{L}}{d\log\tau_u^{-1}} > \frac{\frac{\gamma^2}{4}\tau_u^{-1}\tau_v^{-1}}{D_1D_2(z^*)^2}\left[(1-z^*)^2 + G(z^*)\frac{d\log z^*}{d\log\tau_u^{-1}}\right],$$

where

$$G(z) := \gamma^2\tau_u^{-1}\tau_v^{-1}(1-z) - (1-5z)$$

There are two cases to consider. If $\gamma^2\tau_u^{-1}\tau_v^{-1} \geq 1$, then $G(z) > 0$ for all $z \in (0,1)$, and so the fact that $\frac{d\log z^*}{d\log\tau_u^{-1}} > 0$ implies $\frac{d\log\mathcal{L}}{d\log\tau_u^{-1}} > 0$.

If instead $\gamma^2\tau_u^{-1}\tau_v^{-1} < 1$, then $G(0) < 0$, and so we proceed as follows. We substitute $\frac{d\log z^*}{d\log\tau_u^{-1}}$ from (E.13), and use the fact that $\nu(z) \geq 2$ to get the following bound that applies when $G(z^*) < 0$:

$$\frac{d\log\mathcal{L}}{d\log\tau_u^{-1}} > \frac{\frac{\gamma^2}{4}\tau_u^{-1}\tau_v^{-1}}{D_1D_2(z^*)^2}\left[(1-z^*)^2 + \frac{G(z^*)}{2}\frac{4z^*(1-z^*)\frac{\tau_u\tau_v}{\gamma^2}}{4z^*\frac{\tau_u\tau_v}{\gamma^2} + (1-z^*)}\right], \quad \text{if} \quad G(z^*) < 0.$$

After some algebra, we can rewrite this bound as

$$\frac{d\log\mathcal{L}}{d\log\tau_u^{-1}} > \frac{\frac{\gamma^2}{4}\tau_u^{-1}\tau_v^{-1}(1-z^*)}{D_1 D_2(z^*)^2}\Big[1+z^*-\frac{1}{2}\frac{4z^*(1-z^*)\frac{\tau_u\tau_v}{\gamma^2}}{4z^*\frac{\tau_u\tau_v}{\gamma^2}+(1-z^*)}\Big], \quad \text{if} \quad G(z^*)<0.$$

It is easy to see that the term in square brackets is positive, proving that $\frac{d\log\mathcal{L}}{d\log\tau_u^{-1}} > 0$ in this second case as well. Thus, we have in general

$$\frac{d\log\mathcal{L}}{d\log\tau_u^{-1}} > 0.$$

For the effect of fundamental uncertainty, the proof is simpler. In the next section, we prove that $\frac{d\log\mathcal{V}}{d\log\tau_v^{-1}} > 0$. Given that we also previously showed $\frac{d\log\mathcal{I}}{d\log\tau_v^{-1}} < 0$, equation (E.15) implies that

$$\frac{d\log\mathcal{L}}{d\log\tau_v^{-1}} < 0$$

as desired. ■

Volatility in NREE. Finally, we analyze how volatility $\mathcal{V}$ varies with noise τ_u^{-1} and fundamental uncertainty τ_v^{-1}.

Substitute the expressions for β_u, β_v from (E.3) into the expression for volatility in (E.11) in order to rewrite volatility as

$$\mathcal{V} = \underbrace{\Big(\frac{D_1}{D_2}\Big)}_{=\beta_v}\tau_v^{-1/2}\Big[1+\tau_u^{-1}\tau_v^{-1}\Big(\frac{\gamma}{2}\frac{1-z^*}{z^*}\Big)^2\Big]^{1/2}$$

$$\text{where} \quad D_1 := 1+\frac{\gamma^2}{2}\tau_u^{-1}\tau_v^{-1}\frac{1-z^*}{z^*}, \quad D_2 := 1+\frac{\gamma^2}{2}\tau_u^{-1}\tau_v^{-1}\frac{1-z^*}{z^*}\frac{1+z^*}{2z^*}.$$

For reference, we also define the additional variable

$$D_3 := 1+\tau_u^{-1}\tau_v^{-1}\big(\frac{\gamma}{2}\frac{1-z^*}{z^*}\big)^2$$

We start with noise τ_u^{-1}. Differentiating, using (E.16), and simplifying the result, we

have

$$\frac{d\log \mathcal{V}}{d\log \tau_u^{-1}} = \frac{\frac{\gamma^2}{2}\tau_u^{-1}\tau_v^{-1}(\frac{1-z^*}{z^*})^2}{D_1D_2D_3} \times \left[\frac{D_1D_2 - 2D_3}{4}\left(1 - \frac{2}{1-z^*}\frac{d\log z^*}{d\log \tau_u^{-1}}\right) + \frac{D_3}{z^*}\frac{\gamma^2}{4}\tau_u^{-1}\tau_v^{-1}\frac{d\log z^*}{d\log \tau_u^{-1}}\right]$$

There are two cases to consider. In the first case, if z^* is not too close to 1, then[18]

$$D_1D_2 - 2D_3 = -1 + \gamma^2\tau_u^{-1}\tau_v^{-1}\frac{1-z^*}{z^*} + \left(\frac{\gamma^2}{2}\tau_u^{-1}\tau_v^{-1}\frac{1-z^*}{z^*}\right)^2\frac{1+z^*}{2z^*}$$

is positive, which implies by the bounds (E.14) that $\frac{d\log \mathcal{V}}{d\log \tau_u^{-1}} > 0$. In the second case, suppose z^* is close to 1. We rewrite the derivative as

$$\frac{d\log \mathcal{V}}{d\log \tau_u^{-1}} = \frac{\frac{\gamma^2}{2}\tau_u^{-1}\tau_v^{-1}(\frac{1-z^*}{z^*})^2}{D_1D_2D_3} \times \left[\frac{D_1D_2 - D_3}{4}\left(1 - \frac{2}{1-z^*}\frac{d\log z^*}{d\log \tau_u^{-1}}\right) - \frac{D_3}{4}\left(1 - \frac{2}{1-z^*}\frac{d\log z^*}{d\log \tau_u^{-1}}\right) + \frac{D_3}{4}\frac{\gamma^2\tau_u^{-1}\tau_v^{-1}}{z^*}\frac{d\log z^*}{d\log \tau_u^{-1}}\right]$$

Note that

$$D_1D_2 - D_3 = \frac{1+z^*}{2z^*}\left(\frac{\gamma^2}{2}\tau_u^{-1}\tau_v^{-1}\frac{1-z^*}{z^*}\right)^2 + \gamma^2\tau_u^{-1}\tau_v^{-1}\frac{1-z^*}{z^*}$$

is strictly positive, and so

$$\frac{d\log \mathcal{V}}{d\log \tau_u^{-1}} > \frac{\frac{\gamma^2}{2}\tau_u^{-1}\tau_v^{-1}(\frac{1-z^*}{z^*})^2}{4D_1D_2}\left[\frac{\gamma^2\tau_u^{-1}\tau_v^{-1}}{z^*}\frac{d\log z^*}{d\log \tau_u^{-1}} - \left(1 - \frac{2}{1-z^*}\frac{d\log z^*}{d\log \tau_u^{-1}}\right)\right] \tag{E.17}$$

Now, plugging in $\frac{d\log z^*}{d\log \tau_u^{-1}}$ from (E.13), the term in square brackets of the above expression

[18] A sufficient condition for $D_1D_2 - 2D_3 > 0$ is that

$$z^* < \frac{\gamma^2\tau_u^{-1}\tau_v^{-1}}{2(\sqrt{2}-1) + \gamma^2\tau_u^{-1}\tau_v^{-1}}.$$

can be written:

$$\frac{\gamma^2\tau_u^{-1}\tau_v^{-1}}{z^*}\frac{d\log z^*}{d\log\tau_u^{-1}} - \left(1 - \frac{2}{1-z^*}\frac{d\log z^*}{d\log\tau_u^{-1}}\right)$$
$$= \frac{(1-z^*)\left[2-\frac{\nu(z^*)}{2}\right] - \left[4z^*\frac{\tau_u\tau_v}{\gamma^2} + 1 - z^*\right]z^*\left(\frac{\nu(z^*)}{2}-1\right)}{4z^*\frac{\tau_u\tau_v}{\gamma^2}\left((1-z^*)+z^*\frac{\nu(z^*)}{2}\right) + (1-z^*)\frac{\nu(z^*)}{2} + z^*(1-z^*)\left(\frac{\nu(z^*)}{2}-1\right)}$$

For $\nu(z^*)$ close enough to its lower-bound of 2, this expression is also strictly positive. Thus, we have proved that

$$\frac{d\log\mathcal{V}}{d\log\tau_u^{-1}} > 0$$

for any parameters such that either (i) z^* is far enough from its upper-bound of 1; or (ii) $\nu(z^*)$ is close enough to its lower-bound of 2.

Regarding fundamental uncertainty, since $\frac{d\log z^*}{d\log\tau_u^{-1}} = \frac{d\log z^*}{d\log\tau_v^{-1}}$, we have

$$\frac{d\log\mathcal{V}}{d\log\tau_v^{-1}} = \frac{d\log\mathcal{V}}{d\log\tau_u^{-1}} + \frac{1}{2} \tag{E.18}$$

Using (E.17) and using $\frac{d\log z^*}{d\log\tau_u^{-1}} > 0$, we have that

$$\frac{d\log\mathcal{V}}{d\log\tau_u^{-1}} > -\frac{\frac{\gamma^2}{2}\tau_u^{-1}\tau_v^{-1}\left(\frac{1-z^*}{z^*}\right)^2}{4 + 4\frac{1+z^*}{2z^*}\left(\frac{\gamma^2}{2}\tau_u^{-1}\tau_v^{-1}\frac{1-z^*}{z^*}\right)^2 + 2\frac{\gamma^2}{2}\tau_u^{-1}\tau_v^{-1}\frac{1-z^*}{z^*}\frac{1-z^*+4z^*}{z^*}} > -\frac{1}{2}$$

Combining this result with (E.18), we have that

$$\frac{d\log\mathcal{V}}{d\log\tau_v^{-1}} > 0$$

as desired. ■